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OR

Driver Machine Interfaces for EMU/DMU, Locomotives and driving coaches - Functional and system requirements associated with harmonised Driver Machine Interfaces

Interfaces conducteur/machine sur les rames automotrices électriques/diesel (EMU/DMU), les locomotives et les cabines de réversibilité - Exigences fonctionnelles et exigences "système" vis-à-vis des interfaces harmonisées conducteur/machine
Schnittstelle Triebfahrzeugführer-Maschine für Elektro- und Dieseltriebwagen (EMU/DMU), Lokomotiven und Steuerwagen - Funktions- und Systemanforderungen für eine harmonisierte Schnittstelle Triebfahrzeugführer-Maschine

UNION INTERNATIONALE DES CHEMINS DE FER
INTERNATIONALER EISENBAHNVERBAND
INTERNATIONAL UNION OF RAILWAYS

Leaflet to be classified in Volume:

VI - Traction

Application:

With effect from 1 June 2009
All members of the International Union of Railways

The provisions in this leaflet are applicable to new designs of the rolling stock concerned.
This leaflet does not replace European legislation (TSI, including specific/national cases).
For upgrading and renewal of existing designs, UIC Leaflets 641 and 651 apply respectively.

Record of updates

1st edition, June 2009	First issue

The person responsible for this leaflet is named in the UIC Code

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Summary

This leaflet

- defines the operating elements on EMU/DMU, locomotives and driver cabs of driving coaches.
- sets out the rules associated with the functional requirements for driver cabs of EMU/DMU, locomotives and driving coaches.

1 - General

The functions and design associated with harmonised Driver Machine Interfaces comprise all the operationally relevant interfaces of the rolling stock concerned.

This specification describes the functional and system requirements associated with harmonised Driver Machine Interfaces for EMU/DMU (see List of abbreviations - page 131), locomotives and related driver cabs of driving coaches. This sets out all the functional requirements for the cab-equipped rolling stock concerned.

Objectives of this leaflet	Use	Verification
Define harmonised requirements in order to avoid project specific engineering of common rolling stock functions.	Minimize the scope of interpretation for solutions that meet the requirements, identify deviations/ non-compliance of functional requirements	Target/actual comparison "specification ← → manufacturer's solution".
Define functional requirements in a precise manner, so that functional changes or restrictions due to technical changes/ innovations are avoided or minimized. Functional changes are acceptable for the case of proven cost-effective LCC (see List of abbreviations - page 131) advantages (e.g. reduction of faults and training expenditure) or mandatory regulation (i. e. TSI including specific cases). Comprehensive physical interface-specifications for interchangeability, independent of the supplier, are the target for future standards.	An unambiguous definition of consistent functional interfaces (standardisation) enabling innovative design at module level: old components can be replaced with new ones easily.	• Costs for - spare parts and - maintenance and - repair. • Compliance with regulations and directives.
Retain known operational procedures for the driver, while trying to simplify them further.	Reduce training expenditure.	Training costs.
Avoid operating errors through largely self-explanatory and fault-tolerant functions and a high degree of automation and diagnostics.	Increase reliability.	• MTBF • Experience and User Feed-back

More than 15 years of European research studies have demonstrated that a standardised desk configuration is a key issue for all European drivers, irrespective of the specific type of rolling stock. Therefore the rolling stock specific deviations should be limited to a minimum. A set of now commonly shared conclusions and design principles have been provided by recent projects (e. g. EUCAB) and they have been considered in the course of the elaboration of the *UIC Leaflet 612 series* (see Bibliography - page 134).

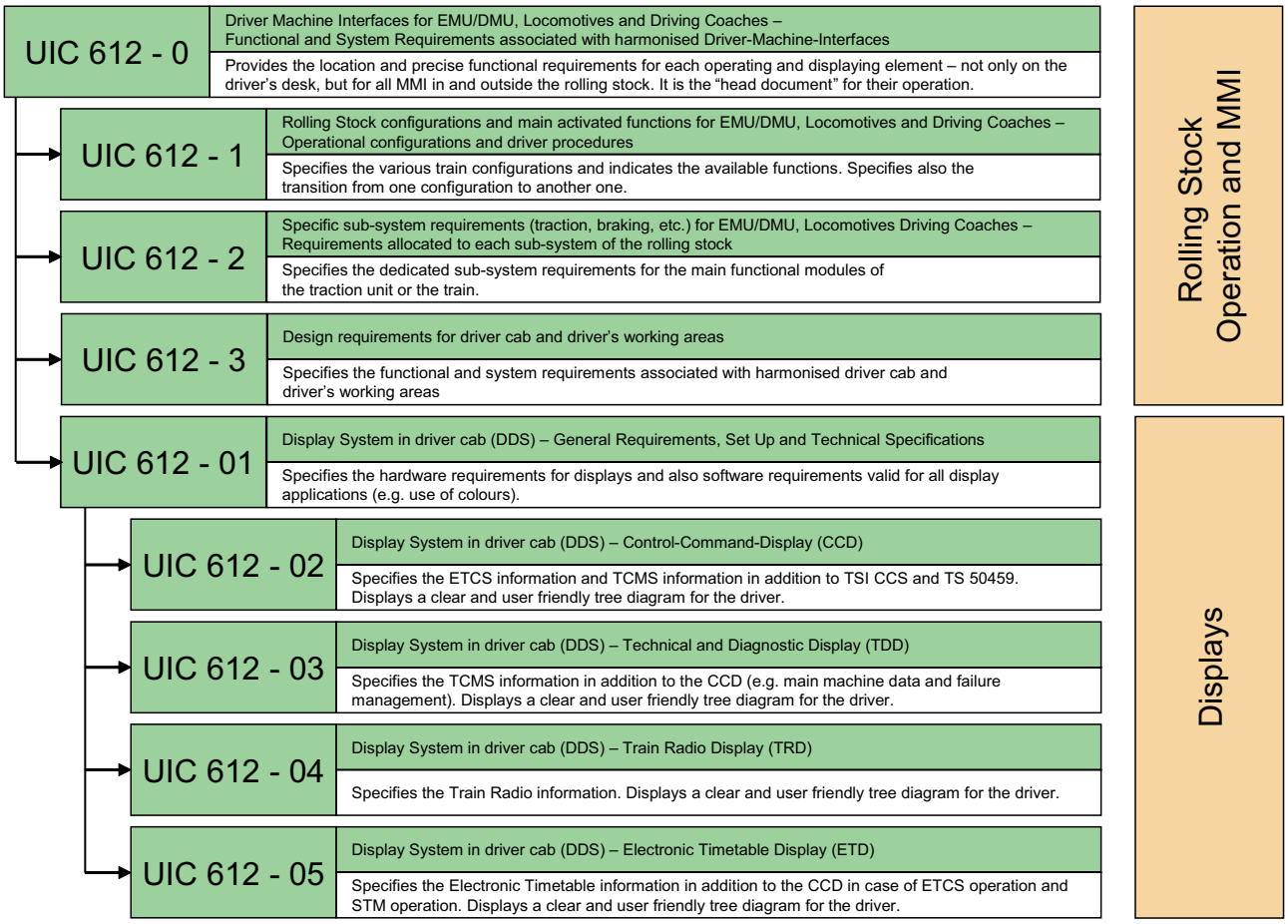


Fig. 1 - Hierarchy of UIC Leaflet 612 series

2 - Definitions

pos. xxx	Refers to a Driver Machine Interface shown in Appendix J - page 88.
Running (V > 0)	Detection in case of increasing speed from V > (1...3) km/h
Standstill (V = 0)	Detection in case of decreasing speed from V < (3 - Δv) km/h with 0 < Δv < 2,5 km/h
Immediate	As soon as possible, i.e. maximum duration t ≤ 1 s, if not specifically or otherwise defined.
T > 0	Traction power increasing from 0 kN
T = 0	T = 0 kN
Driving coach	A coach is a non-tractive vehicle in a fixed or variable formation capable of carrying passengers. A coach may be equipped with a driver cab. Such a coach is then referred to as a driving coach (see *CR RST TSI* - see Bibliography - page 134).
Cab	Refers to the compartment which is equipped with actuators and instruments for the driver to control EMU/DMU, locomotives and driving coaches in the train from and which is located at least at one end of a vehicle.
Engine room	The Engine room for EMU/DMU and locomotives is the area or compartment or cabinet where drive units (electric or diesel) and/or auxiliary units are situated or the equivalent for driving coaches (e. g. cubicle for ETCS on-board equipment and air cocks).
Automatic brake (Service brake and emergency brake)	Air brake system for the whole train, for example controlled by the Brake Pipe (BP - see List of abbreviations - page 131) , in compliance with TSI (including specific/national cases), EN and UIC.
Electropneumatic (EP) brake	See *CR RST TSI, section 4.2.4.1.2.2*.
Dynamic brake	Brake force generated on EMU/DMU and locomotives.
Parking brake	Brake used to prevent parked rolling stock moving under the specified conditions taking into account location, wind, gradient and rolling stock loading state until intentionally released.
Direct brake	Local brake on locomotives and optionally for EMU/DMUs and driving coaches, where the brake force is directly commanded by the driver, e.g. by means of TCMS (see List of abbreviations - page 131). The brake force will be controlled from 0 kN to the maximum brake force designed.
Holding brake	See *EN14478* (see Bibliography - page 134)
STM	See *CCS TSI* (see Bibliography - page 134).
Brake flip switch	Switch that provides flip position(s) for pulse selection as well as stable position(s) for permanent selection of a switch position.

Display system	Consists of the display units (max. 4 per main desk), the display control unit and the internal and external interfaces to the TCMS.
Display control unit	Control of the complete display system and distribution of the display images regarding the defined concepts in normal operation and in case of failure
Display unit	Hardware to display the defined display images with circumferential keys for driver interaction and selection of functions offered by these keys (hard and soft keys).
Display image	Visible image with a defined content in a display unit.
Redundancy images	Display image with merged content out of two regular display images, which shall be displayed on the remaining display unit (e. g. in case of failure of one display unit).

Explanation of the tables below:

For more complex operating elements, the symbols in the following table are only represented schematically. The detailed design constraints (minimum and maximum of dimensions, application forces and angles) are contained in Appendix K - page 97.

Device [*reference to shape as shown in Appendix K*]	**Symbol/ indication + name**	**Positions and function** **S** = stable (fixed) position (the device remains in the position after its application "brake switch") **U** = unstable (= dynamic/spring loaded) position (the device does not remain in the position after its application, it moves automatically in the former stable position "flip switch") **I** = impulse (bi-stable, e. g. push-button or flip switch) (digital operation between two states "on"-"off") Any angle of actuation indicates a position or range referring to the vertical position of the switch.	**Explanations** **Mandatory preconditions** **Miscellaneous**	**Operational constraint for** (see definitions below table)				
				Master Cab status: Defined by driver's ID-card and travel direction (see pos. 033-036)	**MCB**	**v=0**	**v>0**	**MT**

MCB = main circuit breaker (function depends on status of the MCB)
MT = multiple traction (function also valid on slave EMU/DMU and locomotives)

Operational constraints linked to

1. **Master cab status - Defined by Driver's identity (ID) card (pos. 036) and Travel Direction (pos. 033 - 035):**

   - no entry: **function shall be available** in a EMU/DMU, locomotive or driving coach in "switched on" configuration and all higher configurations (see *UIC Leaflet 612-1*), irrespective of the activation of the driver cab (driver's ID card is not inserted - pos. 036 - or cab is not activated by the driver's ID card override button - pos. 301).
   - F, N, R: **function shall only be available in an activated driver cab** (driver's ID card is inserted - pos. 036 - or cab is activated by the driver's ID card override button - pos. 301. A Travel Direction may be chosen in addition - pos. 033 - 035) = "In service"-configuration and "driving"-configuration (see *UIC Leaflet 612-1*).
   - F, R: **function shall only be available in an activated driver cab** (driver's ID card is inserted - pos. 036 - or cab is activated by the driver's ID card override button - pos. 301) **and with a chosen Travel Direction** (F - forwards - pos. 033, R - reverse - pos. 035) = "driving"-configuration (see UIC *UIC Leaflet 612-1*).
   - 0: **function shall only be available in a non-activated driver cab** (driver's ID card is removed).

2. **MCB (main circuit breaker):**

   - no entry **function shall be available**, irrespective of whether the MCB is closed or not.
   - X function shall only be available after closing the MCB.

3. **V (speed):**

   - no entry **function shall be available**, irrespective of the vehicle speed.
   - V = 0: X function shall only be available in case of vehicle at standstill.
   - V > 0: X function shall not be available in case of vehicle at standstill.

4. **MT (multiple traction-signals transmitted by train-wide communication):**

   - no entry function shall **only** be available on the **master unit**.
   - X function shall be available on the master unit and all slave units (i.e. transmission of the command/message via train-wide communication).

3 - Structure of the operating areas, general information and basic principles

o 3.1 - General requirements

The following requirements are valid as guidelines for operating the vehicle and thus also for the layout of the driver cab:

- The driver must devote maximum possible attention to signals and obeying their aspects. Therefore the driver must be able to carry out all operational control actions without being distracted from observing the track system and signals. He shall not operate other secondary or non-essential control equipment. Therefore the driver must not interact with the rear panel (see Appendix E - page 59) other than at standstill.

- Furthermore, he must at all times be able to monitor clearly, unambiguously and immediately the brake pressure (pressure information for brake pipe (BP) pressure (pos. 007 or TDD - see *UIC Leaflet 612-03*) and brake cylinder pressures (pos. 008 or TDD - see *UIC Leaflet 612-03*), the information pertaining to signal aspects or to monitoring train control systems (driver cab signalling, CCD - see *UIC Leaflet 612-02*) and the timetable (ETD - see *UIC Leaflet 612-05*) during the running of the train under all lighting and operating conditions (see List of abbreviations - page 131).

Visual and audible information that is not essential for driving should only be given if it has an impact on operation (state of EMU/DMU, locomotives and driving coaches or train, malfunction etc.). A strict standard shall be implemented here in order to avoid unnecessary distraction. This also requires manufacturer information not to be marked on components in the driver cab and display images that are not essential to functions to be omitted (e.g. superfluous information in the display screens).

Regarding visibility and location of signals, the cab design shall comply with *UIC Leaflet 612 series*, TSI and EN (see Bibliography - page 134). The standardised driver desk may be installed on the left, on the right, or in a central position in the driver cab.

All driver's seat positions shall allow for unrestricted and unobstructed operation when seated, it shall not prevent operation in standing position and it shall be possible to open the side doors or side windows without obstruction at any time. Control operations when standing shall be possible from the main desk without restrictions.

Operating functions of minor importance (e. g. light switches), for rare specific operating cases (several test buttons) or for maintenance purposes (e. g. indication of specific data) may be integrated in the DDS (e. g. TDD - see *UIC Leaflet 612-03*) if their operation is only requested in an activated cab (= "In-service" and "driving" configuration - see *UIC Leaflet 612-1*).

Basic rule: TCMS shall detect failures and any prohibited driver action on interfacing DMIs in order to differentiate these from genuine driver commands.

In the event of a failure a message shall be displayed on the TDD (see *UIC Leaflet 612-03*) and the function of the signal shall be suspended until it is reinstated by the TCMS (e.g. by realising the next fault-free operation or any other criteria depending on the individual case). If this malfunction appears frequently, the TDD failure management shall advise the driver of such occurrences. For the diagnostic system requirements see *UIC Leaflets 612-2 and 652*.

The structure of *UIC Leaflets 612-0, 612-1, 612-2, 612-3, and 612-01 to 612-05* is demonstrated in Fig. 2.

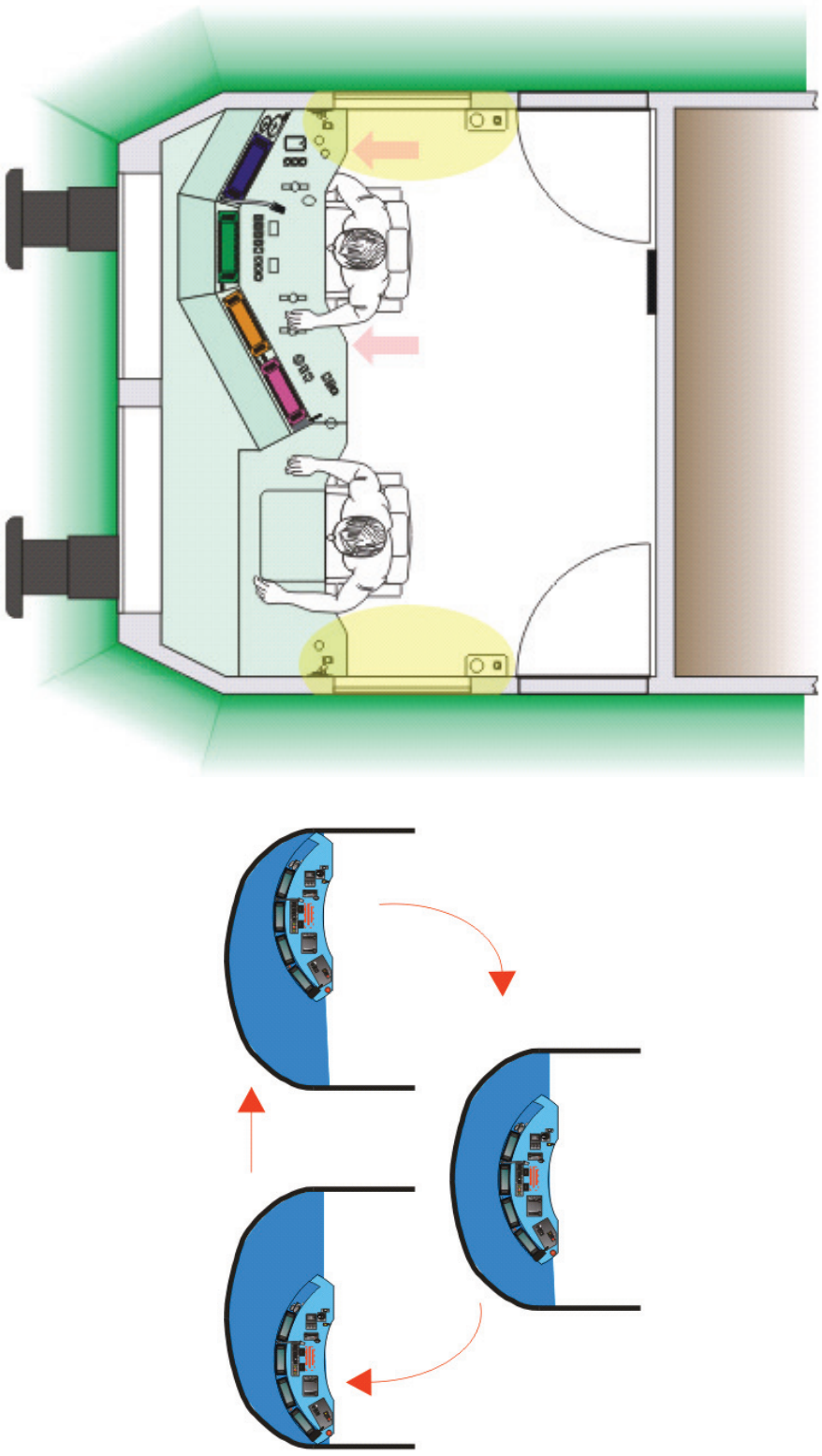


Fig. 2 - Locomotive cab with driver's desk on the right hand position
(The driver's desk is standardised, but the location of the desk is optional)

Document Hierarchy (colour-indication see Fig. 2 - page 9):

	UIC Leaflet 612-0	**Driver Machine Interfaces for EMU/DMU**
		1. General
		2. Definitions
		3. Structure of the operating areas, general information and basic principles
		Appendix A - page 28: Operating and display elements on the driver's desk
		Appendix B - page 52: Operating and display elements below the driver's desk
		Appendix C - page 55: Operating and display elements in the cab at further specific locations
		Appendix D - page 56: Auxiliary driver's desks
		Appendix E - page 59: Operating elements and equipment located in the cab, but not accessible to the driver while driving
		Appendix F - page 66: Operating elements grouped together on each unit and not necessarily located in the cab; storage of equipment
		Appendix G - page 75: Operating and display elements outside the EMU/DMU, locomotives and driving coaches
		Appendix H - page 82: Markings
		Appendix I - page 84: Radio remote-control desk
		Appendix J - page 88: Operating elements (overview of Driver-Machine Interfaces) positions
		Appendix K - page 97 (informative): Specification of the operating elements
		Appendix L - page 116 (informative): Automatic Speed Control (ASC - see List of abbreviations - page 131)
		Appendix M - page 122: Examples for connecting to earth electric EMU and locomotives
		Appendix N - page 124: Specific cases - N.1 "DB, ÖBB" - N.2 "SNCF" - N.3 "Trenitalia"
		Appendix O - page 127: Guideline for the revision of *UIC Leaflet 612 series*: Human factors aspects and standardisation benefits
		List of abbreviations
	UIC Leaflet 612-1	**Rolling Stock configurations and main activated functions for EMU/DMU, locomotives and driving coaches**
	UIC Leaflet 612-2	**Specific sub-system requirements (traction, braking, etc.) for EMU/DMU, locomotives and driving coaches**
	UIC Leaflet 612-3	**Design requirements for driver cabs and driver's working areas**
	UIC Leaflet 612-01	**Display System in driver cabs (DDS): General Requirements, Set-up and Technical Specifications**
	UIC Leaflet 612-02	**DDS: Control-Command-Display (TDD)**

	UIC Leaflet 612-03	**DDS: Technical and Diagnostic Display (TDD)**
	UIC Leaflet 612-04	**DDS: Train Radio Display (TRD)**
	UIC Leaflet 612-05	**DDS: Electronic Timetable Display (ETD)**

3.2 - Requirements for DMI-elements

3.2.1 - Environmental conditions

O 3.2.1.1 - General

The operating elements shall be designed to withstand adverse ambient conditions and excessive mechanical stress (e.g. drinking, eating, smoking at the driver's desk, the effects of forces exerted during operation in case of collisions (e. g. in shunting operations) and situations requiring rapid reactions). Their operability shall not be affected by such factors.

O 3.2.1.2 - Obligatory requirements

Criterion in relation with driver machine interfaces	**Verification (TSI is mandatory - additional standards to be fulfilled, provided there is no conflict with the TSI - see Bibliography - page 134)**
General	*EN 547-3* (other prEN in preparation)
Vision (view angles)	*UIC Leaflet 651 / prEN 15xxx / RST TSI*
Optical quality of windscreen	RST TSI, *EN 15152* (derived from *UIC Leaflet 651*)
Maximum required actuating forces (= to be considered for applicability by weaker users)	*EN 894, DIN 33401, EN 1005-1, EN 1005-2, EN 1005-3*
Maximum possible actuating forces (= to be considered to prevent breakage/malfunction of devices due to force of very strong people)	*DIN 33411-1, DIN 33411-2, DIN 33411-3, DIN 33411- 4 (Physical Strengths of Men), EN 1005-1, EN 1005-2, EN 1005-3*
Salt spray test	*EN 50021-SS, EN 50155*
Damp heat	*EN 50155/IEC 68-2-30*
Insulation test	*EN 50124*
Mechanical impact due to ballast	For front windows: *EN15152 part 6.2.1*
Ambient condition (temperature, moisture, …)	*EN 50155*: - 40...+70 °C (= temperature class TX)
Lighting inside the cab (illumination)	*EN13272, prEN 15xxx*
EMC compatibility	*EN 50121-3-2; EN 61000-4-4*
Shock and vibration	*EN 50125-1/EN 50155/EN 61373*
Noise	*Noise TSI*
Pollution (in the cab)	No increase of pollution by application of any device
Interruption of voltage supply	*EN 50155, section 3.1.1.2, class S2*

Classification of chemically active substances	*EN 60721-3-5 5C2*
Classification of contaminating fluids	*EN 60721-3-5 5F2 (electric) 5F3 (Diesel)*

3.2.1.3 - Recommended requirements

Criterion in relation with driver machine interfaces	Verification (TSI is mandatory – additional standards to be fulfilled, provided there is no conflict with the TSI)
Protection class, inside the rolling stock, inside the cab.	*EN 60529:* IP 42 is recommended for the operating element itself and IP 44 is recommended when this element is integrated inside the cab.
Protection class, inside the rolling stock, outside the cab.	*EN 60529:* IP 54 is recommended.
Protection class, outside the rolling stock.	*EN 60529:* IP 65 is recommended.

o 3.2.2 - Colour identification

3.2.2.1 - General

All colours shall be matt in order to avoid reflections and shall comply with the referenced RAL according to ISO/EN.

Operating elements or operating equipment of minor operational importance (e.g. light switches, socket covers), operated infrequently in specific conditions (several test buttons) or for maintenance purposes (e.g. shut-off cocks for condensate collection tank) may be coloured in white, black or metal finish (not coated).

3.2.2.2 - Black/grey

Operating elements that shall be operated at the driver's desk **while driving** shall be finished **in black or grey** using the range of finishes RAL 9004, 9011, 9017 or 7016, 7021, 7024 according to register RAL 840-HR) like all vehicles of this design series, unless the following specifications (see tables in Appendices A - page 28 to I - page 84) request deviations or allow exceptions. Their operation shall not distract the driver from monitoring the track system and signals. It shall therefore be possible to operate them without observing them. RAL 9003 is to be used for the markings and symbols of the operating elements in the driver cab; RAL 9005 is to be used for the base.

3.2.2.3 - Yellow

Operating elements that are normally operated at **standstill only** shall be finished in **yellow** (RAL 1003). Consequently, no operating elements on the driver's desks are affected.

Exceptions:

- The outside air shut-off cocks for the hose couplings of the brake pipe (BP) shall be marked in yellow in order to comply with the colour of the pointer for the BP indication in pos. 007.

- On the driver's desk: ETCS acknowledgement (pos. 015), door-control (pos. 021, 023).

3.2.2.4 - Red

Operating elements that are operated in the **case of faults or emergency** are finished in **red** (RAL 3001). This applies, for instance, to the shut off cocks in the engine room (pos. 511 - 515), to the emergency stop valve (pos. 010), to the line safety switches (LSS) dedicated to ETCS (including STM and ATC) and to elements to be operated by the driver in the event of a fault, if a fault help text is not available (e. g. reset of all displays of the DDS - see List of abbreviations - page 131).

Exceptions:

- The outside end cocks for the hose couplings of the main air reservoir pipe (MP) shall be marked in red in order to comply with the colour of the pointer for the MP-indication in pos. 007 (see point N.2 - page 124).

- The key for the pantograph shut-off valve (pos. 510) and the valve housing shall be designed in red, even if operated regularly in specific networks.

o 3.2.3 - Shape identification

The shape of the handles (enclosed in **[ ]**) of the operating elements, specified in the following tables, is explained in Appendix K - page 97. The shape of pos. 010 is obligatory.

o 3.2.4 - Markings and nominal positions

The meaning, functions and positions of all operating elements, apart from exclusively national applications of control-command on-board systems (STM - see List of abbreviations - page 131), shall be marked with pictograms. Further information (names, details, position number of DMI) is available from the driver's manual only.

The parts of the operating elements visible to the driver (front panels, markings etc.) shall be designed neutrally; i.e. no names or descriptions of manufacturers, brands or companies or similar may be attached to these parts.

If the operating elements described subsequently require a plain text label, the following applies:

- The function "On" is represented by a Roman "I"; otherwise Arabic numerals shall be used,

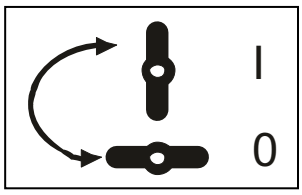
- *ISO 7000* (see Bibliography - page 134) shall be applied to plain text markings (numerals in medium-spaced lettering, texts in close spaced lettering).

All markings and symbols shall be fitted in such a way that their recognition will not be affected by detergents or solvents or by hard-to-remove dirt (e.g. in case of engravings without protective cover).

Marking the position of the shut-off cocks, position indicating line of the "Berne lock" and of the key for the pantograph shut-off valve:

I Open (vertical)

0 Closed (horizontal)



If a cock is mounted on a visible pipe, then the position of the cock shall refer to the direction of the pipe:

- "Open" = in line with the pipe,

- "Closed" = perpendicular to the pipe.

3.2.5 - ETCS and STM's

- The safety-related "override" push button (pos. 013) for suppressing an automatic application of the brakes due to the train-control/command system (e. g. for passing a stop-signal) is designed as a recessed push button for safety reasons. This is to avoid accidentally depressing the override button, which may result in safety-critical situations.

- For ETCS compliant STM any DMI information for use in interoperable rolling stock shall be provided in compliance with the ERA DMI specification.

Specific STM functions should be realised without additional hardware devices in the future, i.e. solely by use of the DDS (CCD, see *UIC Leaflet 612-02*) and pos. 013, 014, 015, 303, 512, 513, 52x.

3.3 - Requirements for the driver cab

O

3.3.1 - Light brightness for illuminated push buttons and indicators in the driver cab

The luminous emission of the non-adjustable illuminated push buttons (pos. 005, 012, 021, 023, 033, 035) in the driver cab shall be (150 ± 15) lx. These indicators and push buttons are not illuminated continuously, but only in case of actuation or demand.

3.3.2 - Main driver's desk

(see Appendices A - page 28 and B - page 52)

3.3.2.1 - Functional allocation of operational elements and integration constraints at the driver's desk level (see Fig. 3 - page 17)

The operational elements at the desk shall be located according to the prescriptions of Fig. 3 / Appendix J - page 88. There are different categories of operational elements.

3.3.2.1.1 - Mandatory elements

- Mandatory elements with fixed location:

  - 001 Train radio display (TRD)
  - 002 Technical & diagnostic display (TDD)
  - 004 Control command display (CCD)
  - 012 Train radio emergency calls
  - 013 ETCS: override (EOA)
  - 014 ETCS: release intervention
  - 015 ETCS: acknowledgement
  - 016 Pantograph / Diesel engine
  - 017 Main Circuit Breaker / Power transmission
  - 019 Automatic Speed Control (ASC)
  - 025 Sanding

- 027 External front lights
- 029 Task (on desk) & driver cab lighting
- 033 Travel Direction "Forwards"
- 034 No Travel Direction: "Neutral"
- 035 Travel Direction "Reverse"
- 038 External warning horn
- 041 Document holders
- 101 Train radio-handset

- Mandatory elements with optional location (pos. 104 - 109, 114 and 115 may alternatively be located on the vertical surfaces to the right and left of the driver knee hole):

  - 036 Driver's Identity card reader
  - 104 Lighting intensity of the instruments
  - 105 Floor & knee space (knee hole) heating
  - 106 Cab air blower rotational speed
  - 107 Driver cab temperature
  - 108 Air-conditioning/ventilation
  - 109 Pressure protection driver cab
  - 114 Windscreen wiper & windscreen washer
  - 115 Windscreen heating

- Mandatory elements with optional location according to the location of the desk in the cab (left or right side):

  - 010 Emergency stop valve with "Emergency stop" function

- Mandatory elements with fixed location only for locomotives and driving coaches:

  - 009 Brake pipe pressure adjuster (overcharge)
  - 020 Combined traction/dynamic brake controller with integrated driver activity control push button
  - 026 Release brake
  - 032a Driver's automatic brake controller (automatic brake)

- Mandatory elements with fixed location only for locomotives:

  - 037 Direct brake

- Mandatory elements with fixed location only for EMU/DMUs and driving coaches:

  - 021 Door control: left doors release & cancel release
  - 022 Door control: forced closing
  - 023 Door control: right doors release & cancel release

3.3.2.1.2 - Optional elements

- Optional elements with fixed location:

  - 003 Only for STM "LZB"
  - 005 Emergency signals (warning lights, may no longer be requested in a new OPE TSI)
  - 006 Electronic timetable display (ETD)
  - 007 BP (brake pipe) & MP (main air reservoir pipe) pressure gauge (option: to be integrated in the TDD)
  - 008 Brake-cylinder pressure gauges (option: to be integrated in the TDD)
  - 015 ETCS: acknowledgement (option: to have a redundant device on the right side)
  - 018 Train Power Supply
  - 024 Train lighting (may be replaced in the target system due to TCMS providing the trigger information - see Appendix A - page 28)

- 028 Instrument lighting (may be replaced in the target system due to TCMS providing the trigger information - see Appendix A)
- 030 Swan neck microphone
- 031 Driver activity control button
- 040 Keyboard

- Optional elements with fixed location for locomotives only:

  - 021 Door control: left doors release & cancel release
  - 022 Door control: forced closing
  - 023 Door control: right doors release & cancel release

- Optional elements with fixed location for driving coaches only:

  - 037 Direct brake

- Optional elements with fixed location for EMU/DMUs only:

  - 009 Brake pipe pressure adjuster (overcharge)
  - 020 Combined traction/dynamic brake controller with integrated driver activity control push button (option: to be replaced by 032b)
  - 032a Driver's automatic brake controller (automatic brake)
  - 032b Combined traction/brake controller
  - 037 Direct brake

O **3.3.2.2 - Functional allocation of operational elements and integration constraints below the driver's desk level**

See Fig. 4 - page 18

Operating elements that are operated while driving but which are not very important for a quick and targeted operation may be located on the vertical surfaces to the right and left of the driver knee hole (see Appendix J - page 88 and Fig. 4).

The foot switches in the driver leg room niche are as important for a quick and targeted operation without distraction from the track system as the equipment on the driver's desk itself. The foot switches for the driver activity control (DAC, pos. 111) and for the external warning horn (pos. 113) are located on the foot support of the niche.

- Mandatory elements with optional location (pos. 104 - 109, 114 and 115 may alternatively be located on the driver's desk in the optional area):

  - 102 Lamp centre console
  - 103 Centre console lighting
  - 104 Lighting intensity of the instruments
  - 105 Floor & knee space (knee hole) heating
  - 106 Cab air blower rotational speed
  - 107 Driver cab temperature
  - 108 Air-conditioning/ventilation
  - 109 Pressure protection driver cab
  - 111 Driver activity control (DAC) foot switch
  - 112 Foot switch for automatic adjustment of the height of the footrest (1 or optionally 2 switches)
  - 113 Foot switch for external warning horn
  - 114 Windscreen wiper & windscreen washer
  - 115 Windscreen heating

View of desk from above

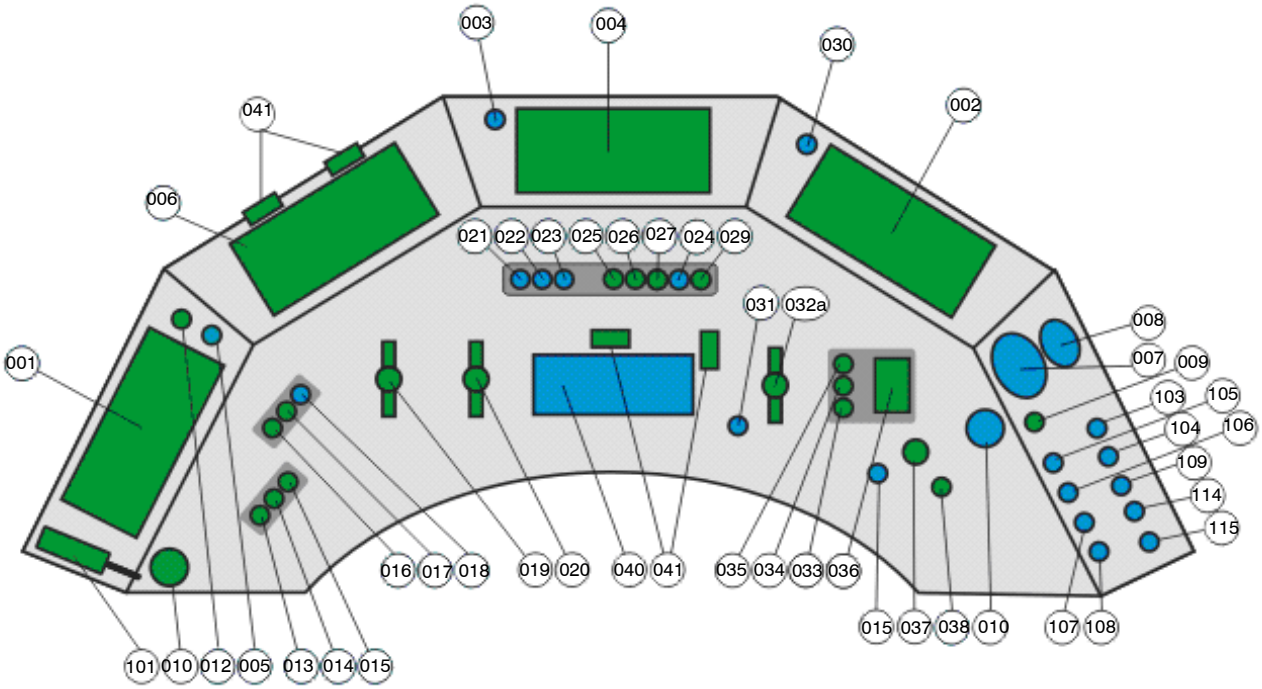


Fig. 3 - Standardised EMU/DMU, locomotives and driving coaches driver's desk

Front view of desk

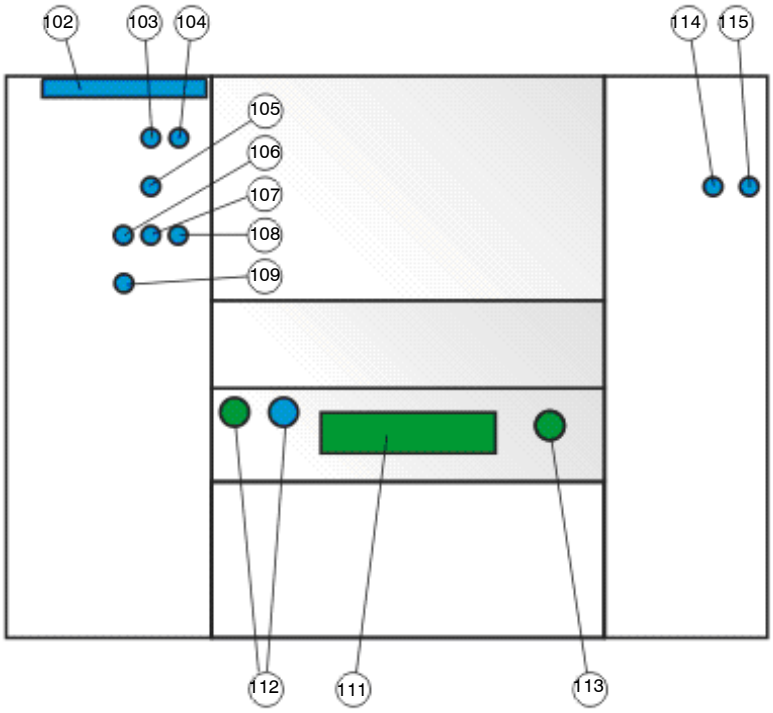


Fig. 4 - Standardised EMU/DMU, locomotives and driving coaches driver's desk

Fig. 3 & 4 show the standardised EMU/DMU, locomotives and driving coaches driver's desk (top and consoles) with maximum integration of equipment (all optional areas are shown), but without integration of heritage ATPsystems (e. g. ATSS, TBL, KVB, SCMT - see List of abbreviations - page 131). Mandatory elements are marked in green, optional elements in blue.

View of desk from above

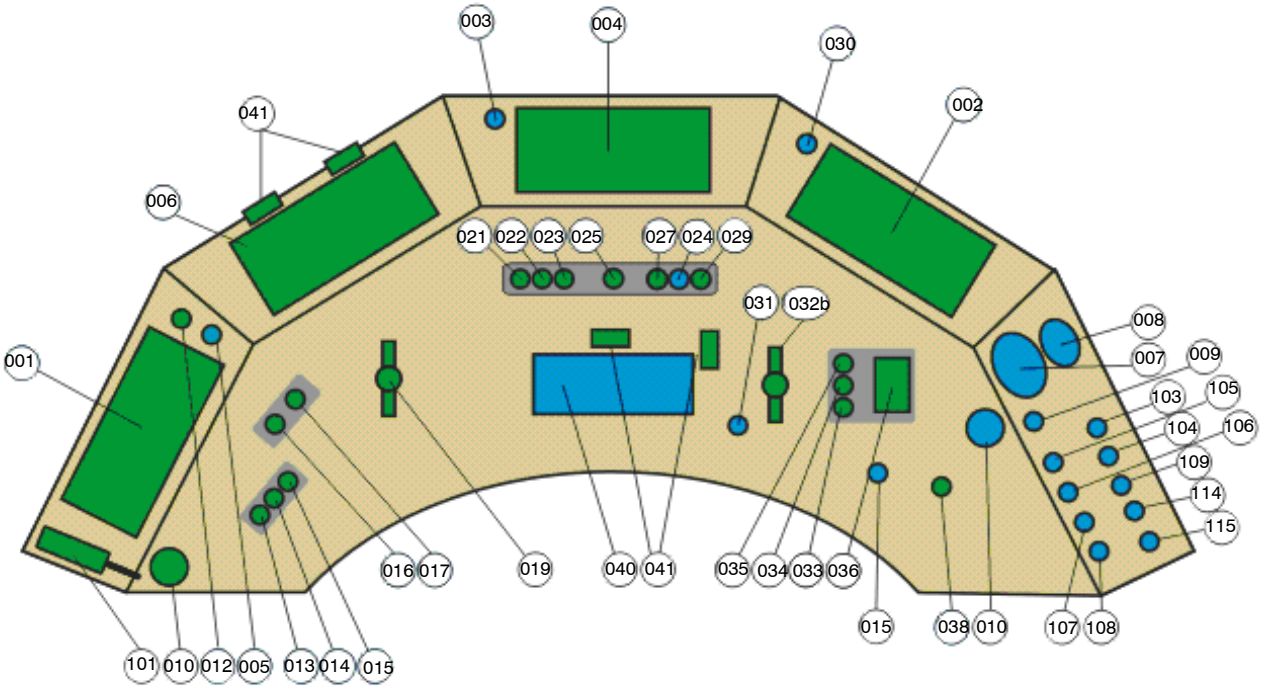


Fig. 5 - Optional standardised EMU/DMU central driver's desk

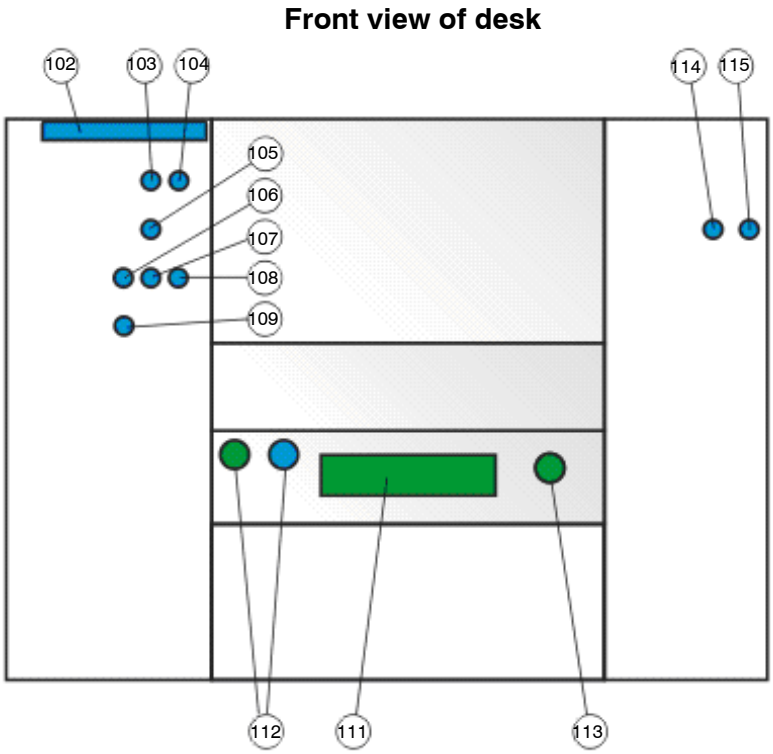
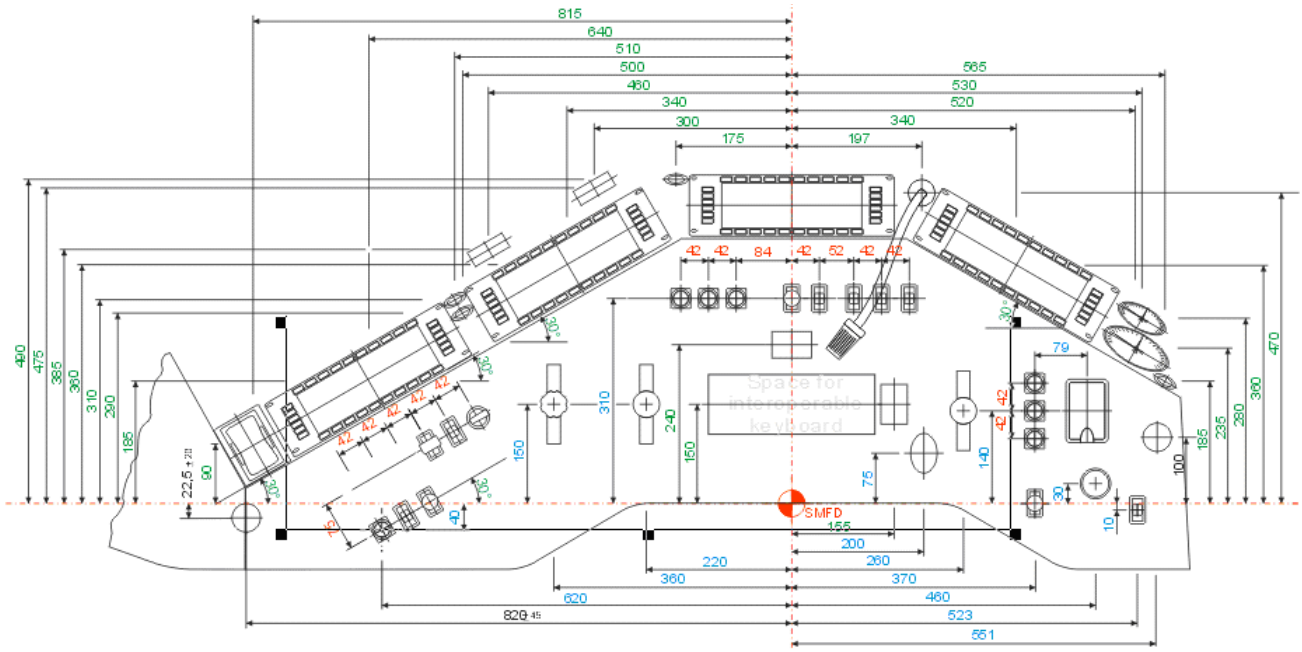


Fig. 6 - Optional standardised EMU/DMU central driver's desk

Fig. 5 & 6 show the optional standardised EMU/DMUs' central driver's desk (top and consoles) with maximum integration of equipment (all optional areas are shown), but without integration of heritage ATP systems (e. g. ATSS, TBL, KVB, SCMT). Mandatory elements are marked in green, optional elements in blue.

The following drawings show the recommended areas of locations of operating elements with the allocated tolerances in order to permit

- minimum distances for the integration of operational elements on the driver's desk in case of limited space in the cab (e. g. non-centre position of the desk) and
- maximum distances for the integration of operational elements on the driver's desk in case of cabs without space constraints (e. g. central desk).



Red values:	tolerance for coordinates: ± 2,5 mm, for angles ± 3°
Blue values:	tolerance for coordinates: ± 30 mm, for angles ± 8°
Green values:	tolerance for coordinates: ± 150 mm, for angles ± 15°

Exception: For pos. 021, 022 and 023 higher tolerances may apply.

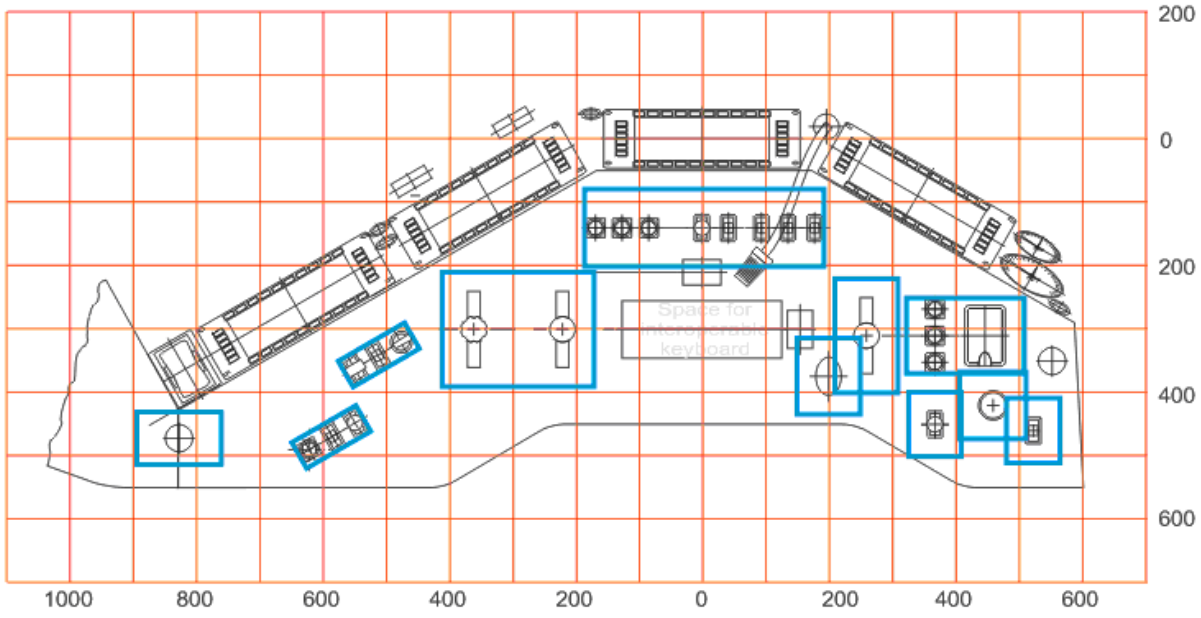


Fig. 7 - Recommended areas for the location of operating elements and distances between them with tolerances relative to the importance of elements for operation

3.3.2.3 - Normal working procedures of a driver at the main desk and the integration constraints derived from these procedures

The driver operates the EMU/DMU, locomotives and driving coaches from a standardised main desk in the cab. The desk may be located in the centre, on the right or on the left. If the desk is located on the right or on the left side of the cab, a fixed seat for an assistant driver has to be provided that offers an equal level of visibility of the track system and signals. Both driver and second man, if they fall within the representative range of body heights (5th to 95th percentile, European female and male population), have to be able to observe route and signals in accordance with the *TSI HS RST* (angle of visibility) from their adjustable seats.

Fig. 3 - page 17 to 8 - page 22 (see also Appendix J - page 88) show the mandatory positions of devices of the standardised main desk for EMU/DMU, locomotives and driving coaches and also the optional areas. No particular rules apply as to whether the driver's desk has to be installed on the left, on the right or in the centre of the driver cab. This is because no preferred track (left or right) has been stipulated for ETCS lines.

In contrast to EMU/DMU, it is recommended that the main driver's desk in locomotives be located close to a side window.

Optionally, if the driver has to operate the locomotive from both side windows (starting and shunting of trains/wagons), where the driver's desk is not within easy reach from the side window, an auxiliary driver's desk can be added (control elements on or close to the side wall of the driver cab).

Regarding all other requirements (especially position and function of operating elements), *UIC Leaflet 612 series* shall apply.

The basic principle for all operators is that only information that is necessary for operation is to be provided for the driver. No other information should be displayed which may disturb the driver or distract his attention.

The upper parts of the driver's desk, the control panels, contain the display level with the display interface and applications as specified in *UIC Leaflets 612-01, to 612-05*. The train radio (TRD, see *UIC Leaflet 612-04*) is located on the extreme left. The display for the electronic timetable (ETD, see *UIC Leaflet 612-05*) is installed to the left of the Control Command Display (CCD, see *UIC Leaflet 612-02*). The ETD may also be used for other tasks, e.g. display for the optimum energy consumption operation mode. The Technical & Diagnostic Display (TDD, see *UIC Leaflet 612-03*) is positioned on the right and it shows standardised images in a defined display hierarchy to the driver and to maintenance staff. Information about BP and MP pressure as well as that about brake cylinders shall also be located on the right.

The holding brake is released automatically if the intentionally commanded traction power exceeds 10 kN per EMU/DMU or locomotive.

Generally, the operating elements for the main circuit breaker / power transmission, the pantograph / diesel engine and the train control systems are only operated in connection with operating the brakes, and not at the same time with the traction / dynamic brake controller (pos. 020). Therefore, they can be easily operated using the free left hand. The operation elements, which are operated both when driving and when braking, shall be arranged on the control panel above the desk central area in the middle, in the case of EMU/DMU, locomotives and driving coaches between the traction / dynamic brake - controller location (pos. 020) and the driver's automatic brake controller (pos. 032) location (see Fig. 3 - page 17), thereby ensuring that they can be operated using either the free right or left hand without problems: these are toggle switches / brake flip switches for sanding (pos. 025), release

for the EMU/DMU, locomotive or driving coach automatic brake (pos. 026), external front lights (pos. 027) and task (on desk) & driver cab lighting (pos. 029).

The traction / dynamic brake controller (pos. 020) is located on the left side of the desk. The operation of this shall be natural, i. e. comfortable, uncomplicated and not requiring considerable body movement. The extraordinary actions demanding more driver concentration can therefore be performed with the free right hand. Almost all left-hand located elements should be operated with the traction / dynamic brake controller being in the neutral position (e. g. pantograph - pos. 016, or main circuit breaker - pos. 017) or not needing a simultaneous change of its position (e. g. ETCS - pos. 013 - 015). Therefore the left hand will not ordinarily be required to operate more than one device at a time.

The push button for the BP overcharge function (see pos. 009) shall be arranged next to the BP pressure indicator or the BP pressure information in the TDD.

Furthermore, an emergency brake device (e. g. push button, pos. 010) shall be provided in the cab close to the driver and the second man so that it can be easily reached from all positions in the driver cab (e. g. in the case of escaping into the engine room aisle before an accident collision) - see Fig. 8 - page 22. For a desk located on the right or on the left, the emergency stop valve with "Emergency stop" function (pos. 010) shall be located in the middle cab position.

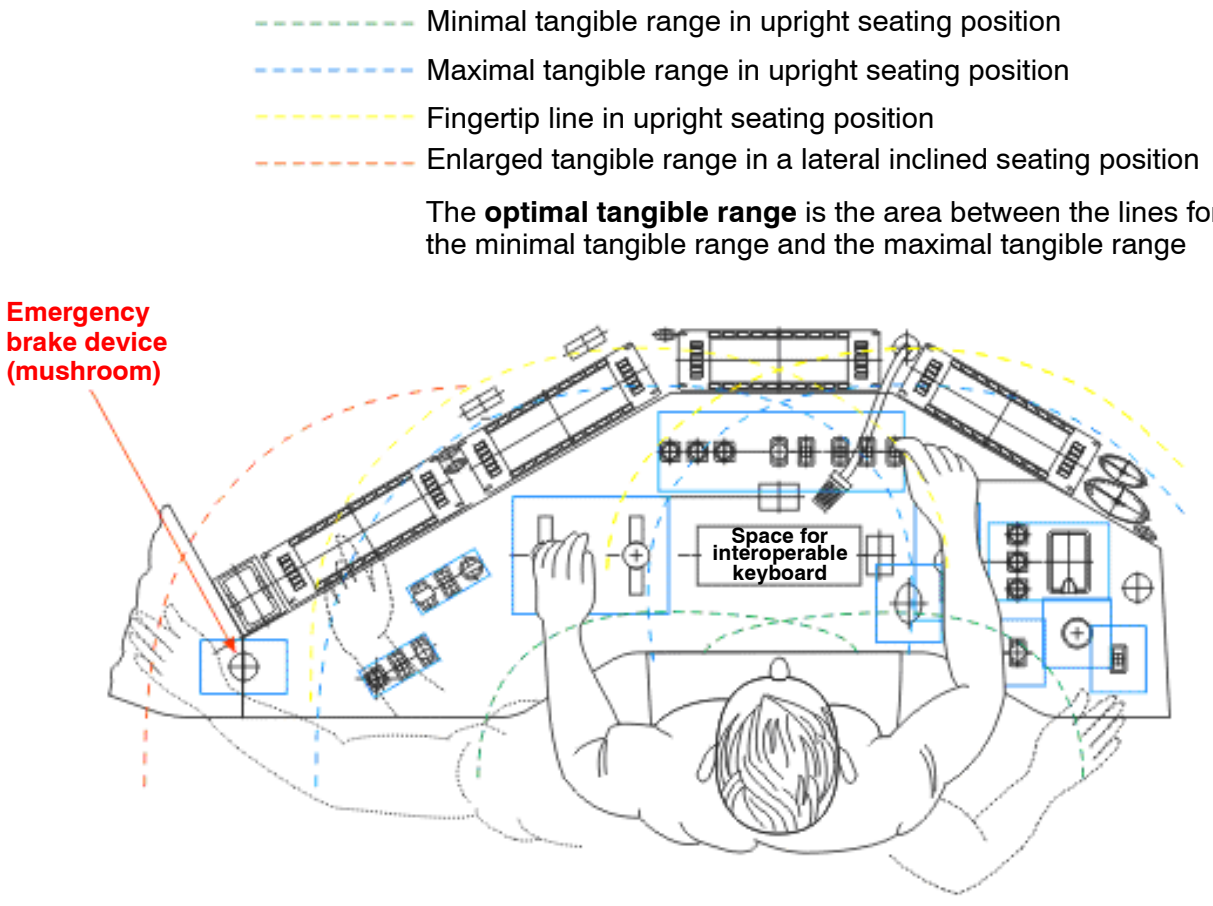


Fig. 8 - Tangible ranges according to the priority and specific functions of the operating elements on the driver's desk

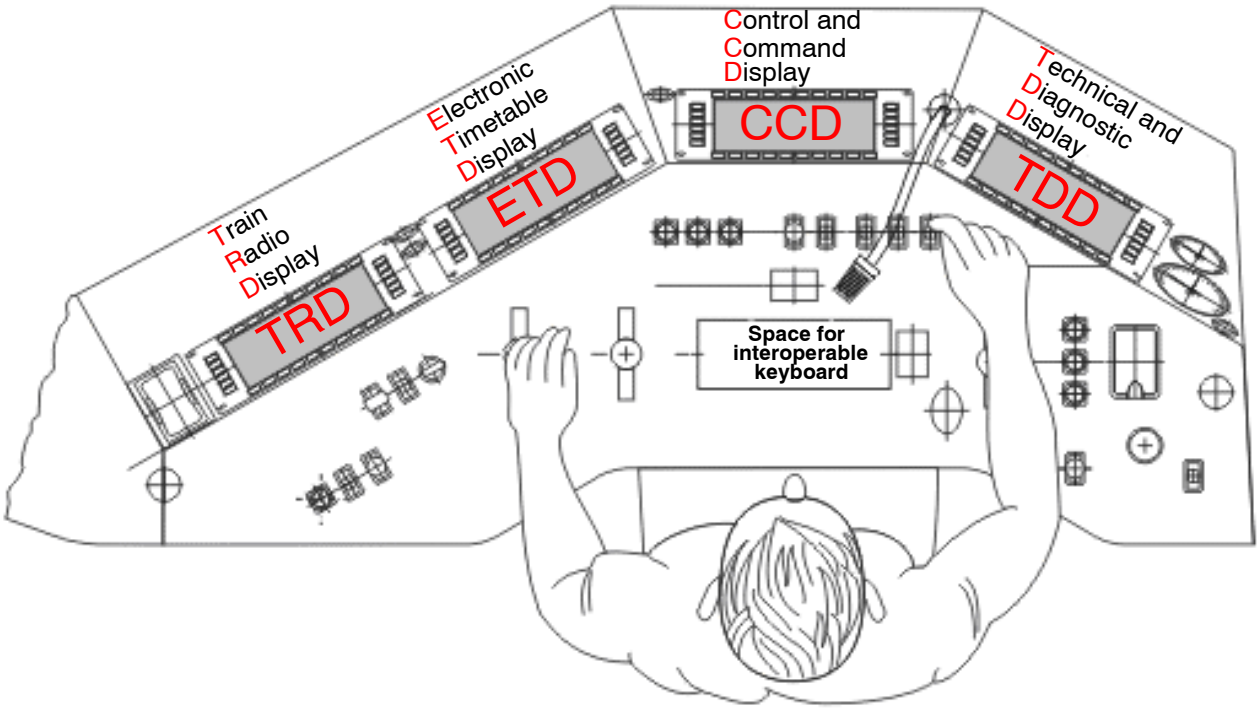


Fig. 9 - Arrangement of the displays on the driver's desk (e. g. Locomotive)

3.3.2.4 - Potential for extension of the driver's desk specification to specific applications

Driver's desks for other traction rolling stock could be inspired from the locomotive desk layout by deleting or modifying elements, depending on the application concerned. This has been the approach for specifying the EMU/DMU desk. No further specific recommendations currently exist for shunting locomotives, tram-trains and on-track machinery (OTM).

However, in case of partial application of *UIC Leaflet 612 series* to other types of rolling stock the following criteria should apply:

- the architecture and images of display units in the cab shall follow the prescriptions of *UIC Leaflet 612-01* in general and of the relevant *UIC Leaflet 612-0x* for the application concerned.

- the location of display units and operational elements in the cab should follow the prescriptions of *UIC Leaflet 612-0* for such elements provided in the cab as far as possible. In cases of major space constraints, such as shunting locomotives, EMU/DMU with front gangway and cab in side position or OTM, the relative positions of these operating elements used in the cab should comply with the layout specified in *UIC Leaflet 612-0* and shall at least provide the same functions as specified in *UIC Leaflet 612* ("same element - same function").

In case of integration of heritage ATP systems (see List of abbreviations - page 131) either the distance between the displays could be enlarged in order to mount ATP driver interfaces between adjacent displays or such heritage components could be installed on top of the display area, respecting the minimum view angles for driving in seating and standing positions.

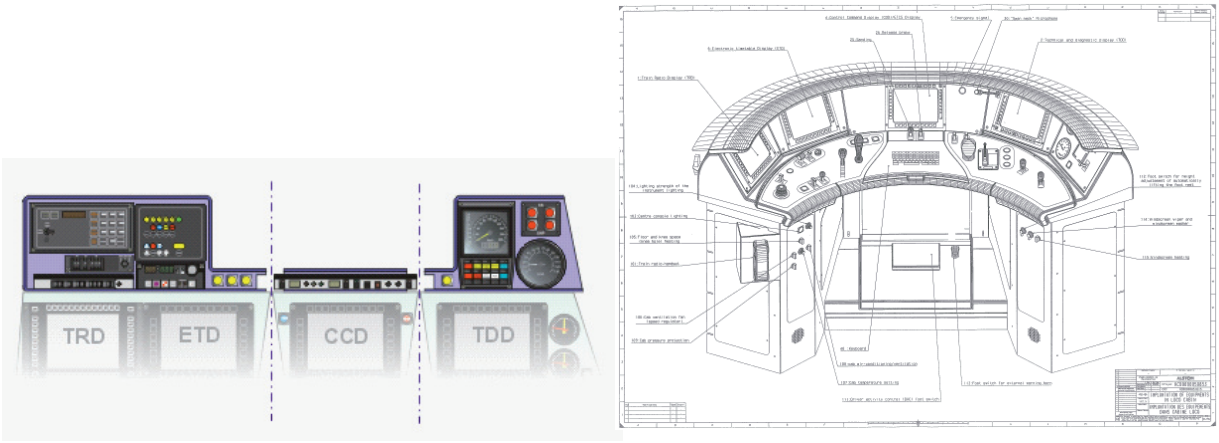


Fig. 10 - Examples of the integration of heritage ATP systems in the driver cab ("on top" or "in between")

The example on the left (see Fig. 10) shows the integration "on top" and allows to keep all the displays (see *UIC Leaflets 612-02 to 612-05*) at close distance. The example on the right provides more space next to the CCD (see *UIC Leaflet 612-02*) for integration of heritage systems and therefore the other displays (see *UIC Leaflets 612-03 to 612-05*) are located in less ergonomic positions.

For desks without heritage equipment the distance between adjacent displays should be minimised:



Fig. 11 - Example of driver's desk without integration of heritage ATP systems

3.3.3 - Driver's seat

See Appendix K - page 97 for requirements.

3.3.4 - Operating and display elements at further specific locations in the cab

See Appendix C - page 55.

The roof area in the cab is only permitted for the loudspeaker (pos. 081).

3.3.5 - Auxiliary driver's desks (only for locomotives as an operator option)

See Appendix D - page 56.

3.3.5.1 - Functional allocation of operational elements and integration constraints

If requested by an operator a control facility (auxiliary driver's desk) can be mounted ergonomically on or close to each driver cab's side wall, thus enabling driving (e.g. shunting and starting), while the driver is standing at the side window. Moreover, if this control facility is requested, an operating element for the DAC (pos. 211) shall be provided below and positioned such to the operating elements as to serve as a hand grip for the driver for the hand not used to operate the auxiliary desk (see Appendix J - page 88).

- Mandatory elements with optional location:

  - 013 ETCS: Override (e. g. EOA)
  - 015 ETCS: Acknowledgement
  - 033 Travel direction "forwards"
  - 035 Travel direction "reverse"
  - 038 External warning horn
  - 051 Second man task light (on desk)
  - 032b Combined traction/brake controller
  - 211 Driver activity control button (DAC)

- Optional elements with optional location:

  - 021 Doors on the left side (left auxiliary desk) - release and cancel release
  - 023 Doors on the right side (right auxiliary desk) - release and cancel release

O **3.3.5.2 - Starting the train and shunting by the driver from the auxiliary driver's desk**

(see *UIC Leaflet 612-1*)

1. Starting (e. g. on slopes) and shunting shall be possible without any kind of restriction.

2. At least, one emergency brake command initiation point shall be within easy reach.

3. The driver shall be able to see the BP pressure and the speed on the main desk from this driving position at the side window. This information (speed and BP pressure) shall be visible in standing position at the auxiliary desk, including by turning the head in standing position, but without leaving position. If the driver cannot observe the speed information on the CDD (see *UIC Leaflet 612-02*), then the traction is automatically cut off as long as the speed is higher than 35 - 40 Km/h and a status message shall be given to the driver in accordance with *UIC Leaflet 612-03*.

4. If the driver is able to use the direct brake on the main desk the driver shall be able to see the brake cylinder pressure indicators or the brake cylinder pressure information on the TDD from his driving position.

3.3.6 - Driver cab's rear walls

See Appendix E - page 59.

Operating elements that are only operated at standstill but which shall exist in every driver cab (such as button for engine room lighting - pos. 302 - or "Heating/Cooling" for food and beverages in the thermo compartment - pos. 320) shall be located on the driver cab's rear wall (see Appendix J - page 88).

- Mandatory elements with fixed location:

  - 301 Override the driver's ID card reader
  - 302 Engine room lighting
  - 303 Isolation of driver's automatic brake controller & ETCS/STM
  - 304 External light
  - 305 ep-brake
  - 306 Manual application of the holding brake
  - 307 Manual release of the holding brake

- Mandatory elements with optional location:

  - 320 Thermo compartment (fridge/stove)
  - 321 Locker for thermo compartment

- Optional element with fixed location:

  - 308 Forced decoupling

3.4 - Requirements for the engine room or inside the rolling stock

See Appendix F - page 66.

Certain operating elements must be located together on the EMU/DMU, locomotives and driving coaches, such as main battery switch / air supply cocks (combined in the "battery valve" - a main battery isolation switch combined with air supply cocks, pos. 508), fault switches (pos. 52x) or fault shut-off cocks (pos. 511 - 515). At the same location, the operation of the distributor (brake mode selection - pos. 507 -, isolating device and release device) shall also be possible (see also point 3.5 - page 27. These elements shall be easily accessible for the driver without disturbing passengers or moving luggage and they shall be protected against public access or use (e. g. in the engine room).

In case of concentrated power (EMU/DMU, locomotives and driving coaches) the operating elements specified in Appendix F shall be assembled on a switch panel in the engine room.

Operating elements which cannot be found in Appendix F (such as line safety switches for protection of low voltage circuits (LSS) and engine protection switches for 3AC auxiliary power circuits (MSS)) shall be fitted in a clearly arranged way at a height of more than 1,30 m., but never higher than 1,80 m. above the floor. They shall be clearly identified and concentrated on no more than two switchboards in the engine room.

There shall be a 230 V/50 Hz socket on at least one switch panel, which may be used provided that the main circuit breaker is closed or that an external power supply (pos. 412 and optionally the train power supply line) is active.

Sanitary facilities are not required on locomotives or in or near the driver cabs of driving coaches. Relevant operational procedures apply (e.g. the use of stationary or on-train facilities).

3.5 - External positions of operating elements on the vehicle (accepted by all operators as an option

See Appendix G - page 75.

If access by qualified staff can be ensured whenever requested operating elements may be situated inside the EMU/DMU, locomotives and driving coaches only (unlike for wagons).

If requested by an operator, the following requirement applies:

Operating elements that request operation in the case of a towed EMU/DMU, locomotive or driving coach shall be located - similar to a wagon - outside on the solebar/underframe. They shall be compatible with coaches equipped with train-wide communication and shall be protected against damage. Examples: brake mode position (pos. 406 - 408), brake distributor valve isolating cock (pos. 402) and release device for the distributor (pos. 401).

These elements shall be easily accessible for the driver, including at high platforms, and protected from areas used or accessed by the public.

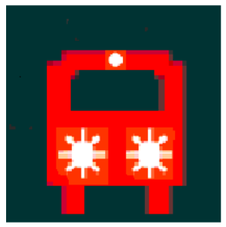
The operating equipment shall be fitted at a height of 0,8 - 1,5 m. above the top of the rail and immediately adjacent to both sides of the longitudinal solebar/underframe (see Appendix G - page 75, Fig. 21. In the case of operational elements located outside - "brake distributor valve isolation cock" (pos. 402) and "distributor release device" (pos. 401) - it is not mandatory to have these additionally replicated inside the vehicle.

3.6 - Guideline for the revision of the *UIC Leaflet 612 series*: Human factors aspects and standardisation benefits

See Appendix O - page 127.

Appendix A - Operating and display elements on the driver's desk

See also point 3.3.2.1 - page 14.

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
001	**TRAIN RADIO DISPLAY (TRD)**		Operator and display device for analogous and/or digital (GSM-R) train radio.		See *UIC Leaflet 612-04* (TRD), for application "train radio" and *UIC Leaflet 612-0* pos. 101.	Available for at least 3 h, independent of cab status and external power supply.				
002	**TECHNICAL AND DIAGNOSTIC DISPLAY (TDD)**		Display of: - TCMS data and selections, - operationally relevant status information, - failure management and help instructions,		See *UIC Leaflet 612-03* (TDD).	F/N/R: all functions available 0: Only "read access", no actions possible				
003	See point **N.1 - page 124**									
004	**CONTROL COMMAND DISPLAY (ETCS display) (CCD)**		Central display for indicating the speed, the train control and TCMS data, also for indicating ETCS parameters.		See ERA DMI specification and *UIC Leaflet 612-02* (CCD)	F/N/R: all functions available 0: Only "read access", no actions possible.				
005	**Most likely this function will be cancelled based on new TSI OPE/ RST** **EMERGENCY SIGNAL** **(Warning lights)** Illuminated push button [R]		**S**	Simultaneous flashing (f = 1 Hz) of the lower signal lamps, upper light is dark. Synchronous flashing of push button and signal lamps after activation.						
006	**ELECTRONIC TIMETABLE DISPLAY (ETD)**		Display of the electronic timetable sheet and other displays (if necessary).		see *UIC Leaflet 612-05* (ETD)	F/N/R: all functions available 0: Only "read access", no actions possible.				
007	**BP** (brake pipe) and **MP** (main air reservoir pipe) **Pressure Gauge**		- Indicator of brake pipe pressure (yellow pointer pointing vertically upwards at a pressure of 5 bar) and main air reservoir pipe (red pointer) - The face diameter of the pressure indicator shall be 100 mm and the background shall be black. - Precision class for the pressure information on the manometer or the TDD: 0,6 according to *EN 837-1*. If chosen, the manometer shall provide a precision class of at least 1,0. - The range of indication shall be 0 to 10 bar. - Precision of indication: ± 0,1 bar. - RAMS: for the event "loss of the pressure value or wrong value (gap > ± 0,1 bar compared with the real value)": failure rate for MP pressure value < 10^{-5} 1/h and failure rate for BP pressure value < 10^{-6} 1/h.		Information can alternatively be provided in the TDD (*UIC Leaflet 612-03*) on request of an operator.					

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
008	**BRAKE CYLINDER PRESSURE Gauge**		- The brake cylinder pressure indicator shall range from 0 to at least 6 bars. - For vehicles with 2 bogies and 1 or 2 brake cylinder pressures: red pointer bogie 1, yellow pointer bogie 2 - The face diameter of the pressure indicator shall be 80 mm and the background shall be black. - Precision class for the pressure information on the manometer or the TDD: 0,6 according to *EN 837-1*. If chosen, the manometer shall provide a precision class of at least 1,0. - Precision of indication: ± 0,1 bar. - RAMS: for the event "loss of the pressure value or wrong value (gap > ± 0,1 bar compared with the real value)": failure rate < 10^{-6} 1/h.		Information can alternatively be provided in the TDD (*UIC Leaflet 612-03*) on request of an operator.					
009	Mandatory for locomotives and driving coaches, optional for EMU/DMU **BRAKE PIPE PRESSURE ADJUSTER** (Overcharge) Push button [Q]		**U**	**<u>Phase 1:</u>** **Short-time pushing (≤ 1 s):** Brake pipe pressure increases automatically on 5,4 bar at a rate of 0,05 bar/s. **<u>Phase 2 (As an operator option, see comments below):</u>** **Continuous pushing:** Brake pipe pressure increases automatically on 5,4 bar at a rate of 0,05 bar/s and continues to increase if the driver is still pushing the button at the above rate. The network-specific limit given in the note below shall not be exceeded. **<u>After end of operation:</u>** Stop time and adjustment gradient according to *UIC Leaflet 541-03* applies.	- This function can be activated only if the BP is charged (5 ± 0,05 bar) and when the Main reservoir Pipe is able to refill the BP. The "overcharge" pressure shall be maintained between 20 s. and 25 s. (holding phase). If the duration exceeds 60 s. then the function is assumed to have failed. When the "overcharge" pressure has faded away, the BP pressure could be higher (≤ 0,05 bar) than before the activation of this function, but after a service brake and a release the nominal pressure of the BP shall be restored (5 ± 0,05 bar). For the status messages in the TDD see *UIC Leaflet 612-03*,	F/R or (N + Soft key "open autom. Brake controller") in the TDD				

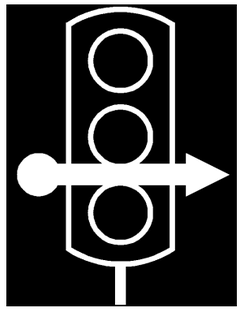
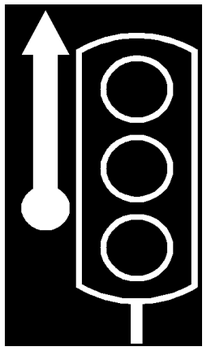
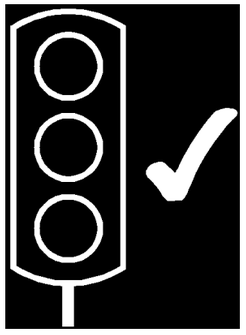
Phase 2 requirements (maximum permitted overcharge is higher than 5,4 bar with a maximum according to the railway undertaking, but not more than 6 bar):
The BP excess pressure over time ideally equals a linear p/t function in phase 2.

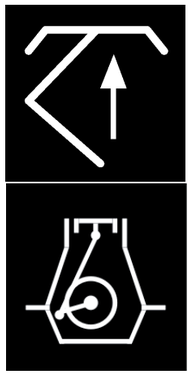
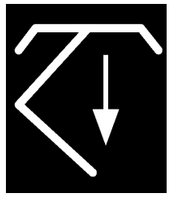
Beginning of the elimination phase:
The "overcharge" of the BP pressure commanded by the driver shall be maintained for a certain amount of time, but at least (20 +5/-0) s, to attain an equalization of flows in the BP and the connected distributors. Only then may the pressure be reduced at the adjustment specified in *UIC Leaflet 541-03* (beginning of the pressure decline phase).

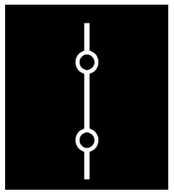
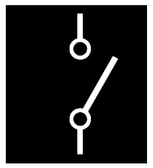
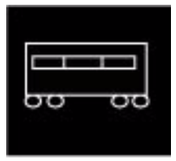
Adjuster memory:
If the automatic brakes are applied during adjustment, the function is stopped and the BP pressure at the start of the braking process shall be kept as reference. After the braking process is fully completed (i.e. application and full release), the BP pressure shall be at least restored to the previous reference value, but the BP excess pressure shall not be higher than 0,2 bar. If this function is stopped or inhibited before its completion, the driver shall be informed automatically in the TDD (see *UIC Leaflet 612-03*).

Requirements in a degraded situation for the backup function, if it exists as an option on the locomotive:
The "overcharge" BP pressure value shall fulfil the UIC pressure gradient value and could be a time-dependent operation as requested by the driver with the same characteristics as phase 2.
If an EMU/DMU is equipped with an automatic brake pipe pressure adjustment function, then the operating element pos. 009 may be deleted.

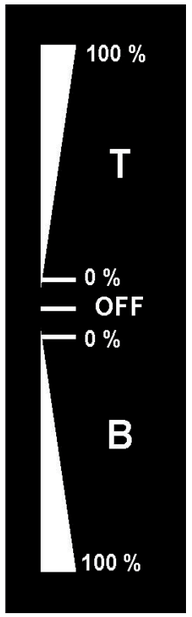
Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
010	**EMERGENCY STOP VALVE** **with** **"Emergency stop" function** Red impact button (carmine red, RAL 3002) with medium fluorescent, white, circular lens [P]		- Operate by pressing the button (impact button): A vertical movement of the red push button shall empty the Brake Pipe to the atmosphere. For a BP pressure of 5 bar, the range of actuation force values shall be 150 - 200 N respectively. - Reset by lifting the button Actuating the emergency stop valve in the driver's cab results in the following functions: - **Direct mechanical venting of the BP by creating a hole directly in the BP. Indirect evacuation is permitted, if the minimum level of safety requested by the TSI CR RST can be demonstrated. At least one of the emergency brake commands shall have a direct impact on the BP.** - **Traction interlock.** - **Locking the feed of the brake pipe.** - **The BP shall be vented via a secondary independent outlet (e. g. driver's automatic brake valve).** - **Depending on the requirements requested by a member state, an infrastructure manager or a railway undertaking, TCMS shall activate automatically dedicated functions (e.g. activate continuous sanding, activate external warning horn, lower all pantographs).** - **Full application of ep-brake.** These functions shall be maintained until the emergency stop valve has been reset. The sufficient condition for full brake release after emergency stop actuation shall be the resetting of the driver's automatic brake controller from braking position to running position.		Automatic application of dynamic brakes is permitted. Any kind of obstruction, related to conception (especially for the air venting) or installation on driver's desk, shall not prevent an emergency brake command. A manual emergency application of the brakes is only displayed via a status message. See *UIC Leaflet 612-2*	Available in all configurations				
012	**TRAIN** **RADIO EMERGENCY CALL** (see also *UIC Leaflet 612-04*) Illuminated push button [R]		**S**	Emergency call via Train-Radio (TRD, see *UIC Leaflet 612-04*) After activation the light in the push button is illuminated (see point 3.3.1 - page 14).	The push-button shall provide a continuous illumination of 0,1 lx Note: Unintentional operation of this push button is prevented by the surround of the push button. The GSM-R specification also requires a minimum application time.					

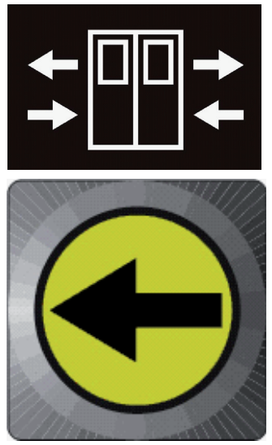
Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
013	**ETCS Override EOA** (possibly also applies to national train control systems) Red push button [Q]		**U > 5 mm**	ETCS: Override (e. g. EOA end of authority, ETCS) - Heritage ATP "LZB/PZB/Indusi": command - Heritage ATP "TVM/KVB": BPFC		F				
014	**ETCS Release Intervention** (possibly also applies to national train control systems) Black lever [B]		**U + 35°**	ETCS: Acknowledgement intervention or If an operating position different from ETCS is needed for a dedicated heritage ATP, e. g. "LZB/PZB/Indusi": free		F				
			0°	Normal position						
			U - 35°	ETCS: acknowledgement intervention						
015	**ETCS Acknowledgement** Lever with yellow ball [K] (only for Trenitalia: yellow lighting push button)		**U + 35°**	ETCS:acknowledgement or If an operating position different from ETCS is needed for a dedicated heritage ATP, e. g. - LZB/PZB/Indusi": vigilant - RSO/KVB": BP(AC)SF		F, R				
			0°	Normal position						
			U - 35°	ETCS: acknowledgement						

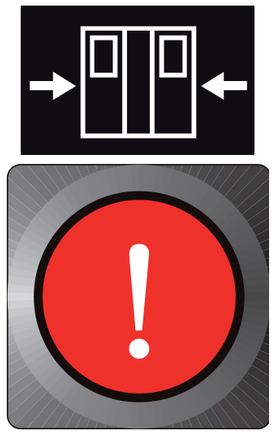
Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
016	**PANTOGRAPH** ***Electric traction*** **DIESEL ENGINE** ***Diesel units*** Brake flip switch [Z]		**U** **+ 17,5° .... + 35°**	- Pantograph up - Start the diesel engine	At **V > 0** the function shall become effective immediately provided that all raising / starting conditions are met at the same time. If one or more raising conditions are not met, the brake flip switch shall again be moved to "Pantograph up" / "Start" after all conditions have been met. Reason: delayed execution while driving may cause the pantograph to be raised / diesel engine to be started when not expected.	F, N, R			see note ←	
			S **0°**	Normal position						
			**U** **- 17,5°**	- Pantograph down/stop diesel engine **and** - Previously automatic switching off of the MCB/power off	In case of previous dynamic braking: operationally requested compensation of braking effort (e. g. by automatic air brake or rheostat), if loss of brake effort is caused by lowering the pantograph / stopping the diesel engine (see pos. 020 and pos. 032a interaction of brakes).					
			S **- 35°**	Emergency power-off (like position U - 17,5° but without software dependence on the master unit)	This function shall become immediately effective by opening an electric LV control circuit without intervention of software. For diesel engines the fuel supply shall be cut off.					
	Choice of current value, choice of local pantograph and separate operation of pantograph and diesel engine are performed by soft key application in the TDD (see *UIC Leaflet 612-03*).									

Operating element Position and [Form]		Symbol/indication		Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
							Master cab status	MCB	v = 0	v > 0	MT
017	**MAIN CIRCUIT BREAKER** ***Electric traction*** **POWER TRANSMISSION** ***Diesel units*** Brake flip switch [B]			**U** **+ 35°**	On	Start-up function only at **V = 0**: The pantograph is raised / diesel engine is started automatically and the main circuit breaker / power transmission switches on if all switch-on conditions are met. At **V > 0** the function shall be effective immediately provided that all switch-on conditions are met at the same time. If one or more switch-on conditions are not met, the brake flip switch shall again be moved to "Main circuit breaker on" / "Power transmission on" after all conditions have been met. Reason: delayed execution while driving may cause the main circuit breaker / power transmission to be switched on when not expected.	F/N/R		see note ←	see note ←	
											
				5 **0°**	Normal position						
				**U** **- 35°**	Off		F/N/R				
018	**TRAIN POWER SUPPLY** (Electrical train heating - ETH) Rotary push button [H]		**O**	**U** **- 45°**	Switch off	Having been switched on, the Train line continues to be supplied irrespective of the chosen direction of travel and is only switched off in accordance with the criteria in *UIC Leaflet 612-2*.					
				S **0°**	Normal position						
			**I**	**U** **+ 45°**	Switch on		F/N/R				

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
019	**AUTOMATIC SPEED CONTROL (ASC)** **v_{target} controller** [see Appendix K]		**V**	In the V-range the target speed V_{target} can be set by the driver: **Normal Operation in Driving configuration:** - 100 % = V_{max} of the master EMU/DMU locomotive or driving coach - V_{target} is allocated linearly between 0 % and 100 %, with the doubled value above 160 km/h. - Change of V_{target}: - 0 - 160 km/h in 5 km/h - steps, - > 160 km/h in 10 km/h - steps **Coupling Master or Washing operation (TDD) in Driving configuration:** - V_{max} = 3 km/h = 100 %. - All target speed values between 0 km/h and 3 km/h can be manually chosen by the driver (linear)	Application of pneumatic brake depends on selected network: no automatic brake pipe control by ASC in Italy and Switzerland. For France the purpose of this function is to maintain the speed selected by the driver in all brake modes (R, P, G) by application of the traction or brake forces. Immediate indication of the selected target speed in the CCD, but delayed application by ASC (TCMS). In order to avoid the unintentional application of ASC while the driver selects a new target speed, the TCMS accepts a target value only after being stable for at least 0,5 s. Optionally, on request of an operator, an ASC-system controlled by the TCMS could be possible. The requirements on this specific ASC-system are given in Appendix L.	**F** "R" **and** V < 50 km/h	X			X
			0 km/h	In position > "0FF", this position is separated from the active section by a tactile threshold.						
			Off	ASC is switched off						

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
020	**Mandatory for locomotives and driving coaches, optional for EMU/DMU** **COMBINED TRACTION / DYNAMIC BRAKE CONTROLLER with integrated Driver activity control (DAC)** Linear switch ("value related to position", position remains if handle is released) (see Appendix K)		**"T"** **S** **5° →** **+ 55°**	Traction power (0 % - 100 %) corresponds to the position of the handle. **Traction power gradients** for **Single traction:** - Increase traction power: 20 % (of the maximum value)/s. but not more than 50 kN/s. - Reduce traction power: 30 %/s. but not more than 75 kN/s. **Multiple traction or push-pull train:** - Increase traction power: 20 %/s. but not more than 22 kN/s. - Reduce traction power: 30 %/s. but not more than 37,5 kN/s. In case of loco not being the first vehicle in travel direction: For T < 100 kN the gradient becomes 20 %/s. but not more than 22 kN/s.	The traction/brake controller is shifted from the "OFF" position via a distinct pressure (tactile) point to the position "0 %". In "0 %", an auxiliary driver's desk can be activated. The traction/brake controller can only be shifted via a pressure point from "0 %" to the T-range after pressing the knob (freeing). In the T-range any traction power can be selected proportionally to position and up to the maximum (100 %). The selected force is also the maximum traction power in ASC operation.	F/R	X			X
			"OFF" **S** **0°**	- Traction power reduction to 0 kN in 1 s. - Brake force reduction to 0 kN according to the valid gradient (see *UIC Leaflet 612-2*).	To move into the traction function mechanical lock (push button on handle, e.g.) shall be actively (consciously) unlocked, in order to avoid unexpected traction command (due to driver error).					
			"B" **S** **5° →** **- 55°**	Dynamic brake force (0 % - 100 %) is proportionate to the position of the handle. **Dynamic Brake force gradients** for all EMU/DMU and locomotives in the train apply as follows, whether single or multiple traction: Rate of **Increase** in dynamic brake force: - independent of the position of the driver's automatic brake controller and except for emergency brake: o **R/P/G: 30 kN/s.** - emergency brake o **R: 20 kN/s** o **P: 10 kN/s** o **G: 5 kN/s** Rate of **Reduction** in dynamic brake force: - **without** automatic brake applied: **20 kN/s** - **with** automatic brake applied, the dynamic brake shall behave like the automatic brake. The characteristic shall not be higher than (depending on brake mode): o **R: 20 kN/s** o **P: 10 kN/s** o **G: 5 kN/s**	In case of applying both the driver's automatic brake controller and the traction/brake controller in braking position, the bigger of the two dynamic brake target values applies according to the maximum permitted wheel/rail adhesion value. For the maximum dynamic brake force and gradient see *UIC Leaflet 612-2*. **For any fading of the dynamic brake in decreasing speed, but at least at V < 10 km/h:** The dynamic brake force shall behave like the air brake. If compensation by air brake is needed, the dynamic brake shall be compensated by the friction brake only of the EMU/DMU or locomotive.					

Operating element Position and [Form]	Symbol/indication	Positions and function	Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
				Master cab status	MCB	v = 0	v > 0	MT
Note: In addition to the DAC pedal, DAC function can be achieved by two sensitive areas of the traction / dynamic brake controller (one on the knob and one on the stick). The DAC pedal (pos. 111), however, has priority over any other DAC interface. Depending on the network (operator option): Double-clicking on the traction/brake controller knob (2 clicks over the entire travel of the knob within 0,8 s.) starts the train length path measurement. When the end of the train is reached, the driver receives an acoustic signal. The double click is only effective in position "OFF" and in the T- and B-ranges, but not in position "T = 0 %" (in order to prevent unintentional activation of this function by mechanically unlocking the T-range: movement of lever is stopped in T = 0 % without pressing the knob, see Appendix K) Optionally for EMU/DMUs, this operating element may be replaced by the pos. 032b operating element.								
021	**DOOR CONTROL** **Left side doors - release and cancel release** Round push button with lamp [Q]		**U**	**First pushing of the button:** - release the left side doors, if train is at standstill. **Second pushing of the button:** (only in case of driving without train crew = driver responsible for closing the doors): - cancel release of the doors **and** - **only for SNCF**: start warning (as long as pushed, at least 5 s) ***Not for SNCF***: each door will close automatically after automatic warning at the door, if no passengers are entering through the doors for t > 3 s. **Meaning of light in the button:** - Dark: all doors are closed **and** not released for opening. - Lit: • V > 0: command for door release has been set by operating pos. 021 or 023, but no door can be opened locally. • V = 0: the door release command will become effective. A door can be opened locally by the door push button on the door. - Flashing (f = 1 Hz): no release for opening is given, but at least one door is not closed.	- For trains **without on-board staff** the push button is only in the driver's desk. - For trains **with on-board staff** responsible for closing the doors, an additional push button or a functionally identical device (key, handle, etc.) is located near the door. Push button "R" (pos. 023, right door) and "L" (pos. 021, left door), if you want to enable the opening of **all** doors.	F/N/R		X for execution of the command	See note at pos. 022	X

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
022	**DOOR CONTROL** **Forced closing** Round push button without lamp [Q]		**U**	Middle push button ("fast closing", e.g. in case of emergency or mistake): - **for all doors**: Cancel immediately the release to open the doors **and** - **only for open doors**: Immediate and simultaneous warning at the doors and closing of all open doors at the same time.		F/N/R		X		X

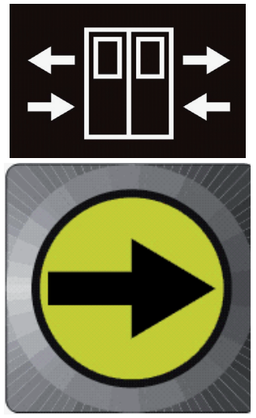
Note:
For pos. 021 - 023: The command remains valid until a new cab in the train is activated **and** another command has been given.

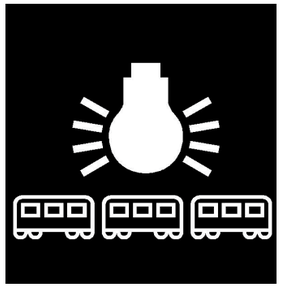
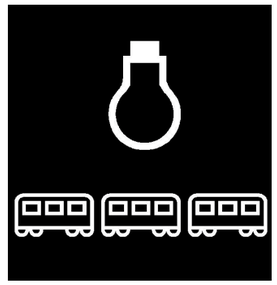
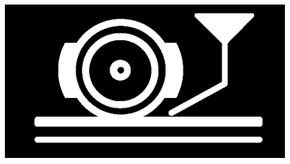
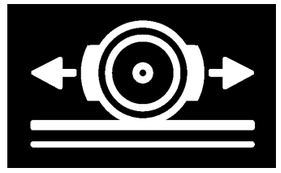
Procedure with on-board staff being responsible for closing the doors
- The driver pushes the "R" or "L" for authority to open (lamp lit);
- Waiting for passengers leaving and boarding the train;
- Open doors remain open;
- "In one action (device to be specified by EUPAX) the member of staff will activate:
  - the cancellation of release to open the doors and,
  - the warning at the doors and,
  - closing (except the door where the member of staff is positioned) (automatic procedure done near the door of a coach).

SNCF procedure used in the suburbs of Paris without on-board staff (manual procedure: warning duration and start of closing is managed by the driver)
- *"The driver pushes the "R" or "L" for release of the doors or there is line-side information (balise indicating that the train has reached the platform, lamp lit)*
- *Waiting for passengers leaving and boarding the train;*
- *Open doors remain open;*
- *The driver pushes "R" or "L" a second time to cancel the door release and activate the continuous warning (warning lasts for as long as the button is pushed; lamp in pos. 021 or 023 dark, if doors are already closed; lamp in pos. 021or 023 flashing, if at least one door is still open);*
- *"The driver pushes the "C" button for immediate closing of the doors (lamp in pos. 021 or 023 dark).*

Procedure without on-board staff (not for SNCF in the suburbs of Paris). Automatic procedure: warning and closing is an automatic sequence
- The driver pushes the "R" or "L" for release to open the doors (lamp of pos. 021 or 023 lit);
- Each door will close automatically after automatic warning at the door, if no passenger passes through the doors for t > 3 s.
- The driver pushes "R" or "L" a second time for cancelling the release of the doors (lamp in pos. 021 or 023 dark, if doors are already closed; lamp in pos. 021 or 023 flashing, if at least one door is still open).

In specific cases the driver is allowed to push pos. 022 for immediate forced closing of open doors.

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
023	**DOOR CONTROL** **Right side doors - release and cancel release** Round push button with lamp [Q]]		**U**	**First pushing of the button:** - release the right side doors if train is at standstill. **Second pushing of the button:** (only in case of driving without train crew = driver responsible for closing the doors): - immediately cancel release to open the doors **and** - **for SNCF only:** start warning (as long as pushed, at least 5 s) **Not for SNCF:** each door will close automatically after automatic warning, if no passenger passes through the doors for t > 3 s. **Meaning of light in the button:** - Dark: all doors closed **and** no release for opening. - Lit: • V > 0: command for door release has been set by operating pos. 021 or 023, but no door can be opened locally. • V = 0: the door release command will become effective. A door can be opened locally by the door push button on the door. - Flashing (f = 1 Hz): no release for opening is given, but at least one door is not closed.	- For trains **without on-board staff** the push button is only in the driver's desk - For trains **with on-board staff** responsible for door closing an additional push button or a functionally identical device (key, handle, etc.) is located near the door. Push button "R" (pos. 023, right door) and "L" (pos. 021, left door) to enable opening of **all** doors.	F/N/R		X for execution of the command	See note at pos. 022	X

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
024	**Option:** **TRAIN LIGHTING** Brake flip switch [B]		**U** **+ 35°**	Train lighting "On"	This function shall be integrated in the TDD (see *UIC Leaflet 612-03*) In addition, TCMS could control the train lighting - for SNCF: automatically "on", OR - based on the ETD or ETCS data, indicating the location of tunnels and - linked to time of sunset. This will prevent the driver from manual operation of this function.	F/N//R				X see also note below
			S **0°**	Normal position						
			**U** **- 35°**	Train lighting "Off"						
025	**SANDING** Brake flip switch [K]		**U** **+ 35°**	Sanding depending on the direction of travel (pos. 033 - "F" or pos. 035 - "R").	*HS RST TSI , section 4.2.3.10* (sanding) applies. See also *UIC Leaflet 612-2, point 6*.	F/R				X see also note below
			S **0°**	No Sanding						
			U **- 35°**	Sanding depending on the direction of travel (pos. 033 - "F" or pos. 035 - "R").						
026	**Mandatory for locomotives optional for EMU/DMUs and driving coaches** **RELEASE BRAKE** Brake flip switch [B] (see Appendix N - page 124, specific case for SNCF)		**U** **+ 35°**	Release of the brakes controlled by the automatic brake of the locomotives	This function allows - The movement of released EMU/ DMU and locomotives against a braked train formation to compress buffers in order to facilitate decoupling or coupling. - The lengthening of a train for precise adjustment of braking. - In case of failed WSP in order to prevent flat wheels.	F/N//R				X
			S **0°**	Normal position						
			U **- 35°**	Release of the brakes controlled by the automatic brake of the locomotives						

Operating element Position and [Form]	Symbol/indication	Positions and function	Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
				Master cab status	MCB	v = 0	v > 0	MT

Activation conditions:

- While the "Release brake" flip switch (pos. 26) of the activated driver cab is being operated, the automatic brake is released by means of a forced evacuation of the brake-cylinder pressure.

- In the service brake area of the BP between 5,0 and 3,5 bar [= full braking] the release signal shall be effective if none of the following inhibition conditions apply.

Inhibition conditions:

- Scenario: The automatic brake has been applied and then the "Release brake" flip switch (pos. 026) has been applied. The "Release brake" function will be deactivated if the driver's automatic brake controller (pos. 032) is shifted into another brake position, until the "Release brake" flip switch (pos. 026) is activated again.

- Release of the brake is cancelled immediately despite an activated flip switch pos. 026, if:

  - the driver's automatic brake controller is shifted from any position to the emergency position while the "Release brake" brake flip switch is being activated,
  - an emergency stop or an automatic application of the brakes is caused by the brake control or the vehicle control system while the "Release brake" brake flip switch is being activated,
  - if the "Release brake" signal is received on the slave locomotive and a rapid, emergency or automatic application of the brakes is detected during the signal,
  - BP < 3,4 bar,
  - after the deactivation the brake cannot be released by activating the "Release brake" push button again while a rapid, emergency or automatic application of the brakes is effective or BP < 3,4 bar.

- In the case of an applied dynamic brake, the brake flip switch is not effective.

After releasing the "Release brake" flip switch (pos. 026), braking shall become effective (immediate refilling of the brake cylinder pressure) according to the current target value (e.g. distributor pilot control pressure).

- Release brake" command shall not become effective in the following cases:

  - application of "pantograph down" (pos. 016) **or**
  - emergency stop valve (pos. 010) activated **or**
  - emergency brake position of the driver's automatic brake valve (pos. 032) **or**
  - BP < 3,0 bar **or**
  - automatic application of the brakes by a train control system (e. g. ETCS, KVB, RPS, ATSS).

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
027	**EXTERNAL FRONT LIGHT** Brake flip switch [B]		**S** **+ 35°**	Full beam head and marker light (LED)	See also pos. 304 *HS RST TSI , section 4.2.7.4.1 and Annex H* applies. No operational rules for loss of light necessary, because certain amount of LED are available and head light is a combination of head lamp and marker light (LED). In case of loss of a head lamp the LED will remain and ensure the 3 white lights for the top of the train.	F, N, R				
			**S** **+ 17,5°**	Dimmed head light and marker light (LED)						
			**S** **0°**	Marker light (LED) For a train with train-wide communication after successful train inauguration the marker and tail lights shall be switched on automatically. The driver can select deviating options in the TDD (see *UIC Leaflet 612-03*) For EMU/DMU, locomotives and driving coaches in single or multiple operation without train-wide communication the driver in the TDD (see UIC 612-03) shall select the light signals or confirm the light signals offered by the TDD. The standard setting is the last confirmed signals.						
			S **- 17,5°**	Dimmed marker light reduced (LED)						
		0	**U** **+ 35°**	No light						
028	**INSTRUMENT LIGHTING** Brake flip switch [B]		**S** **+ 35°**	Lighting on for pos. 007 and pos. 008	Requirements for light: see pos. 104 For optional area 1 this function shall be integrated in the TDD (see *UIC Leaflet 612-03*). In addition, TCMS could control the train lighting - based on the ETD data, indicating the location of tunnels and - linked to time of sunset. This will prevent the driver from manual operation of this function.					
			S **+ 0°**	Off						
			S **- 35°**	Lighting on for pos. 007 and pos. 008						

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
029	**TASK (on desk) and DRIVER CAB LIGHTING** Brake flip switch [B]		**S** **+ 35°**	Task (on desk) light on	Though applying the ETD function in the DDS (see *UIC Leaflet 612-05*), the use of (additional or backup) paper documents will still be needed and have to be read. Requirements for task on desk light: see pos. 039 .					
			U **+ 17,5°**	(optional) Task (on desk) light on						
			S **0°**	Task (on desk) light off						
			**U** **- 35°**	Driver cab lighting on/off	Driver cab lighting as current impulse switch. Requirement for cab light: Luminosity > 500 lx on desk level (see also pos. 212).	Available except for "self protection configuration", see *UIC Leaflet 612-1*.				
030	**Operator option:** **"Swan neck"** **MICROPHONE** (Stalk microphone)		- Announcements in the train. - Communication with an activated passenger emergency call station. - This is available as an alternative to pos. 101 digital (GSM-R) and analogous train-radio communication, if pos. 030 has been switched on after incoming call. Then pos. 081 (loudspeaker) is used. Alternatively pos. 101 can be used for all of the above mentioned functions.		Integrated push button for switching on the microphone. Pos. 081 provides the loudspeaker function.					
031	**Only for central desk** **DRIVER ACTIVITY CONTROL BUTTON (DAC)** Touch button [XX]		**U**		See pos. 111 and pos. 503. Only requested in case of central desk in order to allow driving in standing position (driver cannot reach the pos. 211). Prior to the other DAC operating elements.	F/R			X	

Tactile handrests do not replace the pedal in the driver cab but complement it. To send the acknowledgement signal, the driver can activate either system.

The acknowledgement signal is sent through a tactile hand rest. The device surface protrudes 1 mm from the surface of the handle.

Position of the tactile hand rest on the driver desk:

- The tactile hand rest shall be placed on the surface of the driver's desk (see desk drawing with measurements);
- The device shall be able to turn 30° around its own axis on the desk surface.

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
032a	**Mandatory for locomotives and driving coaches, optional for EMU/DMU** **DRIVER'S AUTOMATIC BRAKE CONTROLLER** **(automatic brake)** [attachment 2]		**> 5 bar** **U**	"Filling (charging) stroke" - function: see also specification below. In case of a passenger alarm or unintentionally open door - the filling stroke will be suppressed, - in this case only, this position is used for cancelling the traction interlock (see pos. 506) or inhibiting the passenger alarm application (see pos. 516).	**Assessment criteria:** According to *UIC Leaflet 541-03* it is necessary to increase the pressure in a 400 litres reservoir linked to the BP of the EMU/DMU, locomotive and driving coach at the opposite side of the unit, which must be increased from 3,5 bar to a pressure value that corresponds to the running BP pressure minus 0,1 bar in a period < 5 seconds.	F, N, R See also note below.				X (ep-command)
			5,0 bar **S**	Running position.	The driver shall be able to control the dynamic brake separately and independently by pos. 020.					
			S	Braking position: According to *UIC Leaflet 541-03*.		F, N, R See also note below, for slave units see also pos. 303.				
			S	Braking position: 6 positions for stepwise choice of BP-target values.						
			3,5 bar **S**	Maximum brake application position according to *UIC Leaflet 541-03*.						
			SOS **S**	Emergency brake position (high integrity method of applying the "automatic brake" by emptying the train automatic air brake pipe) See *UIC Leaflet 612-2*.	Actuating the emergency brake position in the driver cab shall result in the functions as specified for pos. 010. Any kind of obstruction related to design (especially) for the air venting or installation on the driver's desk shall not prevent an emergency brake command. Emergency Brake position of these devices must be self locked by a mechanical holding catch or a system with an equivalent reliability level. Unlocking this position is only possible by an intentional action on the part of the driver. A manual emergency application of the brakes is only displayed via a status message in the TDD (see *UIC Leaflet 612-03*). See note below.	Available in all configurations.				

Operating element Position and [Form]	Symbol/indication	Positions and function	Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
				Master cab status	MCB	v = 0	v > 0	MT

Activation criteria:

- the driver's automatic brake controller shall be activated ("unlocked") and can thus perform the "Control" and "Feeding" functions if the MP-pressure is > 5,5 bar AND the EMU/DMU, locomotive or driving coach is in the "In service" configuration (see *UIC Leaflet 612-1*).

- driver's ID card is inserted (pos. 036) or ID-reader is overridden (pos. 301), and

- for locomotives, illuminated push button driver's automatic brake valve off (pos. 303) is not actuated;

- there shall only be one activated driver's automatic brake valve in the train and, thus, only one that can be fully operated. In case of no train-wide communication on this function, deactivation of further automatic brake valves must be done by the driver by operating pos. 303. The emergency brake position shall be active.

Braking parameters:
By moving the automatic brake controller backwards the automatic brake will be activated by the reduction of the pressure of the Brake Pipe (BP) under 5 bars ± 0,05 bars. As the BP is the pneumatic control line for the complete train the automatic brake will be activated in the complete train. On the EMU/DMU, locomotive and driving coach, the pressure of the BP will be observed by a pressure sensor. The reduction of the BP pressure under 5 bar will be converted into a set point value for the Dynamic Brake of the EMU/DMU or locomotive: Nominal BP release pressure = 0 %, Full Service Brake = 100 %, linear interpolation in between the first step of braking and the full service brake referring to brake pipe pressures in the BP.

Gradients for the dynamic brake force, if **only** the automatic brake is applied **(pos. 032a is applied without pos. 020):**

- **Increasing** brake force:

  - Brake mode R: **30 kN/s**
  - Brake mode P: **20 kN/s**
  - Brake mode G: **5 kN/s**

- **Decreasing** brake force:

  - Brake mode R: **20 kN/s**
  - Brake mode P: **10 kN/s**
  - Brake mode G: **5 kN/s**

Gradients for the dynamic brake force if the automatic brake is applied together with the dynamic brake controller **(pos. 032a and pos. 020 applied in parallel):**

- See the requirements at pos. 020.

V < 6…10 km/h: Reduce dynamic brake to 0 kN while compensating with the air brake or control the dynamic brake force as the air brake.

Filling stroke function:

- Filling stroke function is activated only, if the BP-pressure < 4,65 bar before setting pos. 032a into filling stroke position.

- During the filling stroke, the maximum BP pressure achieved shall have no detrimental effect on brake components.

- At the end of a filling stroke (pos. 032a in running position), the pressure in the whole BP shall not exceed 5,4 bar.

- The elimination of the overcharge of the BP will be carried out after pos. 032a is set in running position. Then the pressure shall be reduced at the adjustment specified in *UIC Leaflet 541-03* (beginning of the pressure decline phase), see pos. 009.

- The activation delay of this function shall not be greater than 60 seconds to re-establish the automatic characteristic of the brake pipe. If the activation delay of this function is greater than 60 seconds due to system default or driver error the function shall be inhibited and a message shall be displayed on the TDD (see *UIC Leaflet 612-03*) telling the driver what to do.

Specific interaction with the driver's automatic brake controller:

During the application of the driver's automatic brake controller the following requirements shall apply:

- During automatic brake commanded by the driver's automatic brake controller (pos. 032a) and with the traction / dynamic brake controller (pos. 020) in a braking position the dynamic brake force on each individual EMU/DMU or locomotive corresponds to the maximum of the BP reduction command and the dynamic brake request, while respecting the maximum permitted wheel/rail adhesion value. (For example: a dynamic brake force effective prior to the application of the automatic brake is not reduced by additional pneumatic braking (BP reduction), but is maintained.)

- For a maximum speed V ≤ 15 km/h, it shall be possible for the traction power to be increased (pos. 20) in spite of the brakes being applied by the driver's automatic brake controller (pos. 32), i.e. for hill starts or push-up for uncoupling.

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
	Backup driver's brake valve functions: A backup device with full functionality to the main driver's brake valve (driver's brake valve and associated functions) is mandatory if the failure rate of this device is higher than 10^{-6} f/h. If the failure rate of this device is less than 10^{-6} f/h, this backup device is an operator option. According to the RAMS study, it could be possible to use the same automatic driver's brake handle which activates independent electrical contacts dedicated to this backup device, or to use the direct brake handle (e. g. see pos. 037 backup).									
032b	**Optionally for EMU/DMUs** **COMBINED TRACTION-BRAKE CONTROLLER (Monomaster controller)** Rotating pulse switch ("Application-time dependent value") [ ]	**T+**	**U** **+ 40°**	Increase traction power (time-dependent operation)	An applied holding brake will be released automatically, if the traction power becomes T > 10 kN/loco. Reduction of traction power in t = 1 s, if pos. 032b is set in the inverse position across the neutral position (e. g.: cut-off of traction power, if pos. 032b is moved to B-position).	F, N, R See also note below				
		T	**S** **+ 20°**	Traction neutral (maintain constant effort).						X (ep-command)
		0 (T -/B -)	**S** **0°**	Reduce traction power or release brake force.	The traction/brake controller can only be shifted via a pressure point from "0-position" to the T-range (see pos. 020). For any partial release of the brake, the BP-pressure shall be increased prior to a reduction of the dynamic brake.					
		B	**S** **- 20°**	Braking neutral (maintain constant effort).	If pos. 032b is set to "braking" position (B, B+) from traction position, then traction power is reduced immediately (see pos. 020). The first effort levels are done by the dynamic brake alone without brake pipe reduction. After the maximum dynamic brake force has been applied, the BP pressure will be reduced. The dynamic brake force is limited by design, or by the limits given in *UIC Leaflet 612-2* whichever is the more restrictive.					
		B+	**U** **- 40°**	Increase brake force (time-dependent operation.)						X (ep-command)
		SOS	**S** **- 80°**	Emergency brake (see pos. 032a).	This position initiates an emergency brake application (see pos. 010).	available in all configurations				
	Requirements for gradients and specific behaviour see: - Traction: pos. 020. - Automatic brake: pos. 032a ➡ In this case the traction power can be applied only if BP = 5 bar or the train speed is below or equal to 15 km/h. If BP < 5 bar and traction is commanded, then traction is inhibited if the train speed is higher than 15 km/h (see *UIC Leaflet 612-2*). - Dynamic brake: pos. 020 **Assessment, inexhaustibility requirement for a monomaster controller:** - After a sequence of 10 cycles of short time brake applications and releases (i. e. in less than 3 s. each) the 1st step braking force shall be unchanged.									

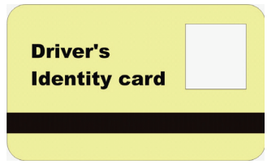
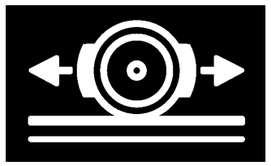
Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
033	**TRAVEL DIRECTION "forwards" (F)** Illuminated Push button [ ]		**U**	Driver's ID inserted: - Activation of the main desk of the cab. - Forwards (F) (vehicle is leading (= master)). - DAC function activated.	- Application will only become valid if the electronic identity (ID) card is inserted (pos. 036) or the ID-reader has been overridden (pos. 301). In driver cabs without this precondition the application of the push button will be ignored. - The travel direction shall be displayed at V = 0 km/h by lighting the relevant push button pos. 033 or 035 according to point 3.3.1 - page 14. - Application of a push button different from the valid one while running (V > 0) shall immediately cause a traction interlock. - New travel direction will become valid if • EMU/DMU, locomotive or driving coach is at standstill (V = 0) **and** • Traction controller is in neutral position ("0").	F/N/R		X see note below		X
034	**NO TRAVEL DIRECTION "neutral" (N)** Push button [ ]		**U**	Neutral (no running direction): - Activation of the main desk of the cab. - With ID inserted: N (cab activated). - With ID removed: 0 (cab not activated).						
035	**TRAVEL DIRECTION "reverse" (R)** Illuminated Push button [ ]		U	Driver's ID inserted: - Activation of the main desk of the cab. - Reverse (R) (vehicle is leading (= master)). - DAC function activated.						

"0": electronic identity card removed

- If the travel direction has been "F" or "R" before opening the battery valve (pos. 508), then the "0" command applies. It means that the driver has to push a travel direction push button "F", "N" or "R" again in the cab with the ID inserted (= "cab occupied") after switching on pos. 508 to make these functions valid.

- *UIC Leaflet 612-1, point 3* provides an overview of the main functions

  - if the cab is activated (ID inserted → travel directions F, N or R) → EMU/DMU, locomotive or driving coach is master, independent from its position in the train;
  - if the cab is deactivated (ID removed → travel direction 0) → cab is not active;
  - when another cab of the same EMU/DMU or locomotive becomes active, the EMU/DMU or locomotive will be in "In-service" or "driving" configuration;
  - when a cab of another EMU/DMU, locomotive or driving coach becomes active, the EMU/DMU, locomotive or driving coach will be in "standby" configuration (slave).

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
036	**DRIVER'S IDENTITY CARD (ID) reader** Fixation for ID-card [ ]		**ID inserted**	Cab activated: - driver related data is provided to TCMS; - TCMS provides the data to all connected sub-systems requesting driver data (e. g. ETCS-JRU); - ID shall remain inserted in any case of dynamic impact; - 10 mm of the card's end shall be uncovered for easy gripping for removal.	The driver's ID card reader shall be located on the right side of the driver's desk within the maximum tangible range in upright seating position (see Fig. 8 - page 22). The driver's ID card dimension shall comply with credit card format. The ID reader function shall be available in any case of introducing dirt/waste. The ID reader shall be able to transmit data without contact in a range of distance of 0 - 50 mm. Override function of ID reader: see pos. 301.			X		X
			ID removed	Only for V = 0: Cab not activated.						
037	**Mandatory for locomotives** **optional for EMU/DMUs and driving coaches** **DIRECT** **BRAKE** **CONTROLLER** [ ]		**S + 37,5°**	Release direct brake (running position).		F/N/R				**X** see note below
			U back to middle position	Release direct brake (time-dependent operation)						
			S 0°	Final position (middle position): no change of previously braking set value.						
			**U back to middle position**	Apply direct brake (time-dependent operation).	The maximum direct brake force shall be at least 30 kN per wheelset at an average wheel diameter. Emergency brake will be commanded by TCMS if - as an operator option, no tail signal is activated on the master locomotive (= locomotive is not operated alone) **and** - V > 50 km/h **and** - T > 4 s. **or** - the direct brake has been isolated in the TDD (see *UIC Leaflet 612-03*).					
			S - 37,5°	Apply direct brake Optionally, if the WSP does not meet the TFR < 10^{-9}/hour for the occurrence of failures and causes the loss of the braking force of the whole locomotive, or if one failure is sufficient for this event to occur, this position is mandatory and the following requirements apply: the Wheel slide protection (WSP) device shall be deactivated after a delay time of t = (1,5 + 0,5) s in this "full application" position of the direct brake handle. It shall be activated again after leaving this position.						

Operating element Position and [Form]	Symbol/indication	Positions and function	Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
				Master cab status	MCB	v = 0	v > 0	MT

Direct brake requirements for slave locomotives:

The direct brake is controlled from outside the slave locomotive by means of command transmission with train-wide communication (e. g. train bus or radio communication) under the following conditions:

- the release command of the direct brake shall be transmitted from the master locomotive: the slave locomotive shall not release the direct brake without prior command from the master locomotive,
- the application command of the direct brake is propagated from the master locomotive, if the initial speed is V < x km/h (depending on network). The brake force applied in the slave locomotive is time dependent according to the presence of the propagated command. This means that if the direct brake controller is used at an initial speed V > x km/h, then the brake force in the master locomotive is higher than or equal to that in the slave locomotive:

DB/NS: x = 10 km/h
Trenitalia: x = 5 km/h
ÖBB/SBB/BLS: x = unlimited

Activation criteria for the direct brake:

Case 1: activated cab:

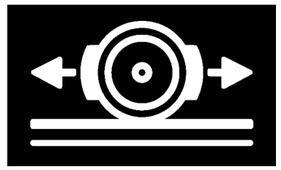
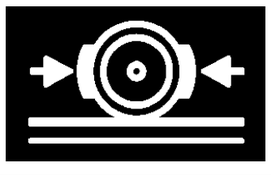
- Battery valve (pos. 508) "on"
- Driver's ID inserted
- Travel direction in "F", "R" or "N"
- The activation of the cab shall not automatically release the direct brake. In order to make the release position effective, the direct brake controller shall be set to the middle position before.

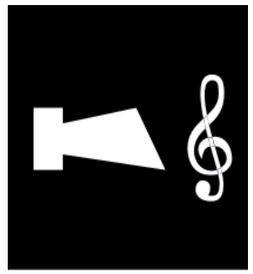
Case 2: deactivated cab (Driver's ID removed and/or slave locomotive):

- Battery valve (pos. 508) "on"
- Travel direction in "0"
- In this case, only braking positions are effective.

Deactivation criteria:

- Battery valve (pos. 508) in "off"-position: The battery valve isolates and bleeds the direct brake → direct brake remains released.

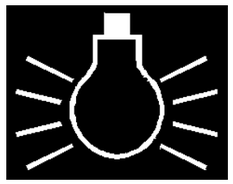
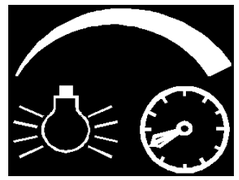
Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
037 backup	**Only for locomotives (as an operator option) only for the direct brake of the main desk as a backup device to the main driver's brake valve, if selected in the TDD (see *UIC Leaflet 612-03*), The direct brake handle shall have the characteristics of an automatic brake controller (time-dependent)** **[ ]**		**S + 37,5°**	Release automatic brake (time-dependent operation).		Main desk: F/R or (N + soft key "brake-test on") in the TDD				X (ep-command)
			U back to middle position	Release automatic brake (time-dependent operation).						
			S 0°	Final position: no change of previously-set value (running position).	These "running" positions of the controller is the pre-condition to activate the automatic brake controllers					
			**U back to middle position**	Apply automatic brake (time-dependent operation).						
			S - 37,5°	Emergency brake position (see emergency braking position of operating component pos. 032). It shall be an electrical initiation of the emergency braking.						
	Activation criteria: Activation in the TDD with the soft key (see *UIC Leaflet 612-03*). **Deactivation criteria:** Deactivation in the TDD (see *UIC Leaflet 612-03*) or application of pos. 032, if the pos. 032 is effective (no default).									

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
038	**EXTERNAL WARNING HORN** Brake flip switch [M]		**U** **+ 35°**	Signal tone high.						
			S **0°**	Normal position.						
			**U** **- 35°**	Signal tone low.						
039	**DESK LIGHT**		If pos. 029 (desk light) has been switched on, then the desk area surrounded by pos. 041 (document holders) shall be illuminated, where the keyboard (pos. 040) or on top of it a paper document (e. g. timetable) may be found. Illumination shall have a uniformity g_2 better than 1:2; the mean horizontal degree of illumination shall be adjustable between 0,3 and 150 lx. The individually adjusted degree of illumination shall be guaranteed, even after an interruption of the supply voltage of any length.		The respective operating elements shall be fixed on the lamp and shall have a form encoded design (e.g. raised points of the button surfaces according to the accompanying illustration) so that they can be found easily. Pos. 039 shall also be applied if LED design is used for the desk light.					

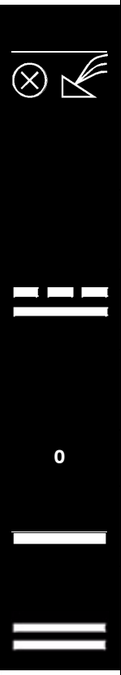
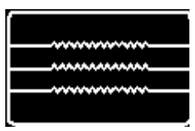
Operating element Position and [Form]		Symbol/indication	Positions and function	Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
					Master cab status	MCB	v = 0	v > 0	MT
040	**KEYBOARD** (see Appendix K)		Activation criteria: - V = 0 **and** - Soft key "keyboard on" in any display calling for driver data input Input is only possible in images which request alphanumerical data entry. The input is immediately shown on the image concerned. Deactivation criteria - V > 0 **or** - Soft key "Keyboard off" in any display calling for driver data input.	The position of the keys on the keyboard shall be flexible according to the activated network. Therefore no hardware constraints are acceptable. Reason: there is no standardised keyboard in Europe. Pending economically viable solutions a fixed keyboard with hard keys according to national provisions is possible.	F/N/R		X		
041	**DOCUMENT HOLDER** [ ]		Fixing of documents and/or manuals on the driver's desk (above and beside the keyboard (pos. 040)).						
101	**TRAIN RADIO-HANDSET** **or** **Headset for hand-free operation, flexibly linked with the driver's seat**		Handset for digital (GSM-R) and analogous train-radio becomes operational if intentionally taken off the support. Alternatively pos. 030 can be used.	See pos. 001 and *UIC Leaflet 612-04* (TRD).	available for at least 3 h, irrespective of cab status and external power supply.				

Appendix B - Operating and display elements below the driver's desk

See also 3.3.2.2 - page 16

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
102	**LAMP CENTRE CONSOLE**		Illumination of the operating elements only pointing downwards on the centre console		Requirements for light: see pos. 104	F/N/R				
103	**CENTRE CONSOLE LIGHTING** Push button [Q]		**S**	Switching the centre console lighting on and off	Requirements for light: see pos. 104	F/N/R				
104	**LIGHTING INTENSITY OF THE INSTRUMENTS** Variable resistor		**Min - Max**	Infinite adjustment of the centre console and pressure gauge illumination Requirements for light: - Pressure gauge: 0,3 - 30 lx - Centre console: 0,3 - 50 lx		F/N/R				
105	**FLOOR AND KNEE SPACE (Knee hole) heating** Revolving switch [E]		**S - 60°**	Niche heating on			X			
			0	**S 0°** Off						
			**S + 60°**	Floor heating on						
			S + 120°	Floor and niche heating on						
			S - 60°	Niche heating on						
106	**CAB VENTILATION FAN (speed regulator)** Revolving switch [E]		**S min. - S max**	Setting the air blower rotational speed in 7 steps			X			
107	**CAB TEMPERATURE SETTING** Variable resistance		**S**	Infinite adjustment of the driver cab temperature in the range 18 °C to 26 °C			X			

Operating element Position and [Form]		Symbol/indication		Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
							Master cab status	MCB	v = 0	v > 0	MT
108	**CAB AIR CONDITIONING VENTILATION** Revolving switch [E]			**S** **- 60°**	only cab air blower, no air-condition						
			0	**S** **0°**	Off						
				**S** **+ 60°**	Air-conditioning **manual**: - Target temperature according to pos. 107 - Cab air blower rotational speed is determined by the position of element pos. 106.			X			
				**S** **+ 120°**	Air-conditioning **automatic**: - Target temperature according to pos. 107, - Cab air blower rotational speed is controlled automatically.						
109	**CAB PRESSURE PROTECTION** Revolving switch [E]		**0**	**S** **0°**	Pressure protection off		F/R	X			
			**I**	**S** **+ 60°**	Pressure protection on	- Quality of air in accordance with the *CR RST TSI* - Pressure deviation gradients in accordance with *EN 14067*					
111	**DRIVER ACTIVITY CONTROL (DAC) foot switch or contact mattress**			**U**	Foot switch or contact mattress for operating the DAC	See pos. 020, 031, 211 and 503 Prior to the other DAC operating elements	F/R			X	
112	**FOOT SWITCH for automatic adjustment of the height of the footrest**			**U**	Mechanical or hydro-mechanical principle: Applying the foot switch of the footrest causes the smooth lifting of the footrest. Electrical principle: One foot switch is dedicated to automatically lifting, another to automatically lowering the footrest.	Mechanical or hydro-mechanical principle: Only one foot switch is requested for unlocking the footrest. - The driver stops the lifting by releasing the foot switch. - The driver pushes the footrest down by pressing the foot switch and applying force on the footrest. Electrical principle: 2 foot switches are requested: - lifting - lowering.					

Operating element Position and [Form]		Symbol/indication		Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
							Master cab status	MCB	v = 0	v > 0	MT
113	**FOOT SWITCH for external warning horn**			**U**	Simultaneous operation of the external warning horns "high" and "low" (by pos. 038 only one frequency can be activated at the same time) Only effective, if air pressure is available.	Pneumatic roller lever valve.					
114	**WINDSCREEN WIPER AND WINDSCREEN WASHER** Revolving switch with push button [Y]			**U (push button)**	Washing	- Effective in all positions of the switch, - At the same time automatic wiping, - After release of the push button the wiping is automatically finished after 5 s.	F/N/R				
				S - 90°	Windscreen wiper interval operation slow (1 wipe/60 s.).						
				S - 60°	Windscreen wiper interval operation medium (1 wipe/30 s.).						
				S - 30°	Windscreen wiper interval operation fast (1 wipe/15 s.).						
			0	**S 0°**	Off						
				S + 60°	Windscreen wiper slow (1 wipe/s).						
				S + 120°	Windscreen wiper fast (2 wipes/s.).						
115	**WINDSCREEN HEATING** Revolving switch [E]		**max.**	**U - 60°**	De-icing with maximum performance.			X			
			0	**S 0°**	Automatic operation.						
			min.	**S + 60°**	Continuous operation with reduced performance (within control of maximum permissible windscreen temperature).	Reason: it must be possible to switch the windscreen heater on in damp tunnels, even in case of higher temperatures.					
			I	**S + 120°**	Off.						

Appendix C - Operating and display elements in the cab at further specific locations

Operating element Position and [form]		Symbol/indication		Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
							Master cab status	MCB	v = 0	v > 0	MT
051	**SECOND-MAN TASK LIGHT** **(on desk)** **Only for locomotives** Flip switch [xx]			**S** **+ 35°**	500 lx	Automatic switch-off of task light (on desk) after 15 minutes, if battery is switched off Located in the second man's place.	Available, except for "self-protection" configuration				
				S **0°**	OFF						
				S **- 35°**	250 lx						
081	**LOUDSPEAKER**			Listening to all kind of announcements, like: - PIS, - acoustic TCMS messages as defined in *UIC Leaflets 612-02 and 612-03*, - communication with staff and passengers, - train-radio (TRD, see *UIC Leaflet 612-04*).		- Set volume automatically according to ambient volume. - Manual adjustment of the relative volume compared to the ambient volume by the driver in the train-radio-display (TRD, see *UIC Leaflet 612-05*) possible. Only for PIS: The volume can be switched off by the driver in the TDD (*UIC Leaflet 612-03*). Preferred location is in the roof of the cab.					
121	**DIAGNOSIS SOCKET**					Interface to be defined by MODCONTROL (e. g. USB). Located on the second man's side or beside the desk.					
122	**ELECTRIC SHOCK-PROOF SOCKET** **230 V / 50 Hz**					Metal design with cover, operative during - battery charge or - external power supply Located on the second-man's side or beside the desk.					
212	**CAB LIGHTING** Push button [Q]		I	For cabs with exterior doors: Switching the cab lights on and off. For other cabs: Switching on and off the light in the compartment or gangway to the cab. Requirements for cab light: > 500 lx (see also pos. 029).		Current pulse control (see also pos. 029). Located close to the exterior doors of the cab (see Appendix J - page 88).	Available, except for "self-protection" configuration (see *UIC Leaflet 612-1*).				

Appendix D - Auxiliary driver's desks (only for locomotives as an operator option)

The operator shall decide whether or not to equip the cab with one or two auxiliary desks and define the location of the auxiliary desk if there is only one. An auxiliary desk can be integrated in the side wall or next to the window added to the main desk. If the main desk is located off-centre in the cab and the functional requirements for the auxiliary driver's desk can be satisfied by using the devices of the main desk (e.g. information display) there is no need to add those devices to the auxiliary desk in parallel. A driver activity control (DAC) device shall be positioned close to the auxiliary desk in the driving position and close to the window and shall be applicable even if the auxiliary desk is not activated (pos. 211).

The following functions shall be available from the driving position on the auxiliary desk:

- Mandatory element with optional location (see point 3.3.5 - page 25):

  - 013 ETCS: Override (e. g. EOA)
  - 015 ETCS: Acknowledgement
  - 033 Travel direction "forwards"
  - 035 Travel direction "reverse"
  - 038 External warning horn
  - 051 Second man task light (on desk)
  - 032b Combined traction/brake controller: emergency brake function shall only be available in the activated cab
  - 211 Driver activity control button (DAC)

- Optional element with optional location:

  - 021 Doors on the left side (left auxiliary desk) - release and cancel release
  - 023 Doors on the right side (right auxiliary desk) - release and cancel release

Preconditions for the activation of an auxiliary desk:

- The driver's ID has been inserted or pos. 301 has been applied

- The combined traction/dynamic brake controller (pos. 020) on the main desk shall be in "OFF" position. If it is not in '"OFF" position and traction power may be applied the traction shall be cut off (traction interlock shall be applied).

- The combined traction/brake controller of the auxiliary desk (pos. 032b) shall be in "0" position.

- The emergency brake function of the auxiliary desk (pos. 032b) shall be available (not isolated).

- In case of more than one driver's brake controller (pos. 032, pos. 037 backup) for automatic brake in the same cab: all the driver's brake controllers shall be in "running position".

- Speed V < 40 km/h.

Activation criteria for auxiliary desk and for operating elements located on the auxiliary desk:

- A travel direction "R" = reverse or "F" = forwards on the auxiliary desk shall be selected.

The following functions of the main desk shall be deactivated if an auxiliary desk is active:

- Traction, as long as the traction/brake controller of the auxiliary desk is operated in Traction or Brake positions.

- "Release" positions of the direct brake controller (pos. 037) or the automatic brake controller (pos. 037 backup) if not located close to or not used when driving at the activated auxiliary desk.

- "Release" positions of the automatic brake controller (pos. 032) as long as the Traction/brake controller of the auxiliary desk (pos. 032b) is activated.

In presence of 2 auxiliary desks in the same cab and for switching auxiliary desk (e. g. from left side window to right side window):

- The activation of one auxiliary desk shall deactivate the other auxiliary desk.
- The changeover shall be made at standstill.

Precondition for the deactivation of an auxiliary desk and for re-activating the main desk:

- Traction/Dynamic Brake controller on the main desk (pos. 020) in "OFF" position

Deactivation criteria for deactivation of an auxiliary desk and for re-activating the main desk:

- Application of travel direction push button on the main desk (only at standstill) or
- Application of Traction/Dynamic Brake controller on the main desk (pos. 020) from the initial "OFF" position. (There shall be no interruption of traction power when changing from the auxiliary desk to the main desk) or
- Application of driver's automatic brake handle on the main desk (pos. 032: braking positions) or
- If the locomotive is switched off (pos. 508 "off") the auxiliary desk shall be automatically deactivated. In this case the main desk is deactivated too. The next time the locomotive is switched on it shall activate the main desk only.
- The emergency brake position ("SOS" position) of the Traction/Brake controller of the auxiliary desk (pos. 032b) still initiates an emergency brake application by acting on the locomotive's safety device brake valve (like DAC), if the cab is activated ("In-service" configuration - see *UIC Leaflet 612-1*). Thus the deactivation of the auxiliary desk shall not inhibit the "SOS" position of the Traction/Brake controller of the auxiliary desk (pos. 032b).

Indication of missing preconditions for activation or deactivation of the auxiliary desk, status message in the TDD in accordance with *UIC Leaflet 612-03* (driver tries to activate or deactivate the auxiliary desk without success):
In case at least one necessary precondition of activating and inactivating the auxiliary desk is not yet fulfilled, this shall be indicated in the cab (e. g. advise the missing actions in the TDD - see *UIC Leaflet 612-03*).

Indication of fault on the auxiliary desk:
In case of auxiliary desk malfunction the fault shall be indicated in accordance with the principles of fault management and diagnosis in the locomotive and the main desk shall not be affected (no downgraded functions of the main desk).

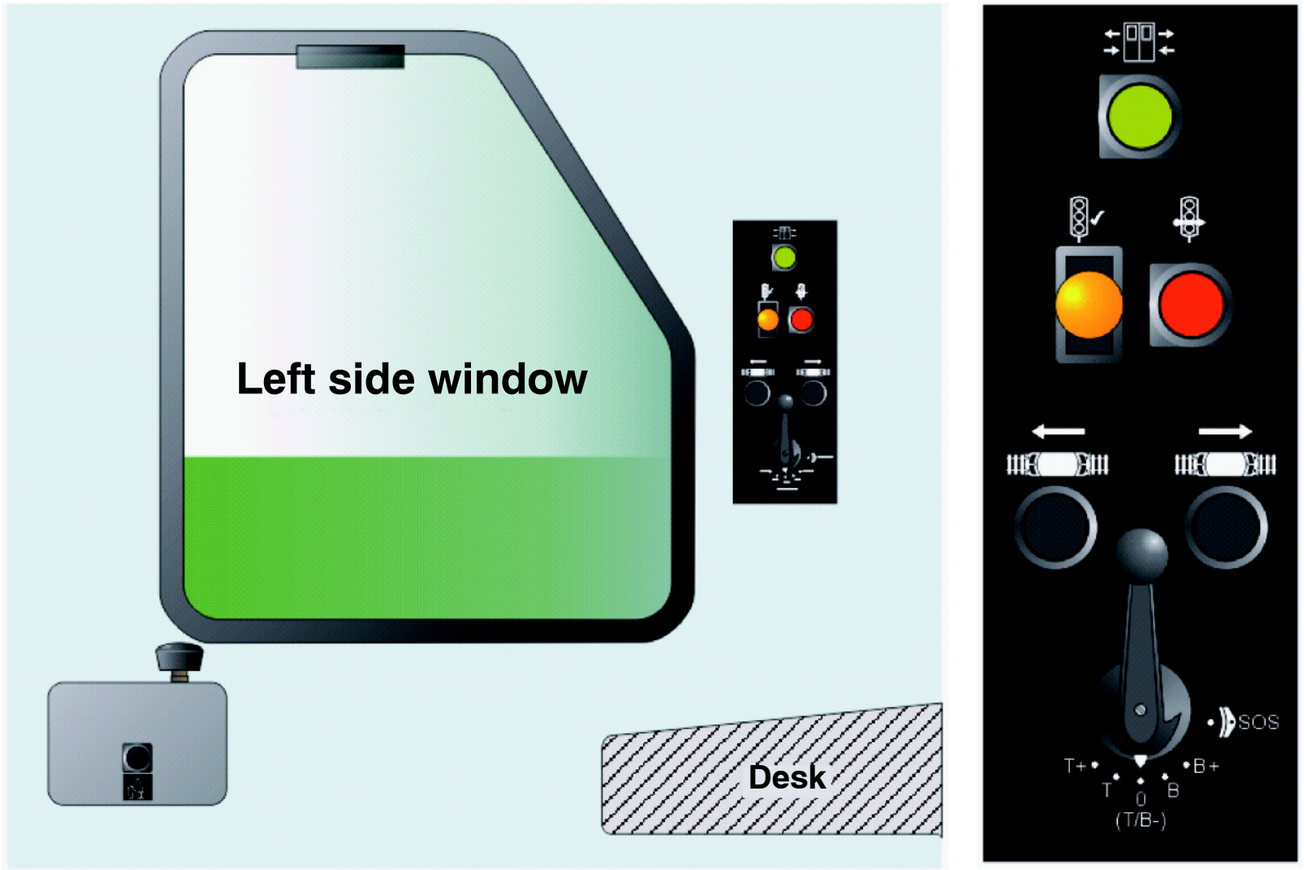


Fig. 12 - Example of an auxiliary driver's desk on the side wall of the cab (example based on Appendix J)

Operating element Position and [form]		Symbol/indication	Position and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
211	**DRIVER ACTIVITY CONTROL BUTTON** **Only for locomotives** **DAC push button** [X]		**U**		See pos. 020, 031, 111 and 503. Prior to the other DAC operating elements, except for pos. 111.	F/R			X	

Appendix E - Operating elements and equipment located in the cab, but not accessible to the driver while driving (e. g. on the driver cab rear wall)

The operating elements described in the following as well as the appliances shall be fixed on the driver cab's rear wall in such a way that they allow for unrestricted operation by the driver when standing, i. e. at a height of 0,8 - 1,5 m above the floor of the cab.

E.1 - Operating elements

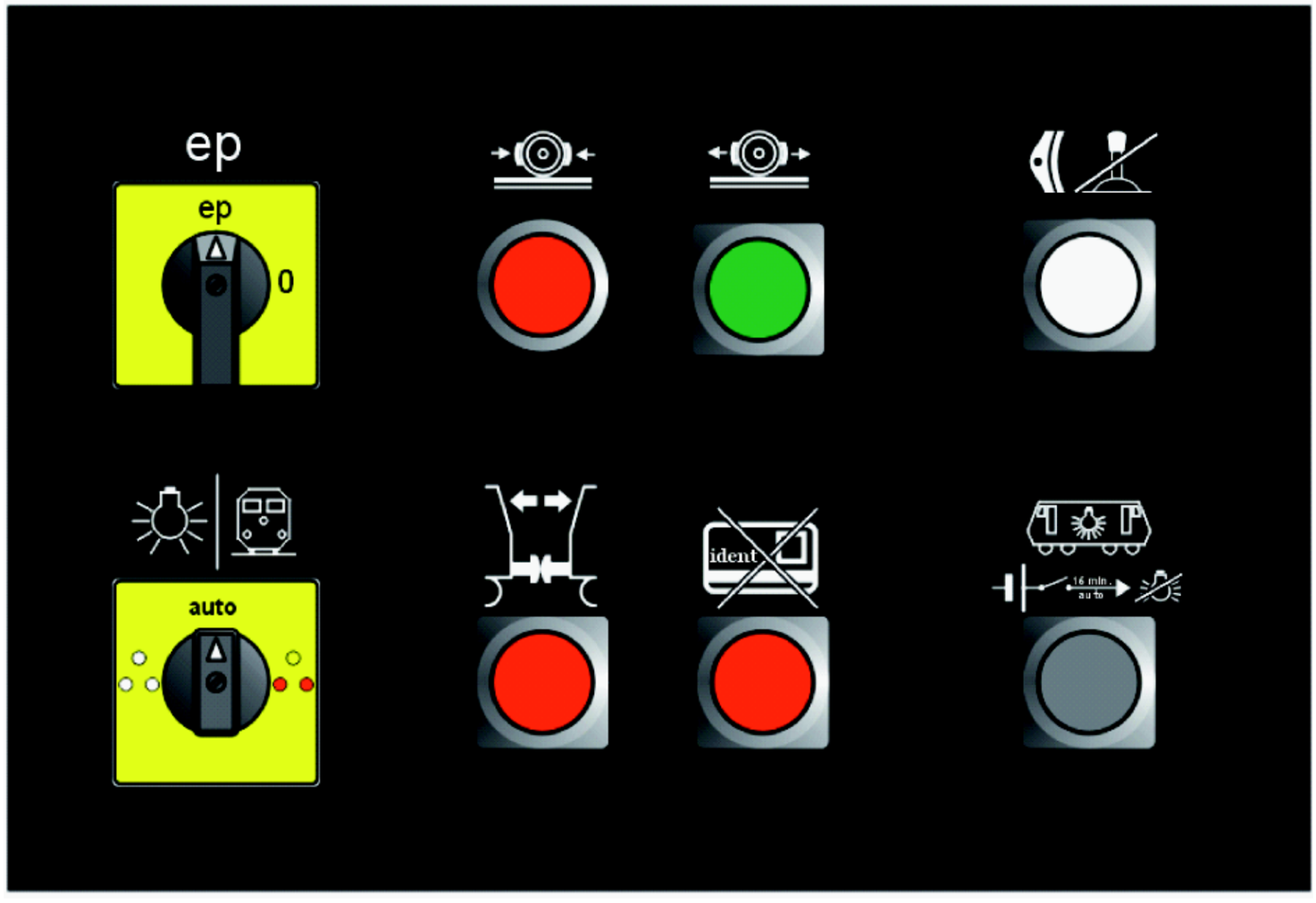
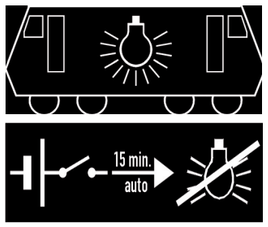
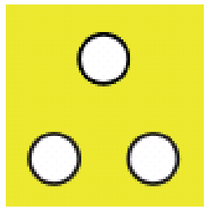
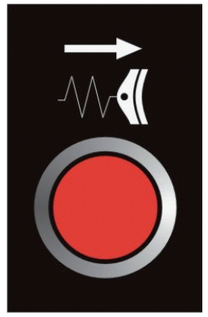


Fig. 13 - Driver cab's rear wall (example based on Appendix J)

Operating element Position and [form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
301	**OVERRIDE the Driver's ID card reader** [R, red push button]		**U**	Activation of push button replaces the insertion of the driver's ID. - In a train the driver's ID must be read prior to the activation of pos. 301 in any cab, which also means that the activation of pos. 301 shall not cancel the driver's ID in any activated cab of the train. - The activation of pos. 301 in a new cab cancels the previous activation of pos. 301 in another cab. - The activation of pos. 301 will be deleted, if the battery is switched off ("parking mode")	Activation sets master cab status "N", deactivation sets "0". See also pos. 036 (ID reader). Possible applications: - For shunting purposes (= operation without driver's data), - Card missing or defective, - Card reader defective.	See note on the left		X		X
302	**ENGINE ROOM LIGHTING** Push button [Q]		**I (on/ off)**	Switching the engine room illumination on and off.	Automatic switching off of engine room illumination after 15 minutes, if battery is switched off (see pos. 502).	Available, except for "self-protection" configuration (see *UIC Leaflet 612-1)*.				

Operating element Position and [form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
303	**Mandatory for locomotives optional for EMU/DMUs and driving coaches** **ISOLATION OF:** **. Driver's automatic brake controller** **. ETCS/STM ("non leading" mode)** Illuminated white flip push pulse button [Q]]		**U**	- Isolating the driver's automatic brake controller despite chosen travel direction (e. g. on the second master locomotive in the case of independently controlled EMU/DMU or locomotives in the train or on the push locomotive in the case of banking). - Switching off/ isolating/ deactivating the train control system without technical and operational restriction (ETCS non-leading mode). - The automatic driver's brake controller can be applied in braking position and the braking force shall be applied too. The release positions of the automatic driver's brake controller are isolated, if this function (pos. 303) is activated. Thus the driver can only apply a brake force. This is to avoid systematic use of the emergency brake by the driver of the second master locomotive.	Note: The illuminated push button also switches the train control system to inactive as it is not operated on the second master or banking locomotive. The train control system shall only be effective on the master vehicle.	F, N, R		X		
	The functions of pos. 303 become effective in the case of: - Applying pos. 303 **and** - V = 0. The functions of pos. 303 remain effective afterwards even if V > 0. After pressing pos. 303 in an occupied driver cab, the push button indicates the effectiveness of the selected functions by means of continuous light and a message is displayed in the TDD ("Automatic brake controller is isolated", see *UIC Leaflet 612-03*). Deactivate: - V = 0 **and** - Applying pos. 303 again. Then the illumination of the push button and the message in the TDD are cancelled.									

Operating element Position and [form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
304	**EXTERNAL LIGHT** Revolving switch [F] Not for SNCF for EMU/DMU : specific case		**S** **- 45°**	Front light manual For function and luminosity see pos. 027.	Marking the front of the train associated with the cab concerned. This function allows the activation of the front light, if the cab is not the master cab.					
			**S** **0°**	Automatic selection of front lights (based on train-wide communication, e. g. Train-Bus) For function and luminosity see pos. 027	Activates the front and tail lights automatically, if train inauguration was successful and confirmed by the driver and the light has not been deactivated in the TDD.	F/N/R				
			**S** **+ 45°**	Tail light manual	Marking the tail of the train associated with the cab concerned.	Available, except for "self-protection" configuration (see *UIC Leaflet 612-1*)				
305	**ep - BRAKE** Revolving switch [F]		**S** **0°**	On (normal position). EP-brake according to UIC-specifications	In long trains due to flow dynamics the transmission times of the BP pressure control signal takes long time. This transmission time can be reduced by means of a decentralized BP pressure control distributed alongside the BP along the rake of wagons: the electro-pneumatic BP control in the train ("EP-brake").	F/N/R				X
			**S** **+ 90°**	EP-brake off This position shall be displayed in the TDD (see *UIC Leaflet 612-03*).						

Operating element Position and [form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
306	**MANUAL APPLICATION of the holding brake or optionally for EMU/DMU of the parking brake** Red-lit push button		**U**	Only effective at standstill: Application of holding brake = local brake force application of the EMU/DMU, locomotive and driving coach (maximum brake force designed applied). Push button shall be lit, if holding brake has been commanded, e. g. by operation of pos. 306 or automatically by TCMS.	- Holding brake replaces the former spring loaded brake for manual application. - Spring-loaded brake is controlled fully automatically (due to R-pressure, see pos. 403). - Indication in the TDD (see *UIC Leaflet 612-03)*. - If the brake force applied by the holding brake is not sufficient to detain the train TCMS shall command an emergency brake application to maintain the train at standstill.			X		X
307	**MANUAL RELEASE of the holding brake or optionally for EMU/DMU of the parking brake** Green push button		**U**	Only effective at standstill: Release of holding brake = full release of the local brake of the EMU/DMU, locomotives and driving coaches. Application commands given apart from pos. 306 remain valid. The holding brake has only 2 effort levels: "Released" (0 kN) or "Applied" (maximum brake force designed).	- Holding brake replaces the former spring-loaded brake for manual application. - Spring-loaded brake is controlled fully automatically (due to R-pressure, see pos. 403). - Automatic release of all holding brakes: the TCMS shall command automatically the release of all holding brakes if the traction power > 10 kN/unit applies.			X		X
308	**DECOUPLING Only for rolling stock with automatic coupler on operator option** Red push button [ ]		**U**	Decoupling the automatic coupler in front of the cab.	Only effective: - at standstill **and** - if no cab activated in the whole train. TFR of the safety-critical event "unintentional decoupling caused by this function" shall be less than 1E-9.	0		X		
320	**APPLIANCES (fridge/ stove)** Flip button [ ]		**S Top**	Heating	For temperate of drinks and snacks.		X			
			S Medium	OFF						
			S Bottom	Cooling						
321	**LATCH OR LOCK HANDLE for appliance's compartment (fridge/stove)** Handle [ ]									

E.2 - Storage facilities

Installation of tools and personal belongings on the rear wall of the driver cab

Personal belongings

- Proposed space on the right rear wall within the EUCAB concept:
  500 mm x 175 mm x 1 200 mm
- It needs to be considered that in this EUCAB concept there is additional space for the driver's belongings in the side cupboards.

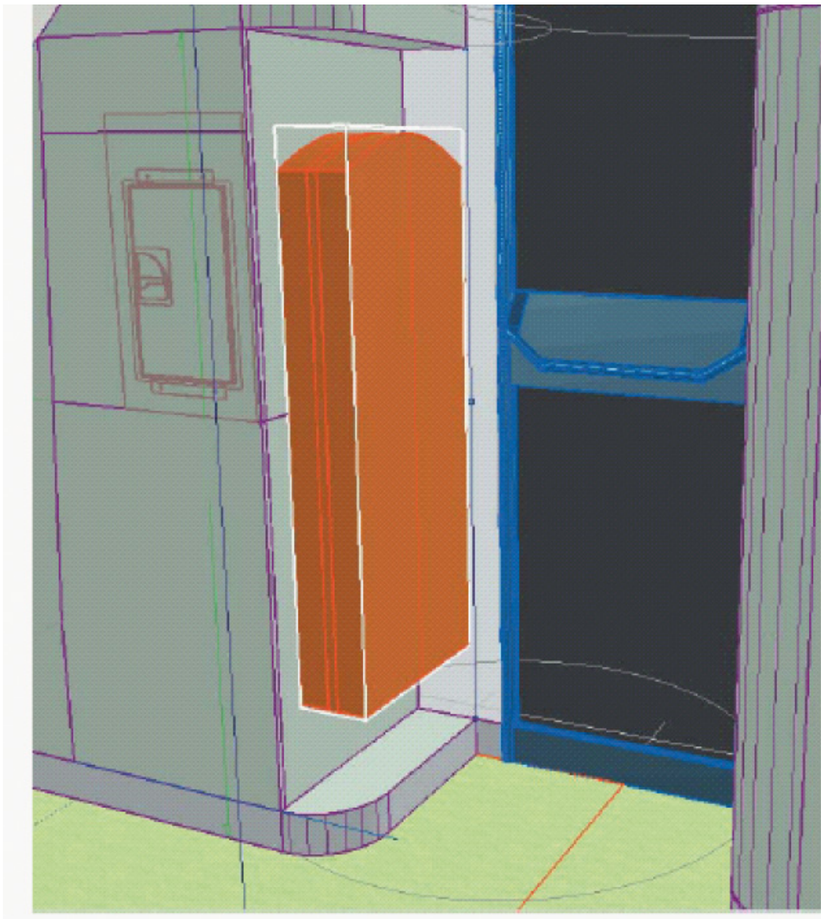


Fig. 14 - Driver cab's rear wall: Storage facilities for personal belongings

Tools, safety equipment

- Preliminary proposed space on the left rear wall within the EUCAB concept:
  510 mm x 175 mm x 1 500 mm.
- Volume includes the following equipment:
  2 fire extinguishers, flag, lamp, first aid kit, helmet, safety waistcoat, gloves.
- This is a preliminary proposal because there is no agreement for a standard set of tools on crossborder trains in Europe.
  The necessary space depends on the amount and volume of tools.

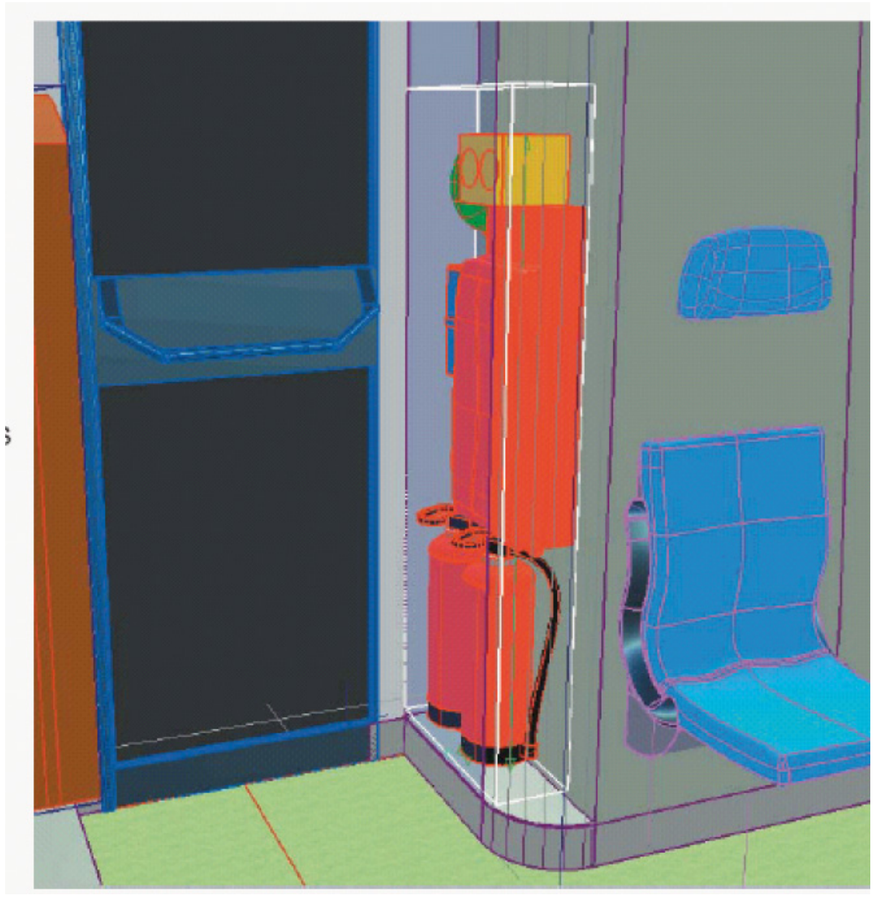


Fig. 15 - Driver cab's rear wall: Storage facilities for tools and safety equipment

Rear wall of the cab

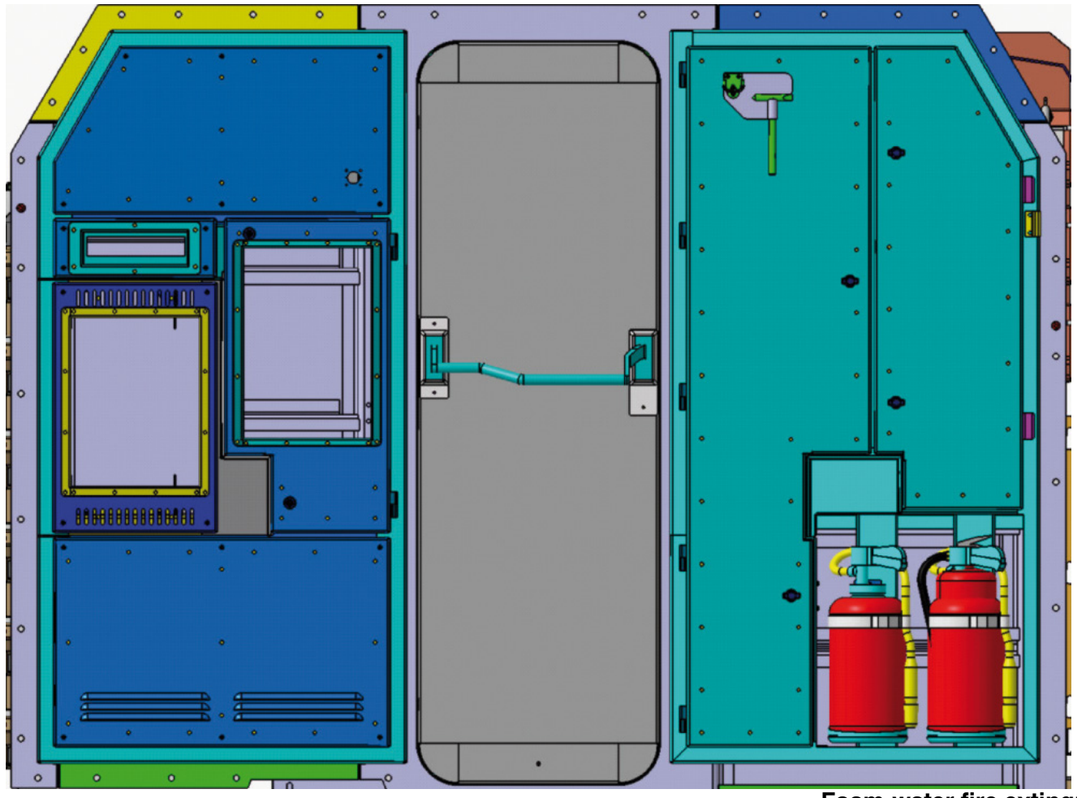


Fig. 16 - Driver cab's rear wall: equipment

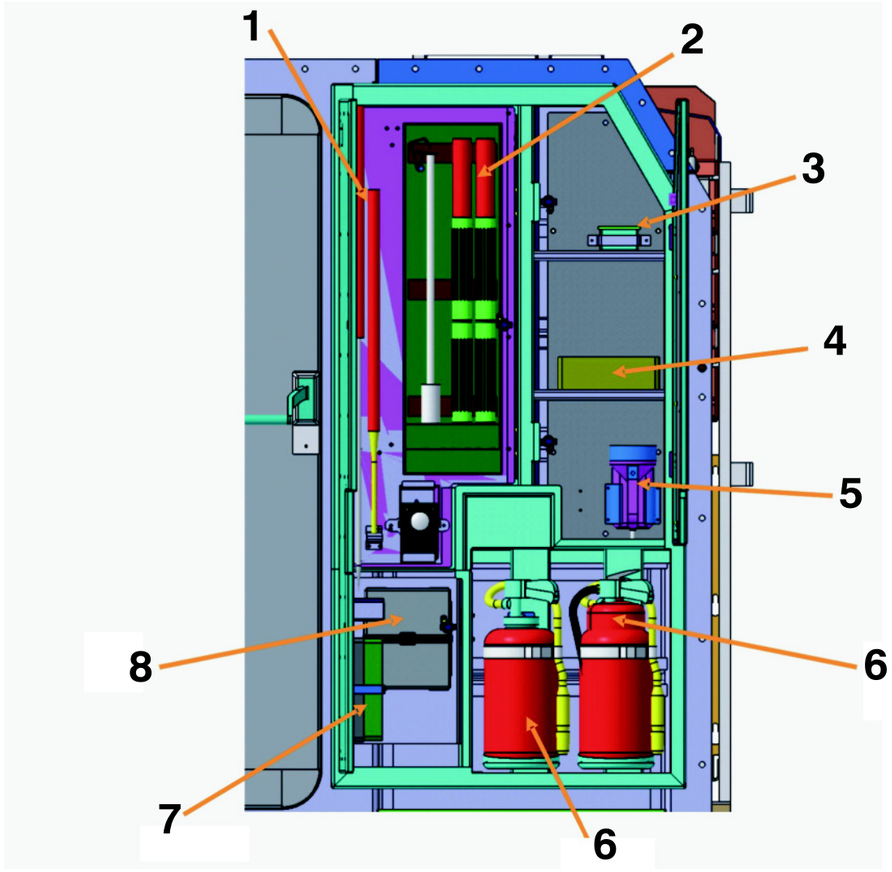


Fig. 17 - Driver cab 's rear wall: equipment provided together in the same place (picture shows an example)

Standard equipment requested by (future) operational requirements

Mandatory in the cab:

6 Foam-water type fire extinguisher.

Optionally in the cab, but mandatory in the train:

7 First-aid kit according to EN.

8 For manned EMU/DMU, locomotives and driving coaches of freight trains: self-rescue mask according to SRT TSI, section 4.7.1.

Voluntary in the train as an operator option, if not requested by the infrastructure register:

1&2 Signalling equipment.

3 Short-circuiting cable.

4 Body cover (accident).

5 Torch (hand-lamp).

Appendix F - Operating elements grouped together on each unit and not necessarily located in the cab; storage of equipment

F.1 - Operating elements

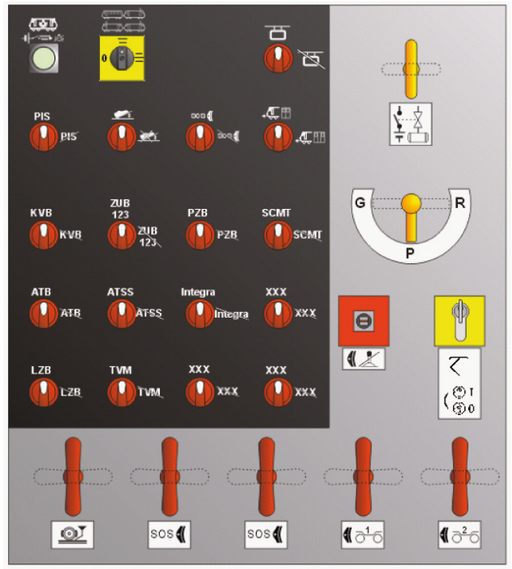
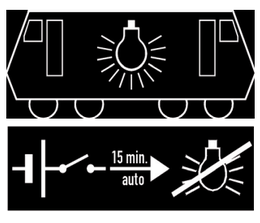
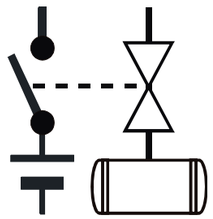
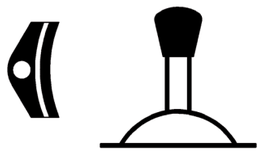
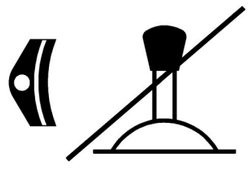


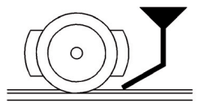
Fig. 18 - Engine room: Operating elements should be provided together in the same place: optionally on different panels (picture shows an example of arrangement of the operating elements)

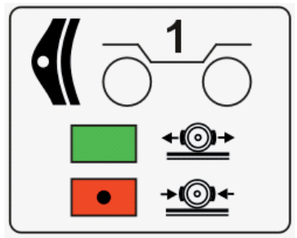
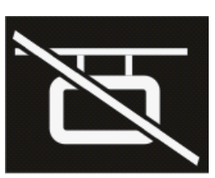
Operating element Position and [Form])		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
501	**PASSENGER INFORMATION SYSTEM** Revolving switch [H]		**S 0°**	PIS switched on.				X		X
			**S + 90°**	PIS switched off.						
502	**ENGINE ROOM LIGHTING** Self-luminescent push button [Q]		**I (on/ off)**	Switching the engine room illumination on and off.	Automatic switch off of engine room illumination after 15 minutes if battery (pos. 508) is switched off (see pos. 302).	Available, except for "self-protection" configuration (see *UIC Leaflet 612-1*).				

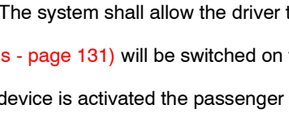
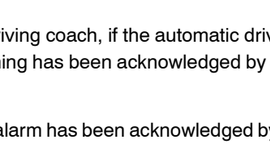
Operating element Position and [Form])		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
503	**FAULT SWITCH for Driver Activity Control (DAC)** Revolving switch [S]		**S 0°**	Operating position (DAC switched on).	See note below.	F/R			X	
			**S + 90°**	DAC switched off.	- DAC automatic application of the brakes: not possible. - Traction lock removed: activation is displayed in the TDD (see *UIC Leaflet 612-03*)					
	The following elements are considered as "DAC operating elements": pos. 020 (knob on top, sensitive area), 031, 211 and 503. **However, every driver control action recognized by the TCMS of the vehicle shall also be valid as a "DAC control operation". The following operating elements, in addition to the "DAC operating elements", shall be evaluated in any case:** - Traction / Dynamic Brake Controller (pos. 020), - Traction / Brake Controller (pos. 032b), - V-target actuator (pos. 019), - Driver's automatic brake controller (pos. 032, in case of backup also pos. 037), - Direct brake controller (pos. 037), - External warning horn (pos. 038), - ATP (e. g. ETCS) operating elements (pos. 013 - 015). The DAC function is specified in CR RST TSI, clause 4.2.9.3.1 and *UIC Leaflet 641*. The DAC test is specified in *UIC Leaflet 612-1, point 4.1.3.3*." If no driver action happens for longer than 2,5 s the DAC display is started. The continuous operation up to 30 s as an acknowledgement is only valid for DAC operating elements. If a DAC operating element is applied for more than 30 s, then the DAC display is started. An automatic emergency application of the brake is displayed via a status message and shall be connected to a fault message in the TDD, which, if possible, identifies the cause and for which applicable help texts are available (see *UIC Leaflet 612-03*).									
504	**TRAIN-WIDE COMMUNICATION Mode selection: (e. g. Train Bus)** Revolving switch [J]		**UIC Train Bus (WTB) or communi-cation system based on TCP/IP**	**S 0°**	Normal position. Automatic mapping of the master with all slaves based on - *UIC Leaflet 556* (TCN) and then - TCP-IP. The master automatically selects the most powerful communication channel that is supported by all units in the train.	Open coupling concept: - After coupling, the master (= server) shall start the mapping according to *UIC Leaflet 556* and TCP-IP and optionally according to heritage communication standards. - All slaves (= clients) report back to the master how they want to communicate with him, because they should understand at least one "language" offered by the master. - Then the master provides in parallel all commands in the languages requested in the train and receives the messages provided by the slaves in the "language" they support. Standardised data profiles for TCP/IP shall be the same as *UIC Leaflet 556* as a minimum.			X		
			0)	**S + 45°**	Off. Only operational for a slave: the slave shall no longer communicate with the master.		**0**				
			Other train-wide communi-cation system	**S + 90°**	Reserved for operator option.					

Operating element Position and [Form])		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
505	**FAULT SWITCH** **"Emergency brake triggered by TCMS"** Revolving switch [S]		**S** **0°**	Operating (normal) position.	An automatic emergency application of the brake is displayed via a status message and shall be connected to a fault message in the TDD, which, if possible, identifies the cause and for which applicable help texts are available (see *UIC Leaflet 612-03*).					X (only message)
			**S** **+ 90°**	Override of automatic application of the brakes by the electronic (brake) control (TCMS).	- Automatic application of the brakes by TCMS not possible (see also *UIC Leaflet 612-2*). - Traction interlock cancelled in case of degraded control of the TCMS automatic emergency brake (see *UIC Leaflet 612-2*). Activation is displayed in the TDD (see *UIC Leaflet 612-03*).					
506	**OVERRIDE of traction interlock in case of an open door** Revolving switch [S]		**S** **0°**	Normal position (Operating position).	If at least one door is open, the traction is cut off. See note below.	F/R		X		
			**S** **+ 90°**	Override of traction interlock in case of open door.	Even if a door is open, traction is possible. Activation is displayed in the TDD (see *UIC Leaflet 612-03*).					
	For unintentionally opened doors (= the train's door loop is opened while in excess of V = 10 km/h, i. e. train running on the line) the following procedure shall apply: - a message is displayed in the TDD (see *UIC Leaflet 612-03*) and - traction interlock is applied, - no further action if the open door alarm has been acknowledged by operating a soft key in the TDD ("open door alarm acknowledged" - see *UIC Leaflet 612-03*) or the driver's automatic brake controller in position "BP > 5 bar" (see pos. 032) within 5 s after acoustic and visible (TDD) indication in the cab. - in case of missing driver acknowledgement an emergency brake is automatically activated.									
507	**"BRAKE MODE"** **Selector** **Handle or operating elements like pos. 406, 407, 408**	**R**		Brake mode "R".	Change of brake mode shall be an intended action and be designed in order to prevent unintentional change of modes (e. g. at V > 0). The brake mode can be operated manually or by remote control. The position of the selector shall indicate the selected brake mode and the TDD (*UIC Leaflet 612-03*) also shall indicate the brake mode selected. Remote control and manual control shall only be possible in V = 0. The selected brake mode position shall be safely mechanically locked and in case of the availability of the remote control, TCMS shall continuously control the brake mode selector position. In case of a default, a message shall be displayed to the driver by TDD (see *UIC Leaflet 612-03*).	Available, except for "self-protection" configuration (see *UIC Leaflet 612-1*).				
		P		Brake mode "P".						
		G		Brake mode "G".						

Operating element Position and [Form])		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
508	**Only for EMU/DMU, locomotives and driving coaches** **PREPARATION SHUTDOWN** **("Energy main switch" → Battery isolation switch function + combination air-supply)** [ ]		**I**	- EMU/DMU, locomotives and driving coaches switched on electrically and pneumatically. - Holding brake shall be automatically applied. - Holding brake will be released by • traction command (pos. 020) or • release position of the "holding brake position" (pos. 307) or • setting the direct brake controller from the middle position to release position (pos. 037).	Coupling of train-wide communication **and** pos. 504 in position other than "0": Slave EMU/DMU or locomotives can be switched on electrically and pneumatically by remote control after successful train inauguration via TDD (see *UIC Leaflet 612-03*). For ETCS actions on the EMU/DMU, locomotives and driving coaches see *UIC Leaflet 612-1*. *UIC Leaflet 612-1*: "switched on" and higher configurations.	Available in all configurations.				
			0	EMU/DMU, locomotives and driving coaches switched off electrically and pneumatically, but able to be switched on remotely by train-wide communication (except for self protection configuration).	Normal position for parking and pulled EMU/DMU, locomotives and driving coaches as a wagon (see *UIC Leaflet 612-1*: "shut-down" configuration)					
	If pos. 508 is switched off and the EMU/DMU, locomotive or driving coach is not in self-protection configuration it shall detect whether its remote control system receives signals from a prepared master EMU/DMU, locomotive or driving coach after its coupling. Via remote control, the battery main switch (pos. 508) shall be set functionally to the "I" state and participate successfully in a train inauguration. The direction setting of the master is adopted. Raising the pantograph and switching on the main circuit breaker shall be commanded manually by the driver on the master EMU/DMU, locomotive or driving coach by actuating the operating elements (pos. 017 or pos. 016 for pantograph only). After uncoupling the train-wide communication (e. g. train bus), the slave EMU/DMU, locomotives and driving coaches shall enter the "shut-down" configuration, i. e. pos. 508 being switched off. Automatic emergency brake application, if pos. 508 is - switched off → BP automatically to be emptied until BP < 0,5 bar has been continuously achieved and then BP to be isolated. An unintentional internal feeding of the BP and R-reservoirs by leakages of connections to MP or reservoirs shall be prevented by automatic venting of the BP (the BP shall not become higher than 1,0 bar). An intentional external feeding shall isolate the BP-venting system to be able to run the EMU/DMU, locomotives and driving coaches as a wagon. - switched on by the battery main switch (pos. 508) when running (V > 0) → BP automatically emptied.									
509	**OPERATOR OPTION** **"Isolate Driver's Automatic Brake Controller"- operating element** **Operated with a "Berne Key"**		**S**	Unlock automatic brake controller. Feeding the BP effective.	Pos. 509 is requested if the failure rate of the main isolation device (pos. 303) for isolating the driver's automatic brake controller is higher than 10^{-6} p/h. This element allows the manual isolation / de-isolation of the driver's automatic brake controller in case of degraded situation.	Available in all configurations.				
			**S**	Lock automatic brake controller. Feeding the BP ineffective.						

Operating element Position and [Form])		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
510	**"CONNECTING HV TO EARTH"-device** Red handle and key See Appendix M		**I** **0°**	Operating position: - Safety loop for pantograph and MCB is closed, - "Connecting-to-earth"-switch is open.	Key cannot be removed (see Appendix M)	Available in all configurations				
			20°	Preparation position: - Safety loop for pantograph and MCB is open. - "Connecting-to-earth"-switch is open. **Option:** - Rely on operational rule "driver checks position of pantograph before connecting to earth" **or** - Monitoring device for pantograph position does not allow further moving of the pos. 510 to the "0" position unless pantographs are lowered.	Key cannot be removed				
			O **90°**	Connecting to earth (Bonding to chassis (car-body) position).	- Key can be removed (possibly without turning). - Required for opening the high-voltage chamber. Therefore this key and all further keys are locomotive-specific and not interchangeable between different rolling stocks.					
511	**ISOLATION COCK** **"sanding"**		**I**	Operating position	See pos. 025 and *UIC Leaflet 612-2*.	Available in all configurations				
			0	Fault. Message in the TDD, see *UIC Leaflet 612-03*.						

Operating element Position and [Form])		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
512 **513**	**ISOLATION COCK** **n° 1 or n° 2** **"Automatic emergency brake valve"**		**I**	Operating position.	An automatic emergency application of the brake is displayed via a status message and shall be connected to a fault message in the TDD, which, if possible, identifies the cause and for which applicable help texts are available (see *UIC Leaflet 612-03*). The valves shall comply with *UIC Leaflet 541-03*.	Available in all configurations				
			O	Fault.	- Brake valve no. 1 or no. 2 dedicated to this cock is pneumatically isolated. So no automatic application of the emergency brake possible - Display with status message in the TDD (see *UIC Leaflet 612-03*). - If the 2 cocks (pos. 512 and 513) are both in "0" position, a traction interlock applies. This traction interlock can be cancelled by operating the override switch "SOS" (pos. 505) (see also *UIC Leaflet 612-2*).					
512b **513 b**	**OPERATOR OPTION** **ISOLATION COCK** **n° 1 and n° 2** **"Automatic emergency brake valve"**		**I**	Operating position.	See pos. 512, 513. This element enables isolation of 1 emergency brake valve only, on vehicles equipped with 2 emergency brake valves. It ensures that at least one emergency brake valve always remains available.	Available in all configurations				
			0-1	Fault brake valve 1.						
			0-2	Fault brake valve 2.						
514 **515**	**OPERATOR OPTION** **Isolation cock** **"Air Brake bogie x"**		**I**	Operating position.	This cock shall be implemented in case of one distributor only on the EMU/DMU, locomotive or driving coach. Air brake of the notified bogie switched off → Display with status message in TDD (see *UIC Leaflet 612-03*).	Available in all configurations.				
			0	Automatic and direct air brake isolated and released.						
516	**ISOLATION OF** **PASSENGER ALARM** Revolving switch with seal [S]		**S** **0°**	Passenger alarm function is available: the passenger alarm signal is transmitted from the coach to the activated driver cab. See specification below	Automatic application of the brakes if pos. 516 is operated while driving Activation is displayed to the driver in the TRD. For the complete specification see: CR RST TSI, ch. 4.2.5.3, *UIC Leaflet 541-6* and *UIC Leaflet 612-04* (TRD)					
			**S** **+ 90°**	- Switching off the passenger alarm function in the event of a fault. - In this position, an activation of the passenger alarm in a coach shall empty the BP of the train.						

Operating element Position and [Form])	Symbol/indication	Positions and function	Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
				Master cab status	MCB	v = 0	v > 0	MT
If a passenger alarm device is activated by a passenger in a coach: - The train's passenger alarm signal is transmitted to the activated driver cab. - A message is displayed in the TRD (see *UIC Leaflet 612-04*) and an acoustic signal is sounding in the cab. - On the driver's initiative, a communication link shall be established between the TRD (see *UIC Leaflet 612-04*) and the place where the alarm(s) was/were triggered. The system shall allow the driver to cancel this communication link on his initiative. - Units designed for operation without staff on-board (other than the driver) shall be equipped with a "call for aid" device for passengers to communicate with the driver in case of emergency. In this case the system shall allow the communication link to be established on the passengers' initiative. The system shall allow the driver to cancel this communication link at his own initiative. - Optionally CCTV (see List of abbreviations - page 131) will be switched on for the area where the passenger alarm has been activated (TDD, see *UIC Leaflet 612-03*). - As long as at least one passenger alarm device is activated the passenger alarm safety-loop is open and therefore *UIC Leaflet 612-04* (TRD) applies. **Train running except when starting from a platform or when the speed is lower than the high speed value:** - In case of an EMU/DMU, locomotive or driving coach, if the automatic driver's brake controller (pos.032a) has been applied in "> 5 bar" position or the traction/brake controller (pos. 032b) has been applied in "T+" position, even following passenger alarms, and if the first passenger alarm while running has been acknowledged by operating the soft key "passenger alarm acknowledged" in the TRD (see *UIC Leaflet 612-04*) within 10 ± 1 second after activation of the (first) passenger alarm, then no further action is required. - After this time, and unless the passenger alarm has been acknowledged by the driver within this time, at least one automatic service brake shall apply. The driver shall be able to override at any time an automatic braking action initiated by the passenger alarm. - To release the automatic service brake the driver shall push the driver's automatic brake controller (pos. 032a) into "BP > 5 bar" position or the traction/brake controller (pos. 032b) shall be applied in "T+" position. **Train running at high speed:** - If the train is running at high speed, activation of the passenger alarm in a coach shall automatically activate a service brake to reduce the speed to below the high speed value. - To release the automatic service brake the driver shall push the driver's automatic brake controller (pos. 032a) into "BP > 5 bar" position or the traction/brake controller (pos. 032b) shall be applied in "T+" position. **Train starting from a platform:** - When the train is stopped at a platform or departing from a platform, activation of a passenger alarm shall lead to a direct application of the service brake or the emergency brake, resulting in a complete stop. In this case and only after the train has come to a complete stop, shall the driver be able to cancel any automatic braking action initiated by the passenger alarm. - **Derogation from the TSI:** As an operator option for specific lines or specific train use (e. g. line running completely in a tunnel, train only running in a big city), this automatic brake application can be overridden by the driver using a soft key in the TDD (see UIC *UIC Leaflet 612-03*). In case of such an override specific requirements apply when the train is running. - The passenger alarm and automatic brake application are reset if there has been a driver acknowledgement **and** V = 0 **and** the passenger alarm device handles in question have been reset to their normal position. - Criteria for a train departing from a platform: A train is deemed to be departing from a platform during the period of time elapsing between the moment when door status is changed from "released" to '"closed and locked" and the moment when the last vehicle has left the platform. This moment shall be detected by an on-board device, and provisions shall be made to trigger it when: • The speed of the train reaches 15 (± 5) km/h, or • The distance covered is 100 m (± 20) m (whichever occurs first).								
52x	**ISOLATING OF STM (specific transmission module)** **For example fault switch of the KVB (French STM)** Revolving switch with seal [S]		**S 0°**	KVB function is available.	Automatic application of the brakes if operated while driving. Activation is displayed in the TDD (see *UIC Leaflet 612-03*).			X		
			**S + 90°**	- Switching off the KVB in the event of a fault, - shall prevent brake intervention by isolated STM.						

F.2 - Additional Equipment

In addition to the equipment mentioned in point E.2 - page 64 the following shall be stored on board of a train:

- On lines with gradients > 3,5 % only or if the EMU/DMU, locomotive or driving coach cannot be immobilised with its parking brake alone on lines with gradients up to 3,5 %: European standardised scotches (see below) or mutual recognition (cross acceptance) of heritage scotches.

- Spare parts and tools may be stored on board of a train as an operator option.

Scotches ("drag shoes")

Type and number of scotches to be stored will be standardised in future. Wooden scotches are preferable due the reduced risk of derailment when passing over points due to forgotten scotches. All European hardwoods are suitable - oak, chestnut, etc. Pine is unsuitable.

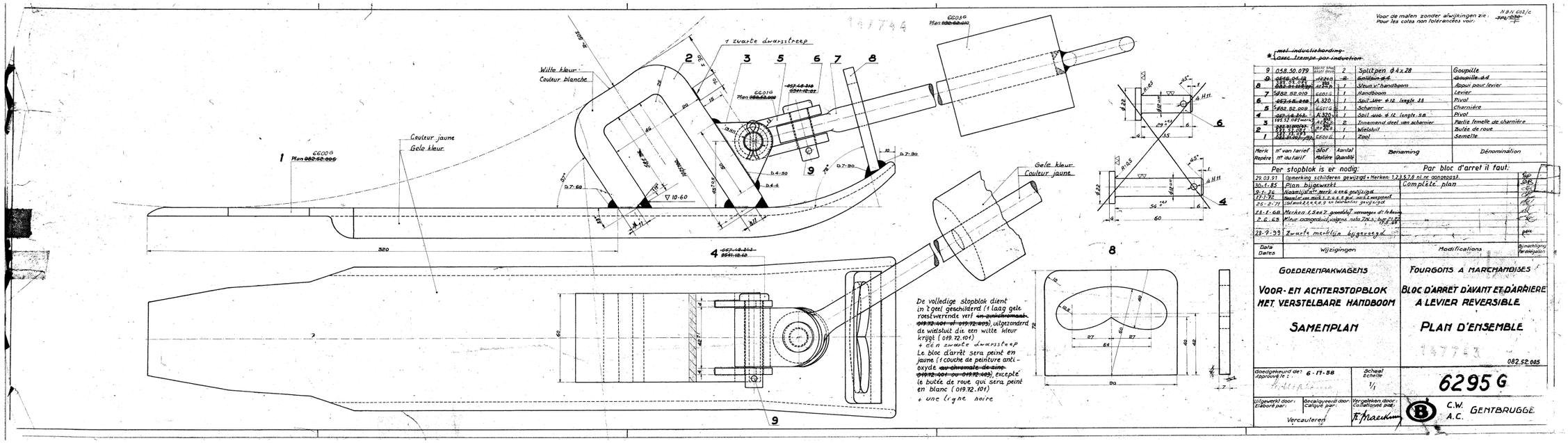


Fig. 19 - Stop block with two-way lever (metallic)

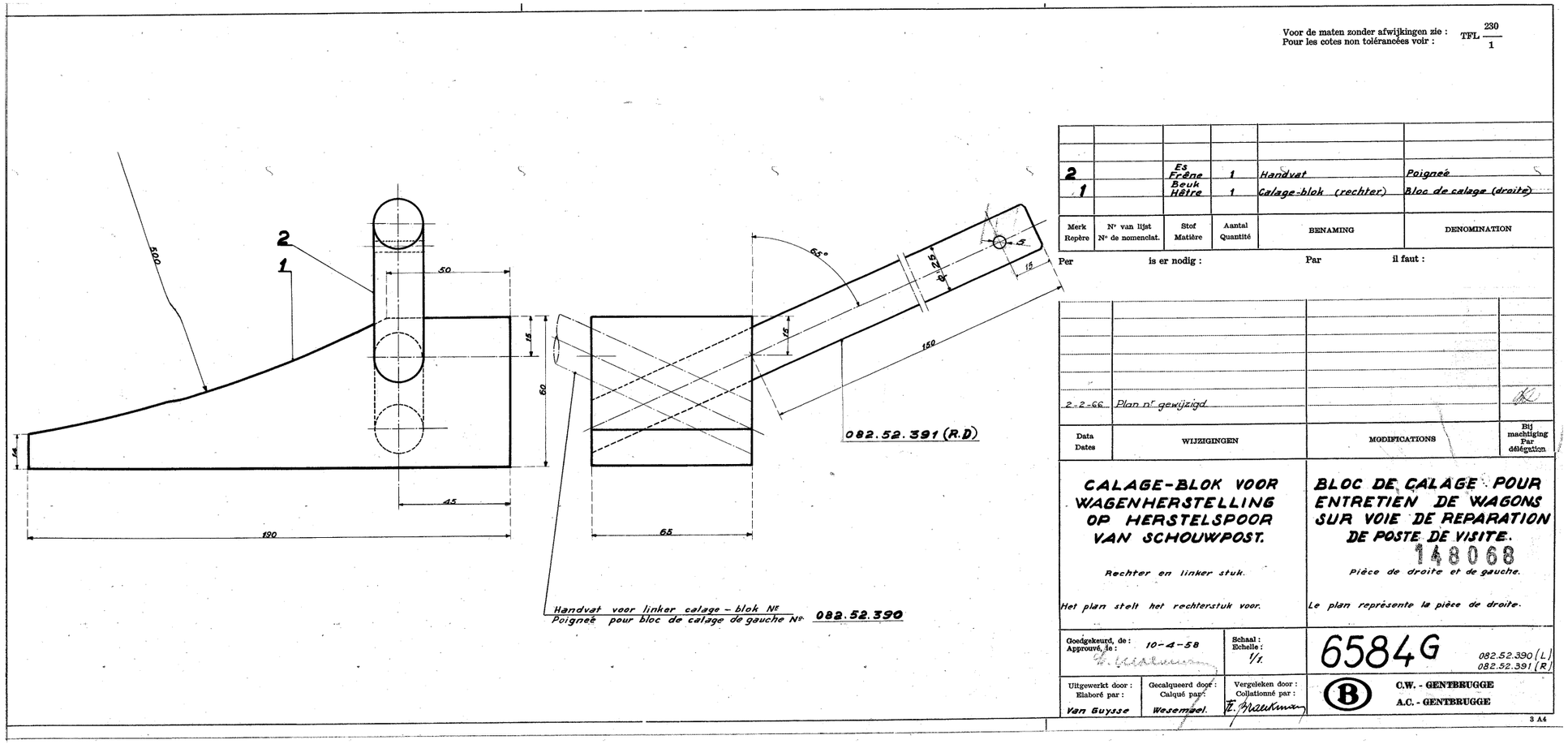


Fig. 20 - Standardised wooden scotches

Appendix G - Operating and display elements outside the EMU/DMU, locomotive and driving coach

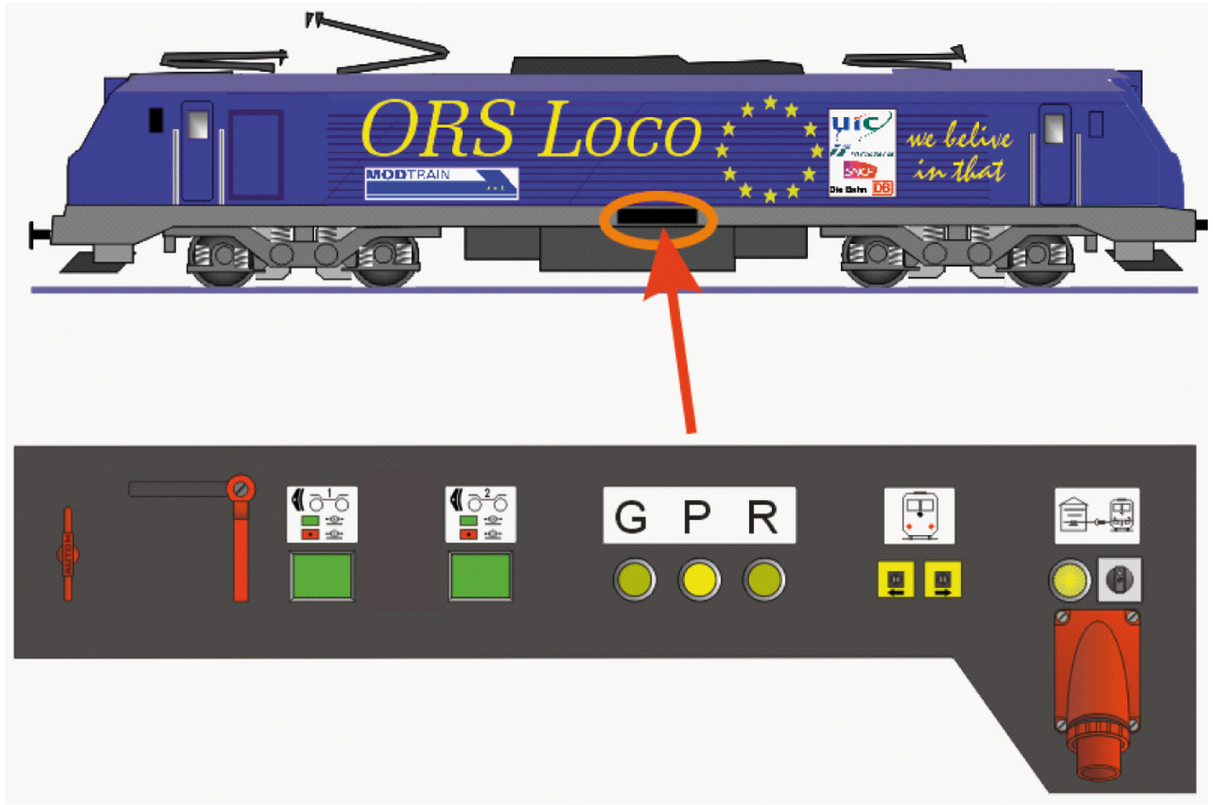


Fig. 21 - Operating and display elements outside the EMU/DMU, locomotive and driving coach (picture shows an example for locomotives)

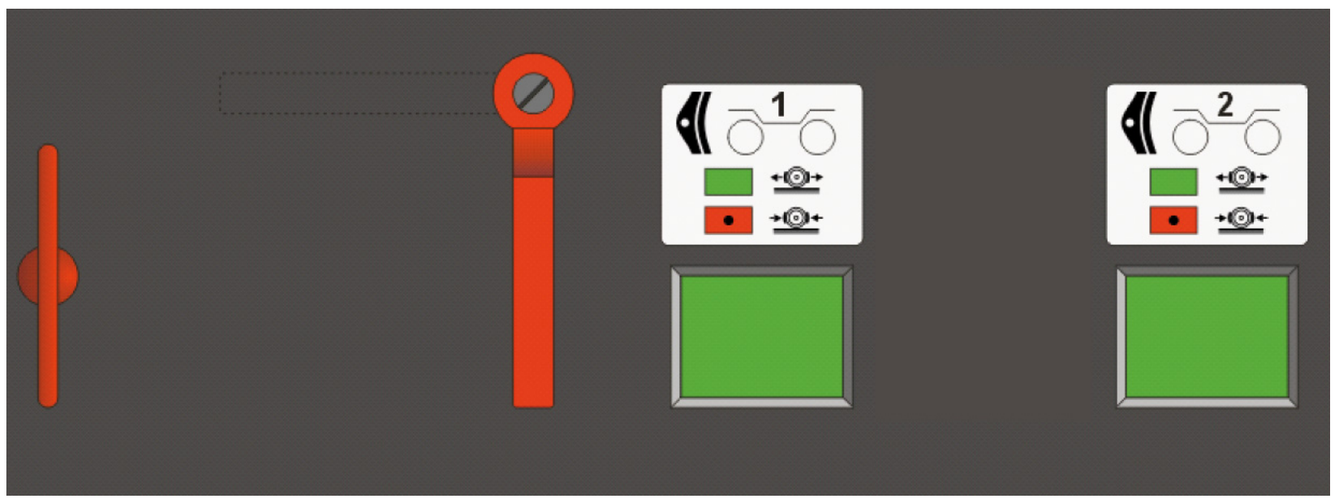


Fig. 22 - Operating and display elements outside the EMU/DMU, locomotive and driving coach: example with 1 distributor

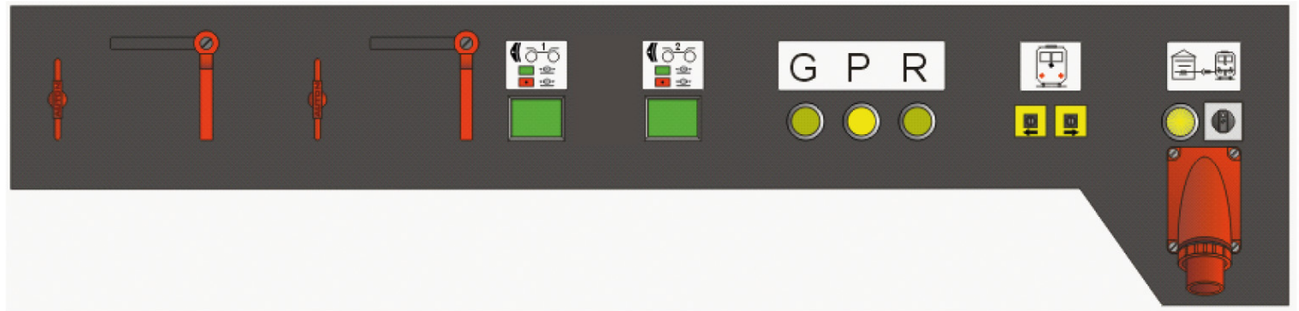
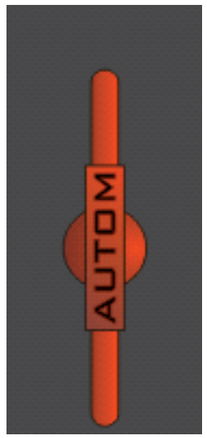
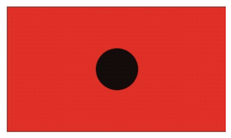
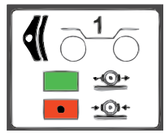
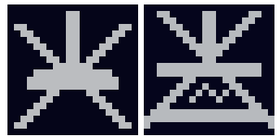
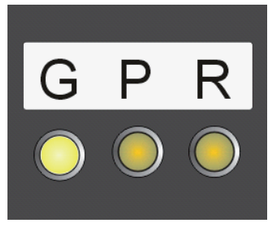
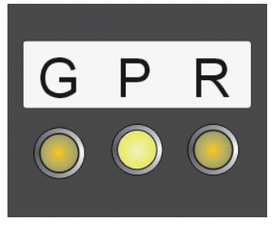
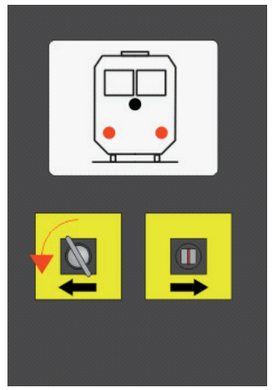
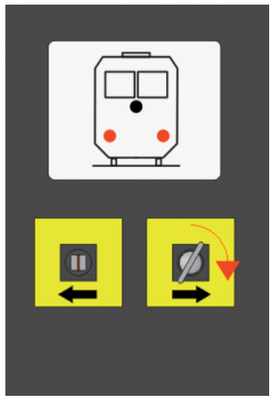


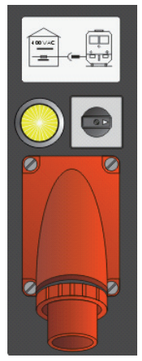
Fig. 23 - Operating and display elements outside the EMU/DMU, locomotive and driving coach: example with 2 distributors

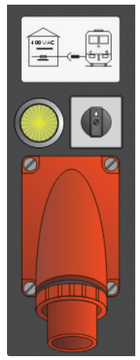
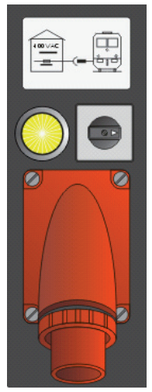
Operating element Position and [form]		Symbol/indication		Position and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
							Master cab status	MCB	v = 0	v > 0	MT
401	**OPERATOR OPTION AUTOMATIC BRAKE RELEASE DEVICE (Device for releasing the distributor valve's pilot chamber)** Pull-Handle [ ]	 (example for the option of automatic function)		Release of brake cylinder pressure caused by automatic air brake.		On both sides of the underframe or inside the rolling stock there shall be a handle for each distributor for releasing the automatic brake by emptying the pilot chamber of the distributor valve and, as an option, to empty the auxiliary air reservoir. As an option, one short application shall empty the chamber totally. Afterwards the distributor valve shall automatically be ready for filling and automatically become fully operational again, if the automatic brake is applied again (given that the automatic brake has not been isolated before). In case of 2 distributors on the EMU/DMU, locomotive or driving coach, a pictogram shall be added to distinguish bogie 1 from bogie 2 (see pos. 514 and 515 as example).	Available in all configurations.				

Operating element Position and [form]		Symbol/indication	Position and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
402	**OPERATOR OPTION AUTOMATIC BRAKE ISOLATING COCK** Revolving cock with handle [S]		**S** **0°**	Automatic brake of the unit (EMU/DMU, locomotive or driving coach or a bogie) available.	On both sides of the underframe or inside the rolling stock there shall be a red cock for isolating and releasing the automatic brake. In case of 2 distributors on the EMU/DMU, locomotive or driving coach, a pictogram shall be added to distinguish bogie 1 from bogie 2 (see pos. 514 and 515 as example). As an operator option, a locked device could be requested to avoid misuse action. A Berne key shall be used for locking and unlocking.	Available in all configurations.				
			**S** **+ 90°**	Automatic brake of unit inactive and released. in this position: - the distributor is isolated from the MP and the BP, **and** - the brake cylinders are emptied; **and** - as an option, the distributor is isolated from the air-reservoir **or** the auxiliary air-reservoir is emptied ➔ In this last case the parking brake shall be automatically locked in emergency release position Except for self-protection and shut-down configurations: A status message on TDD shall be given to the driver (see *UIC Leaflet 612-03*).					
403	Mandatory for locomotives, optional for EMU/DMU: One indicator per bogie or as an option one per wheelset **BRAKE INDICATOR** Flag indicator			"All brakes released" - brake cylinder pressure in all brake cylinders: C < 0,2 bar, **and** - parking brake is being released or fully released (R-Reservoirs are being refilled or supplied).	Except for self-protection and shut-down configurations: the brake status message shall be displayed on TDD (see *UIC Leaflet 612-03*)	Available in all configurations.				
				"Brake applied" At least one air brake or parking brake has been applied. The parking brake force shall compensate the loss of the pneumatic brake force (direct brake, holding brake, automatic brake - see *UIC Leaflet 612-2*), - brake cylinder pressure in at least one brake cylinder: C > 0,4 bar **and/or** - parking brake is not fully released.					

Operating element Position and [form]	Symbol/indication	Position and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
					Master cab status	MCB	v = 0	v > 0	MT
Flag indicators for compressed air brake cylinder and the parking brake function state: - The drop signals for indicating the status of the brake (including the parking brake) shall be fitted on the long sides of the vehicle. - Intermediate positions for the drop signal are not allowed. For each brake indicator the relevant bogie or wheelset shall be clearly identified:   **Degraded situations:** No degraded situation that may provide a brake status different from the drop signal shall occur apart from the following: mitigation by operational rules and staff indicating the downgraded situation by appropriate markings on both sides of the vehicle if the automatic air brake has been manually isolated or the parking brake function has been manually deactivated or released.									
405	**For EMU/DMU:** **Mg-brake** **Or** **Eddy Current-brake**		**0**	Off		Available, except for "self-protect-ion" con-figuration (see *UIC Leaflet 612-1*)				
			I	On						
406 **407** **408**	**OPERATOR OPTION** **Brake mode changeover buttons:** **For example, brake changeover for brake mode "G"** lit push button [ ]		**I**	Brake mode G applies (LED lit).	Selected brake mode shall be available in the TDD, also for slave units (see *UIC Leaflet 612-03*). Change of brake modes is only possible in V = 0. LED shall be lit only if a closely located lamp test button has been operated - as for water supply level indication in coaches.	Available in all configurations, except for "self-protection" configuration.				
				Another brake mode applies (LED for "G" shall remain dark, LED for the selected brake mode can be lit).						
On both sides of the underframe, marked in a manner which is clear for railway staff but not evident to third parties, three yellow illuminating push-buttons shall be available for local choice of braking mode according to the relevant operational rules for the EMU/DMU, locomotive or driving coach in question. The active brake mode shall be indicated on staff request using the lamp test button (i.e. the light indicates the active brake mode). Energy consumption (electric or pneumatic) is only permitted for selecting but not for retaining the selected braking mode. Remote-control selection of braking mode shall also be possible. If there is no train bus application or command in the leading cab for vehicles in the train the brake mode shall be chosen locally on those vehicles. For display requirements see pos. 507.									

Operating element Position and [form]		Symbol/indication	Position and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
409	**Tail signal at the left end of the EMU/DMU or locomotive** square key switch [ ]	 Ex.: Staff applies the Berne key to the left switch.	**U** **30°**	Tail signal is switched on locally at the adjacent end of the unit.		Available in all configurations, except for "self-protection" configuration.				
410	**Tail signal at the right end of the EMU/ DMU or locomotive** square key switch [ ]	 Ex.: Staff applies the Berne key to the right switch.	**U** **30°**	Tail signal is switched on locally at the adjacent end of the unit.		Available in all configurations, except for "self-protection" configuration.				

Operating element Position and [form]		Symbol/indication	Position and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
411	**Indicator for external power supply (shore supply)** white light [ ]		**lit**	External supply via plug available.	A white indicator light (LED) shall be provided immediately next to each plug, indicating the voltage supply of the battery charger powered by an external supply via the plug, irrespective of the side of the vehicle on which feeding takes place.	Available in all configurations.				
412	**3AC 400 V 50 Hz 63 A** **Socket for external power supply (shore supply)** standard socket, with pilot contacts interfacing with TCMS, (see *UIC Leaflets 554-1 & 554-2*, and *prEN* on shore supply) [ ]			When closed, the connector shall be jet-water proof; when plugged, it shall be splash-proof. Both plugs shall be insulated electrically from each other. The indicator lights, which shall be equipped with white illuminated LEDs and the switches, shall be jet-water proof.	CEE connector plugs 3AC 400 V 50 Hz 63 A with cover shall be provided on both sides of the vehicle for charging the vehicle battery from the stationary power supply network. Traction interlock applies as long as a supply cable is connected with the socket.	Available in all configurations.				

Operating element Position and [form]		Symbol/indication	Position and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
413	**Rotary switch for external power supply (shore supply)**		**0°**	External power supply connection switched off.	A revolving switch for switching the external power supply on and off via plug shall be fitted for each plug in order to establish and disconnect the electrical connection without power.	Available in all configurations.				
			**S + 90°**	External power supply connection switched on.						
420	**Emergency release equipment of the parking brake**	Parking brake operational (pin in)		Parking brake operational	For an emergency release of an applied parking brake, a local device at each parking brake unit shall be provided, which may be operated by a UIC standard square socket (Berne key) and shall be accessible whatever the platform height is. See note at pos. 403.	Available in all configurations.				
		Emergency release of the parking brake (pin out)		Parking brake released in emergency: The emergency release status shall be indicated at the parking brake unit concerned, e.g. by a pin indicator (see Appendix H - page 82). After a complete sequence of applying and releasing the air brakes, the parking brake is fully operational again. Pin out: The pin is sticking out at least by 1 cm.					
430	**External doors of the Driver cabs**				A mechanical locking system is requested. The manufacturer's demonstration of long-term security (> 15 years) is a mandatory precondition for a future electronic system. No unauthorised person shall be able to override the system.	Available in all configurations.				

Appendix H - Markings

H.1 - Inside the vehicle

In the driver cab:

- Max. speed (V_{max}),
- Rolling stock identification number (traction vehicle number),
- Identification of cab (1 or 2),
- Vehicle mass and braked weights,
- Instructions for passengers and untrained staff for the activation of emergency brake,
- Maximum number of people in the cab,
- Location of portable equipment (e.g. first-aid kit, self-rescue equipment, signals),
- Emergency exit.

H.2 - On the vehicle (outside)

On the underframe or longitudinal girder of the vehicle the following symbols and markings defined by UIC and, if applicable, warning notices in accordance with health and safety regulations, as well as railway group standards shall be provided:

- Length over buffers, bogie pivot pitch,
- Type of brake,
- Mass of EMU/DMU, locomotive or driving coach, braked weights,
- UIC frame,
- Revision frame,
- Minimum radius value for "valleys" and humps,
- Minimum passable radius of curvature,
- Position and use of lifting points, sand fillers, shut-off cocks and drain cocks, drop signals, connections, ground points and other operating elements,
- Warning notices, e.g. lightning arrow,
- Instructions on parking and hauling the vehicle (see Fig. 24 - page 83).

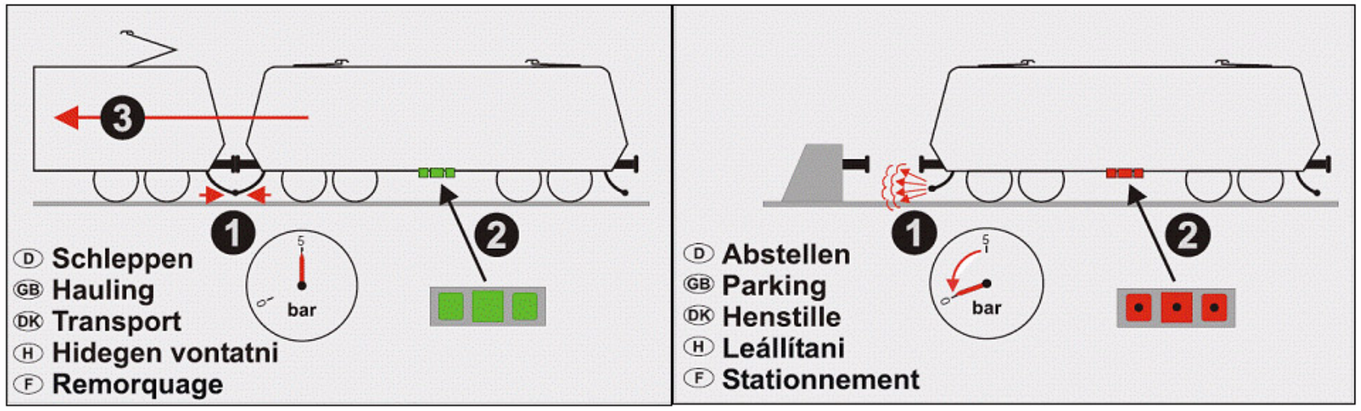


Fig. 24 - Instructions on parking & hauling

Pictograms shall be realised as follows:

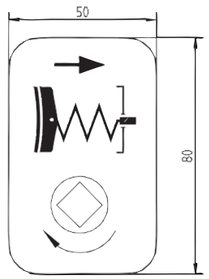


Fig. 25 - Instruction for emergency release of parking brake

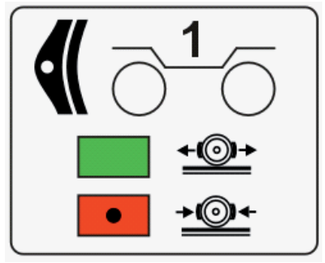


Fig. 26 - Pictogram for the flag indicator of the brake (example for bogie 1)

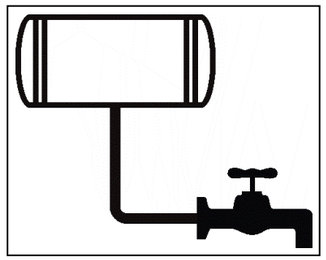
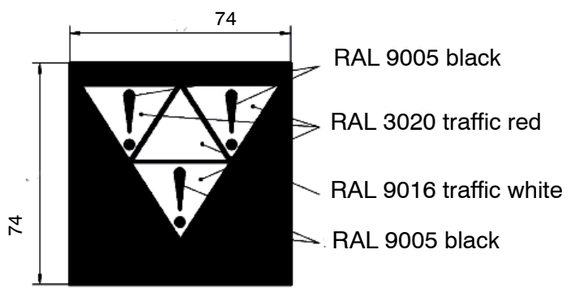


Fig. 27 - Position of the drain cock for the condensate tank

Pushing off and bumping not allowed



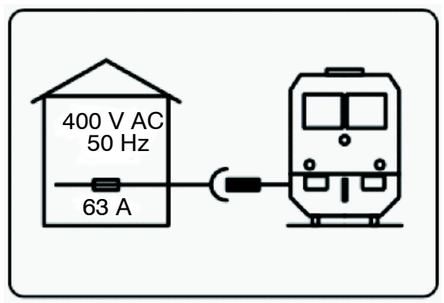


Fig. 28 - Position of external power supply → applies only for 3AC 400 V
(Including information on the required local mains supply)

Appendix I - Radio remote control desk (operator option, only EMU/DMU or locomotives)

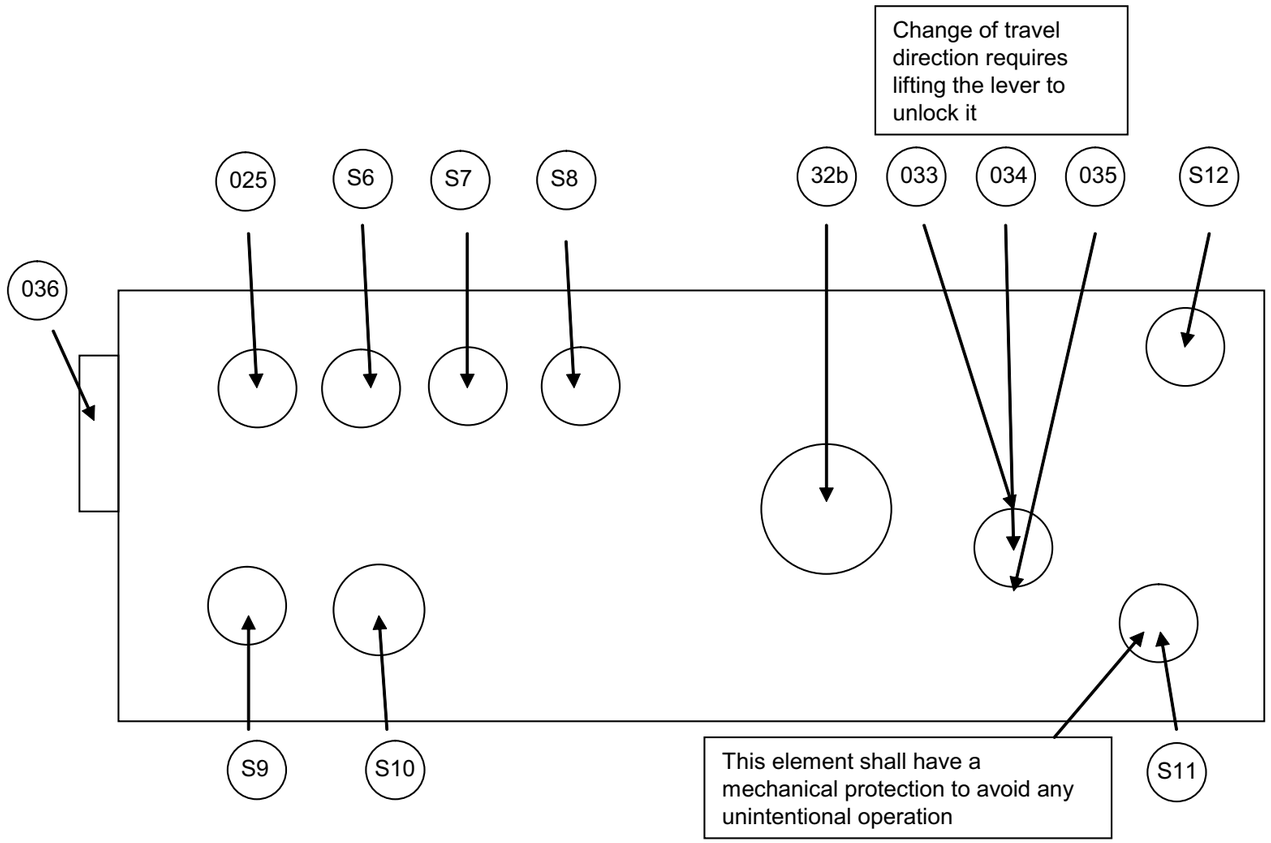


Fig. 29 - Radio remote-control desk

The maximum speed for driving the EMU/DMU or locomotive with the radio remote-control desk shall be selected in the TDD (see UIC Leaflet 612-03). The maximum speed limit authorised for driving the EMU/DMU or locomotive with the radio remote-control system is 40 km/h.

Fault indication:

- In case of malfunction, the indication shall be given in accordance with the principles of fault management and diagnosis in the EMU/DMU or locomotive and there shall be no impact on the use of the main desk.

The following functions of the main desk are deactivated if the radio remote-control desk is active:

- Traction.
- The position of operating elements whose functions depend on the selection of a travel direction shall be ignored in all (non-activated) cabs.
- Release of the automatic brake (pos. 032) if the automatic brake has been applied on the radio remote-control desk (by pos. 032b).

Precondition for the activation of the previously deactivated functions of the main desk:

- Traction/dynamic brake controller (pos. 020) in "OFF" position.

The following actions deactivate the radio remote control desk and re-activate the cab (main desk or auxiliary desk):

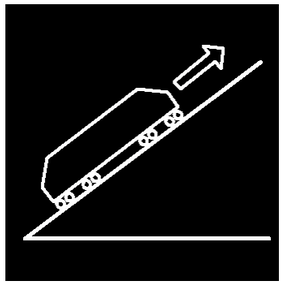
- V = 0,
- Deactivation of the radio remote control desk by removing the driver's ID card (pos. 036),
- TCMS automatically applies the holding brake,
- Activation of a cab (entering "In service configuration", by inserting the driver's ID card (pos. 036) or overriding the ID card reader (pos. 301),
- In the cab the driver shall reset the traction/brake controller (pos. 020 or pos. 032b) to "0" position,
- TCMS will release the holding brake automatically after application of a traction command by the traction/brake controller (pos. 020 or pos. 032b in the cab).

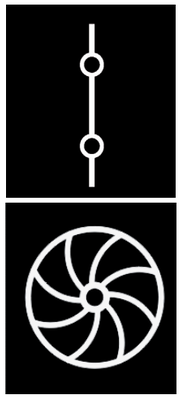
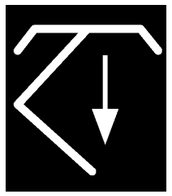
EMU/DMU or locomotive switched off:

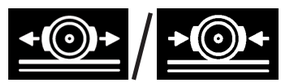
- If the EMU/DMU or locomotive is switched off the radio remote control desk shall be deactivated automatically.

Gradients for traction and dynamic brake force (see pos. 020, 032a):

- The same gradients apply to all EMU/DMUs and locomotives in the train.

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
S 6	**SLOPE STARTING** Flip switch [?]		**U**	The brake-cylinder and BP pressure available is retained, irrespective of the application of the traction/brake controller (pos 032b). This allows application of traction power while brakes are applied.	This function can prevent unintentional back movement on slopes due to a traction command of pos. 032b is releasing the brakes.	F/R		X		X
			S	Pos. 032b becomes fully effective immediately; i. e. the brake follows the command by pos. 032b.						
S 7	**MAINTAIN THE CURRENT SPEED** Flip switch [?]		**U**	- V < 15 km/h **and** - pos. 032b is in "0" position → The speed effective at the moment of application will be retained automatically..		F/R	X		X	
			S	Pos. 032b becomes effective as specified, as soon as pos. 032b is operated in a position other than "0".						

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
S 8	**Electric traction:** **Pantograph and Main circuit breaker** **Diesel-units:** **Diesel motor and power-transmission** Brake flip switch [B]		**U** **+ 35°**	On. The pantograph is raised / diesel engine is started automatically and the main circuit breaker / power transmission switches on if all switch-on conditions have been met.		F/N/R		X		
			0	Normal position.						
			**U** **- 35°**	- Pantograph down / idle (stop diesel engine) **and** - MCB has previously been switched off automatically / power off.	In case of previous dynamic braking: automatic compensation of braking effort necessary (e. g. by automatic air brake or rheostat), if loss of brake effort is caused by lowering the pantograph / diesel engine idling (see pos. 020 and pos. 032a interaction of brakes).	F/N/R				
S 9	**BRAKE PIPE PRESSURE ADJUSTER (overcharge)** **and** **External warning horn** Brake flip switch [?]		**U** **+ 35°**	See pos. 009.		F/N/R				
			0	Neutral position.						
			**U** **- 35°**	See pos. 113. All horns are applied at the same time.						

Operating element Position and [Form]		Symbol/indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v = 0	v > 0	MT
S 10	**Type of air brake selection:** **Automatic brake/direct brake** Brake flip switch [?]		**U** **+ 35°**	Pos. 032b is only effective for the automatic brake.		F/N/R		X		X
			**U** **- 35°**	Pos. 032b is only effective for the direct air brake.						
S 11	**TRACTION INTERLOCK** Flip switch [?]		**U** **+ 35°**	- Traction interlock activated **and** - In V = 0: "Emergency brake due to skipping of the radio remote control desk" is suppressed for 13 seconds.	This function protects the driver from unintentional application of traction, e. g. by operating pos. 032b during coupling. Furthermore it shall provide protection in case of unintentional activation.	F/N/R	X			X
			S	The previously selected position of pos. S 11 remains valid.						
			U **- 35°**	Traction interlock is cancelled.						
S 12	**DECOUPLING** Push button [ ]		**U**	Decoupling the automatic coupler in travel direction.	TFR of the safety-relevant event "unintentional decoupling caused by this function" shall be less than 1E-7.	F/N/R		X		

Appendix J - Operating elements (overview of Driver-Machine Interfaces)

Position of operating and display elements on and below the main driver's desk (mandatory elements are marked in green, optional elements in blue):

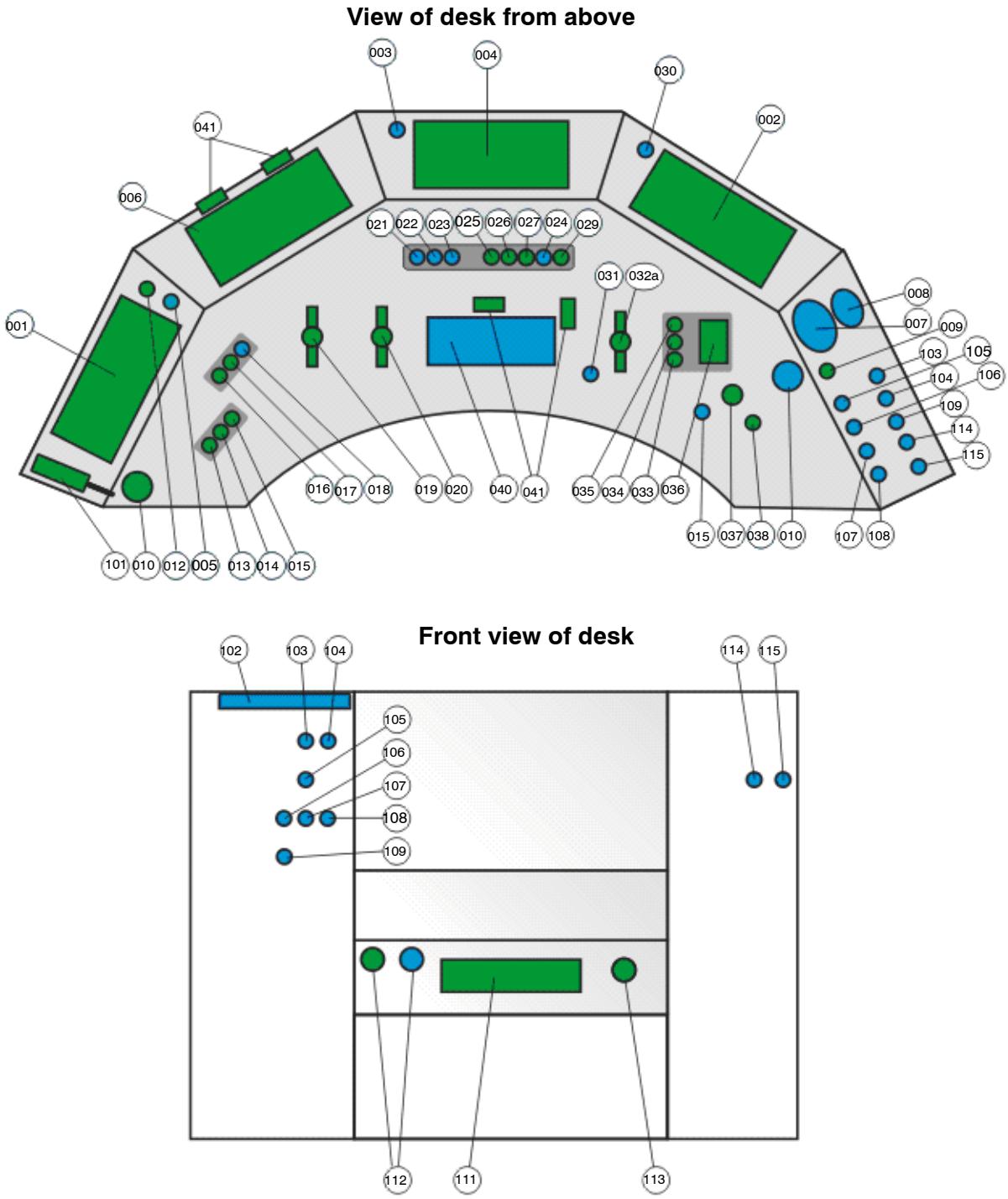


Fig. 30 - Position of operating elements

Position of operating and display elements on and below the main driver's desk for EMU/DMU as an alternative option compared with the solution from Fig. 30 - page 88 (mandatory elements are marked in green, optional elements in blue):

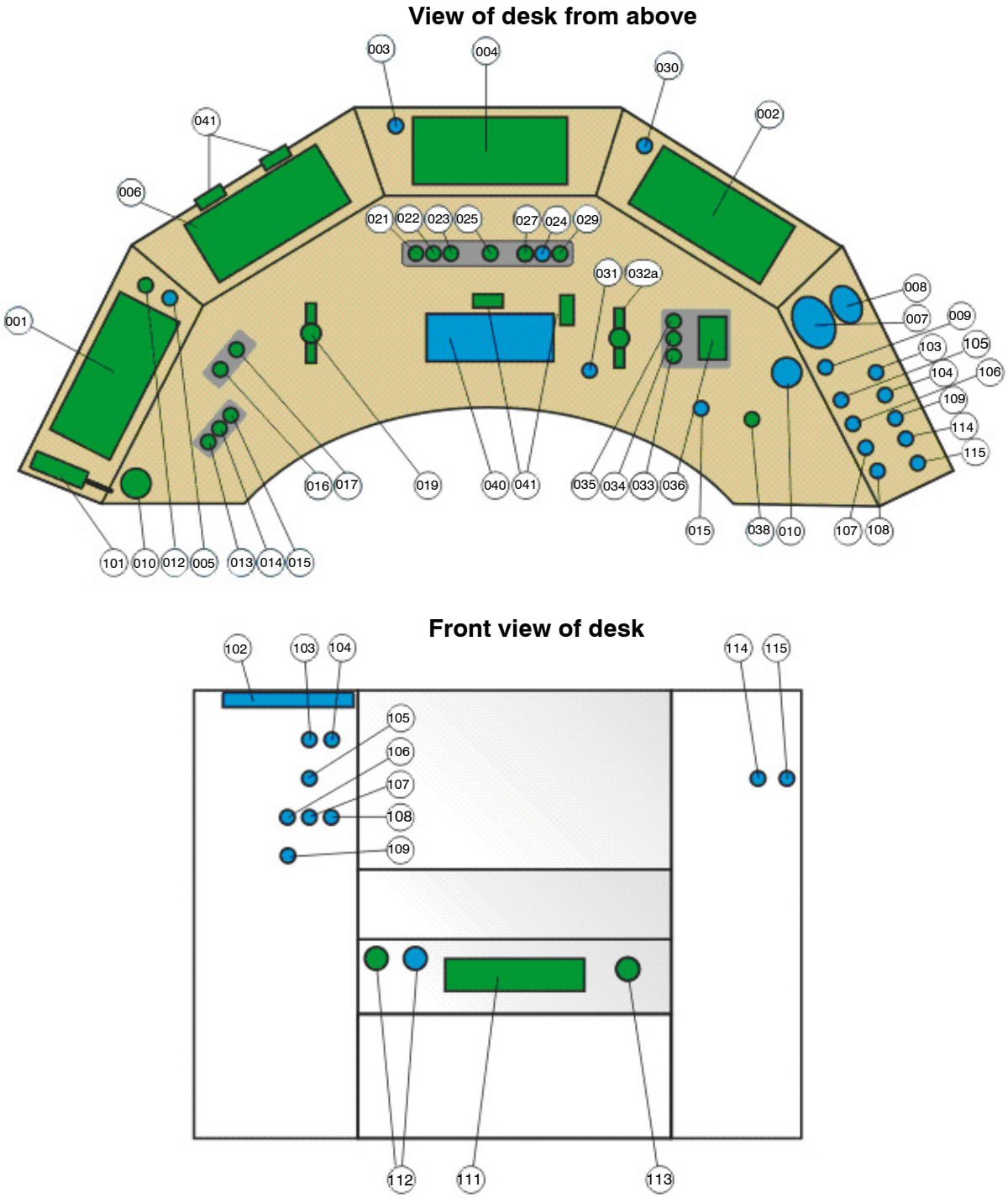


Fig. 31 - Position of operating and display elements for EMU/DMU

The drawing shows the position of MMI of the standardised main desk. However, the shape and dimension of the main shell is dependent on the location (left, centre, right) and integration of the standard desk into a specific cab.

The positions in italics are dependent on optional choices indicated in point 3 - page 8.

001	Train radio display (TRD)
002	Technical & diagnostic display (TDD)
003	*Only for STM "LZB"*
004	Control command display (CCD)
005	*Emergency signals (warning lights)*
006	*Electronic timetable display (ETD)*
007	*BP & MP pressure gauge (option: to be integrated in the TDD)*
008	*Brake-cylinder pressure gauges (option: to be integrated in the TDD)*
009	*Brake pipe pressure adjuster (overcharge)*
010	Emergency stop valve with "Emergency stop" function
012	Train radio emergency calls
013	ETCS: override (e. g. EOA)
014	ETCS: release intervention
015	ETCS: acknowledgement
016	Pantograph/ diesel engine
017	Main Circuit Breaker/ power transmission
018	*Train Power Supply*
019	*Automatic Speed Control (ASC)*
020	*Combined traction/brake controller with integrated DAC push button*
021	*Door control: left side doors - release & cancel release*
022	*Door control: Forced closing*
023	*Door control: right side doors - release & cancel release*
024	*Train lighting (can be replaced in the target system due to TCMS providing the trigger information - see Appendix A - page 28.*
025	Sanding
026	*Release brake*
027	External front light
028	*Instrument lighting (can be replaced in the target system due to TCMS providing the trigger information - see Appendix A.*
029	Workspace (on desk) & driver cab lighting
030	*"Swan neck" microphone*
031	*DAC button*
032a	*Driver's automatic brake controller (automatic brake)*
032b	*Combined traction brake controller*
033	Travel Direction "Forwards"
034	No Travel Direction "Neutral"
035	Travel Direction "Reserve"

036	Driver's Identity card reader
037	*Direct brake controller*
038	External warning horn
039	*Desk light (not indicated in the drawings Fig. 30 - page 88 and 31 - page 89. Position according to Appendix A, depending on the location of the lamp)*
040	*Key board*
041	Document holders
101	Train radio handset
102	*Lamp center console*
103	*Centre console lighting*
104	*Intensity of instrument lighting*
105	Floor & knee space (knee hole) heating
106	Cab ventilation fan (speed regulator)
107	Cab temperature setting
108	Cab air-conditioning/ventilation
109	Cab pressure protection
111	Driver activity control (DAC) foot switch
112	Foot switch for automatic adjustment of the height of the footrest
113	Foot switch for external warning horn
114	Windscreen wiper & windscreen washer
115	Windscreen heating

On the second man's side or next to the desk (not indicated in the drawing):

121	Diagnosis socket
122	Electric shock-proof socket 230 V/ 50 Hz

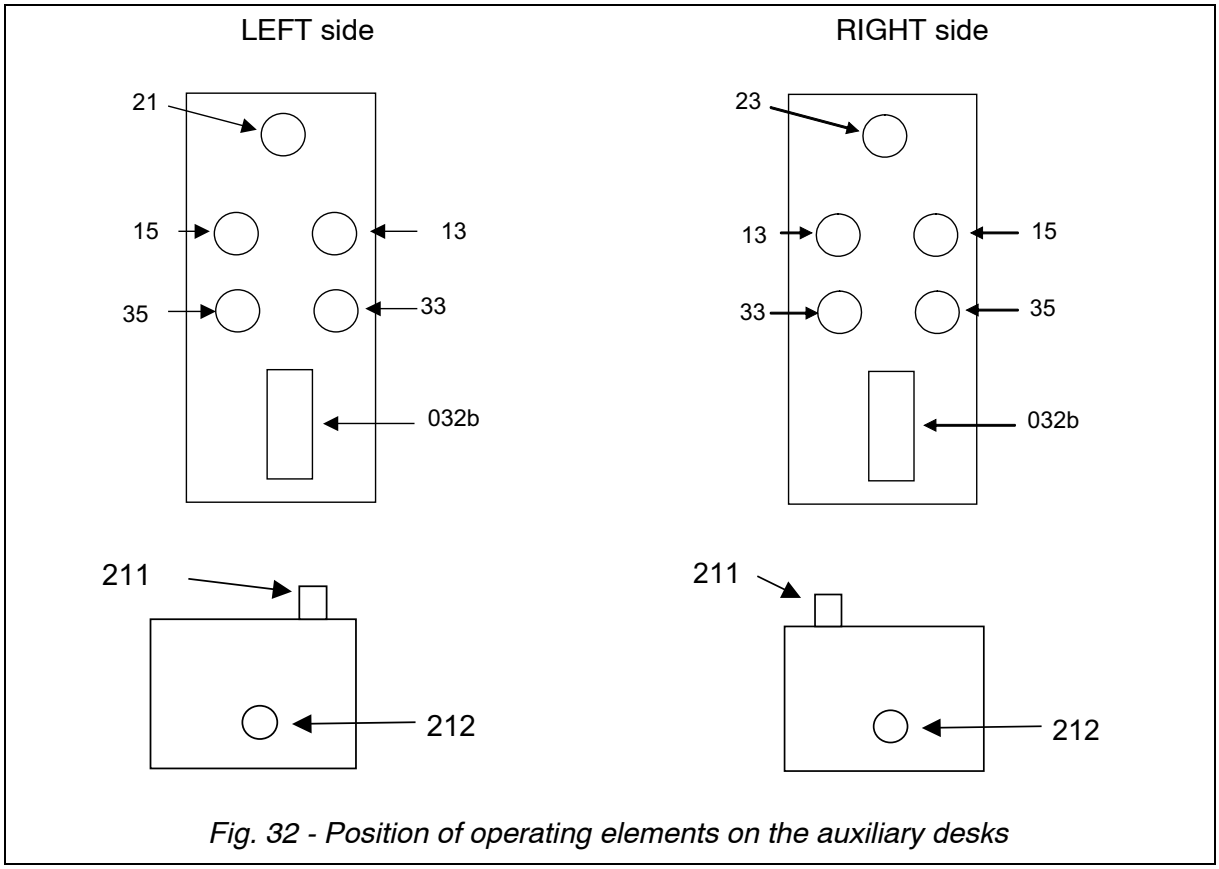


Fig. 32 - Position of operating elements on the auxiliary desks

Operating elements close to the auxiliary desk at the second-man side or, if these main-desk elements are not within reach of the driving position, at the side window (pos. 037, 038, see Appendix A - page 28

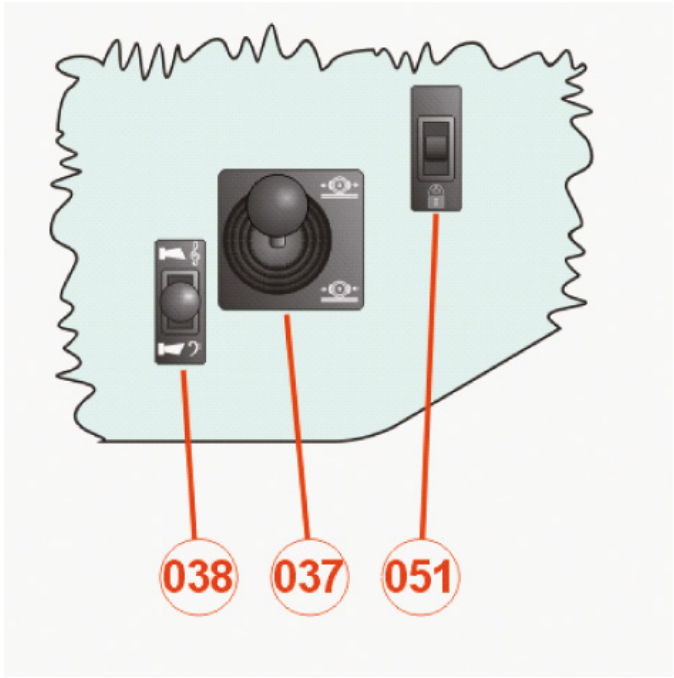
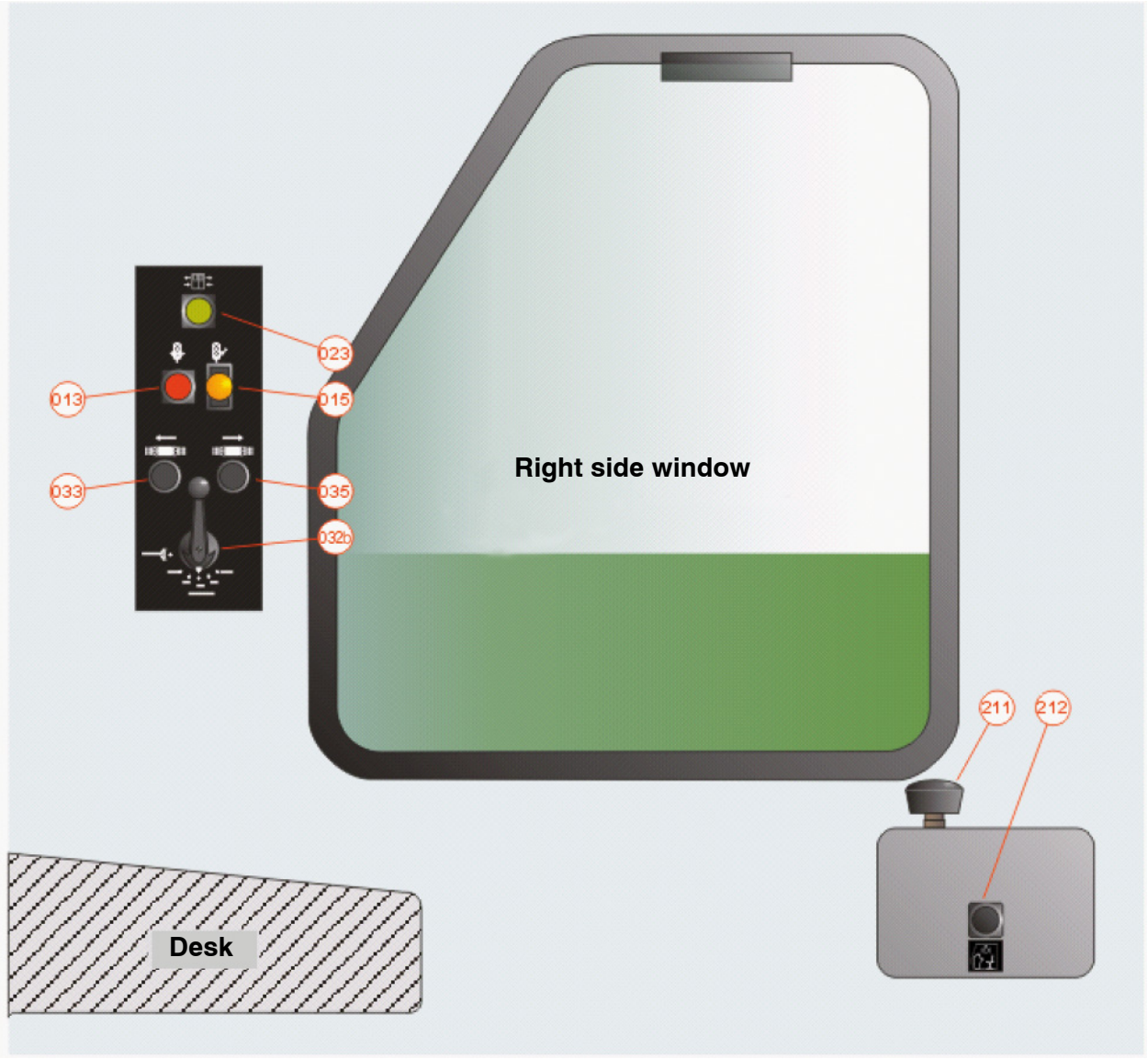


Fig. 33 - Auxiliary desk (example for right side position)



013	ETCS: Override (e. g. EOA)
015	ETCS: Acknowledgement
021	*Doors on the left side (left auxiliary desk) - release and cancel release*
023	*Doors on the right side (right auxiliary desk) - release and cancel release*
033	Travel direction "forwards"
035	Travel direction "reverse"
037	*Direct brake*
038	External warning horn
051	Second man task light (on desk)
032b	Combined traction/brake controller
211	Driver activity control button (DAC)
212	Cab lighting

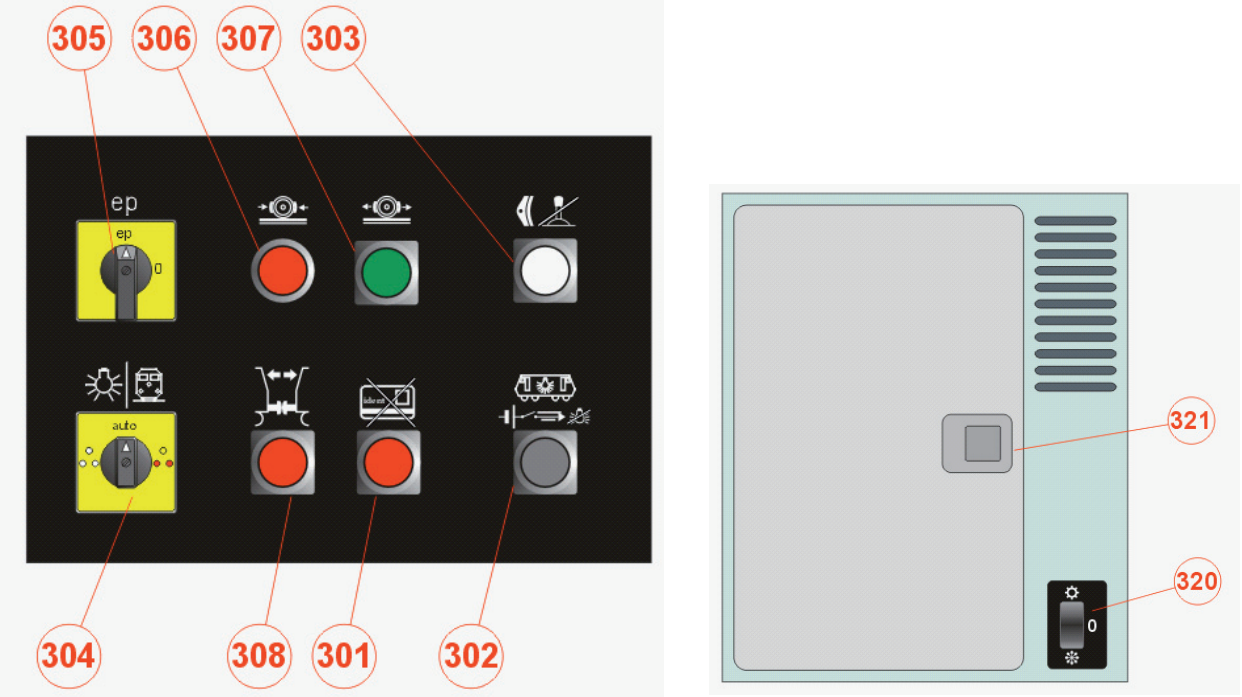


Fig. 34 - Position of operating elements on the driver cab's rear wall operation board

301	Override the driver's ID card reader
302	Engine room lighting
303	Isolation of driver's automatic brake controller & ETCS/STM
304	External light
305	ep-brake
306	*Manual application of the holding brake*
307	*Manual release of the holding brake*
308	*Forced decoupling*
320	Thermo compartment (fridge/stove)
321	Locker for thermo compartment

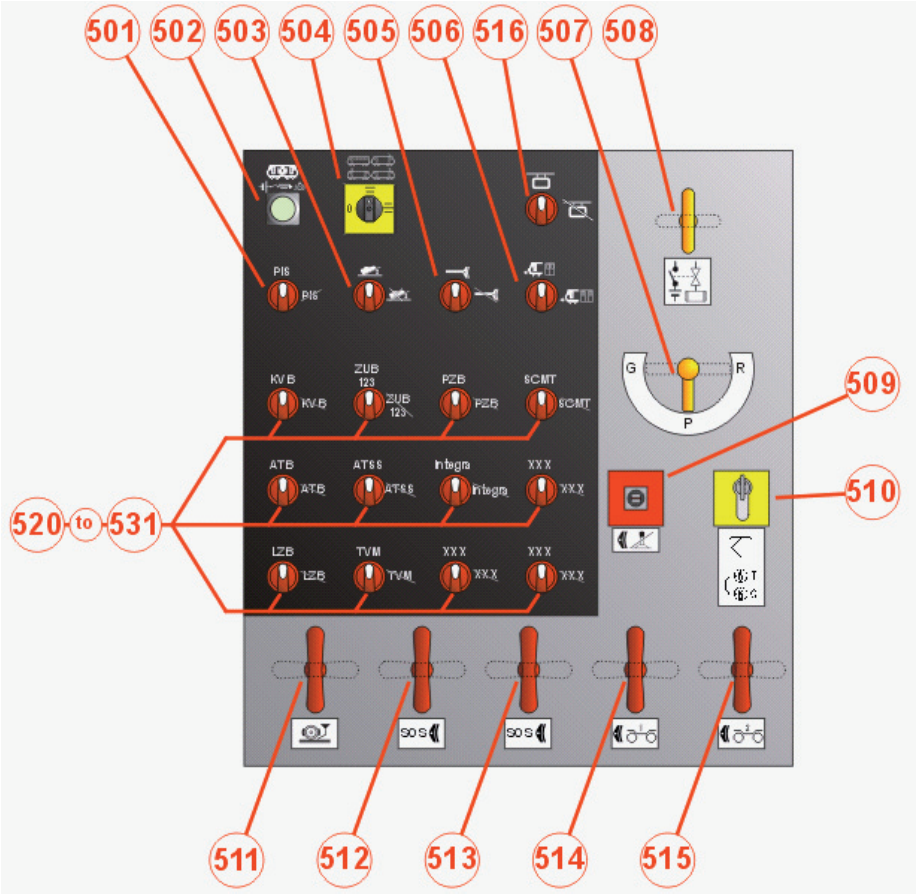


Fig. 35 - Position of operating elements in the engine room

501	Passenger Information System (PIS)
502	Engine room lighting
503	Fault switch for driver activity control (DAC)
504	*Mode selection: Train bus selection*
505	Fault switch "TCMS emergency brake"
506	*Override of traction cut-off in case of an open door*
507	Impulse valve "Brake Mode"
508	Preparation/shut-down
509	Impulse valve "Isolate driver's automatic brake controller"
510	Key lock "Pantograph"
511	Isolation cock "sanding"
512	Isolation cock "automatic emergency brake valve 1"
513	Isolation cock "automatic emergency brake valve 2"
514	Isolation cock "Air Brake bogie 1"
515	Isolation cock "Air Brake bogie2"
516	Isolation switch for passenger alarm function
52x	Isolation of ETCS-STM

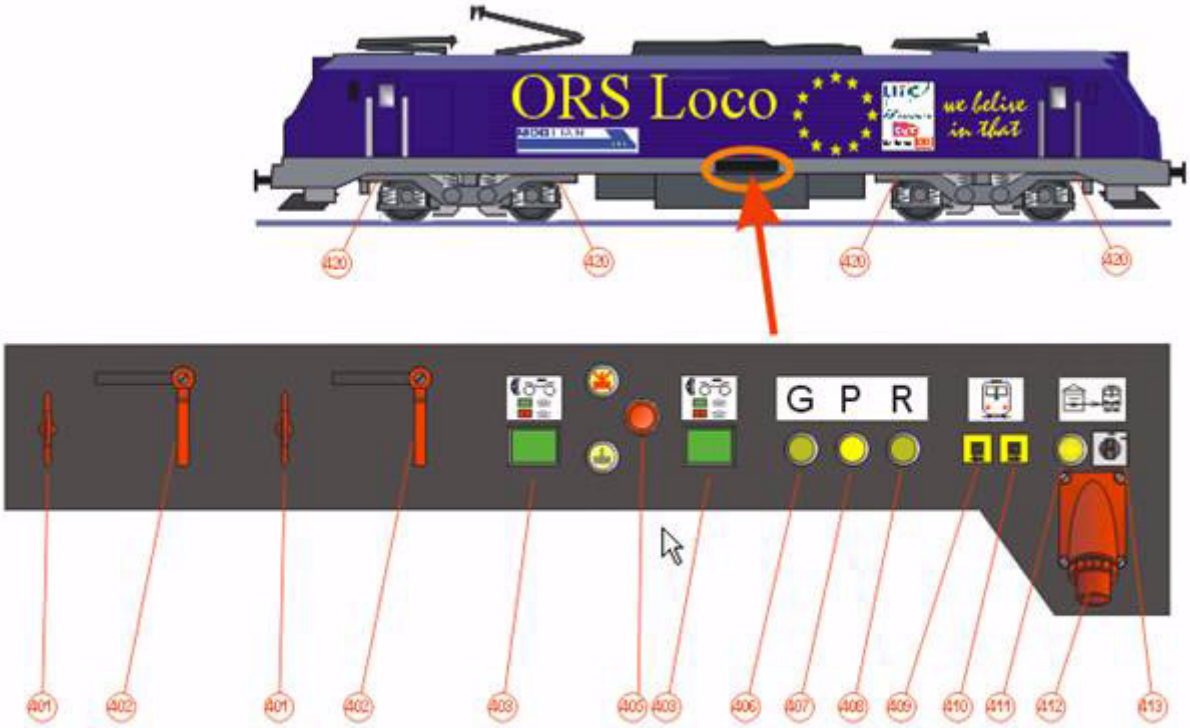


Fig. 36 - External position of operating elements which are also accessible for operating staff not authorised to enter EMU/DMUs, locomotives and driving coaches

401	Brake release device
402	Brake isolating cock
403	Brake indicator
405	*Mg-brake or eddy current brake*
406	Brake changeover for brake mode "G"
407	Brake changeover for brake mode "P"
408	Brake changeover for brake mode "R"
409	*Tail signal at the left end of the EMU/DMU or locomotive*
410	*Tail signal at the right end of the EMU/DMU or locomotive*
411	*Indicator for external power supply*
412	*Socket for external power supply (shore supply)*
413	*Rotary switch for external power supply*

Appendix K - Geometric forms, operation angles, forces, specific and system wide characteristics (RAMS and LCC) of the operating elements and the driver's seat

Specification of the operating elements

The angles, forces, height of levers, function and associated positions (with permitted tolerances) shall be obligatory. Other parameters may be recommended.

K.1 - Automatic brake controller (Pos. 032a)

Working angles (0° = vertical in relation to the desk surface / + stands for forward direction starting from the vertical position) and forces (see also Fig. 37 - page 98):

Position	Angle	Force (Newton)
Filling stroke	+ 40°	↑40
Running position	+ 32°	
		↓↑20
1A (last release notch)	+ 20° [a]	
		↓↑10
1B (first application notch)	+ 15° [a]	
		↓↑10
2	+ 10°	
		↓↑10
3	+ 5°	
		↓↑10
4	0°	
		↓↑10
5	- 5°	
		↓↑10
6	- 10°	
		↓↑10
7	- 15°	
		↓↑10
Full service brake, 3,5 bar	- 22°	
		↓↑60
Emergency brake, 0 bar	- 40°	

a. The positions 1A and 1B are at the opposite ends of a large slot (not separated by a notch). The lever is held in the position 1A or 1B by friction.

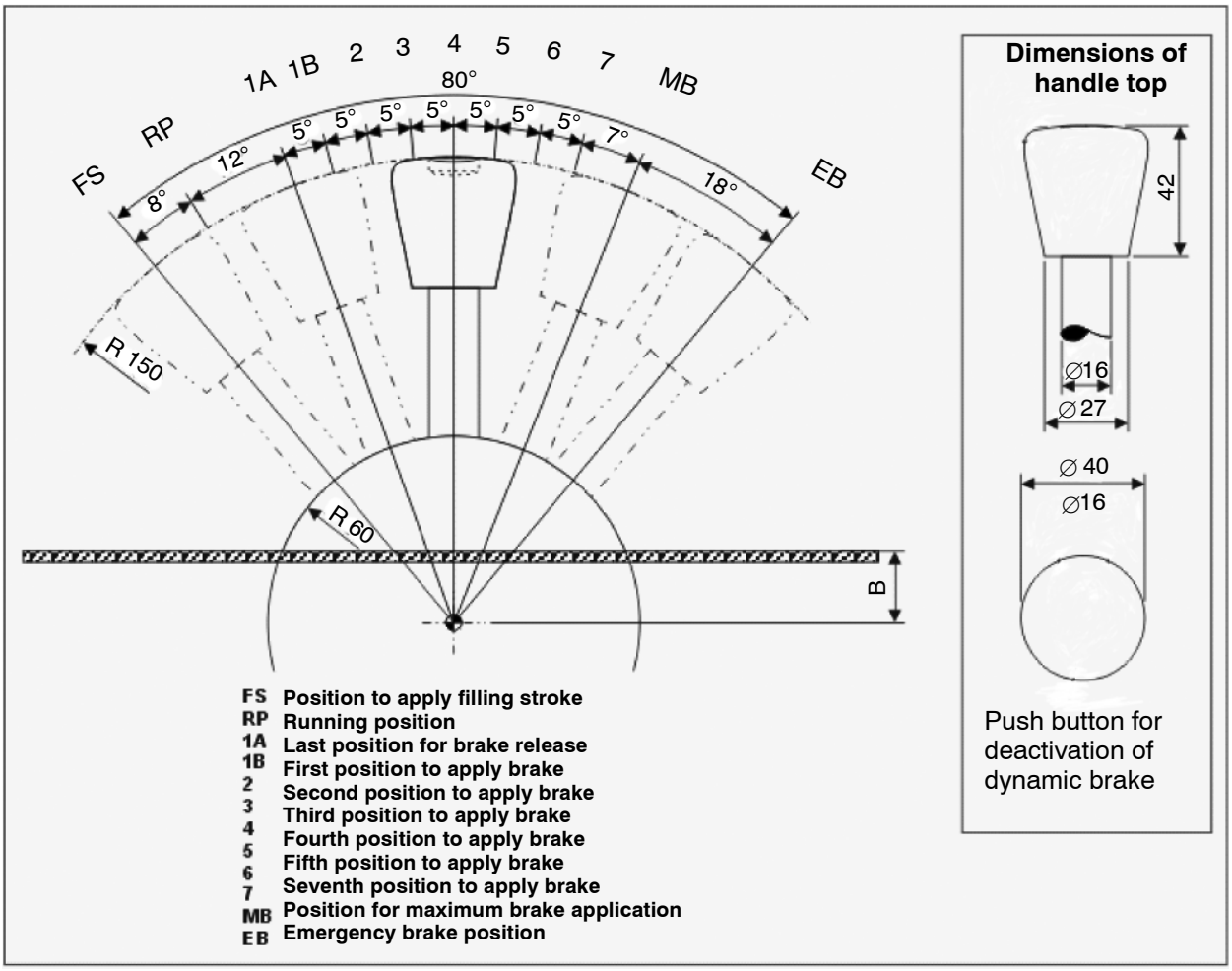


Fig. 37 -

Distance B requires a minimum distance of 20 mm and a maximum of 24 mm to allow proper desk integration.

Permitted deviations for distances:	5 %
Permitted deviations for angles:	1°
Permitted deviations for forces:	± 1N

Reliability

The FIT rate of the mechanical parts (levers, handles, bushings, gears - if any) shall not be greater than 100.

Avaibility

It shall be possible to replace a defective Automatic Brake Controller Module on the EMU/DMU, locomotive and driving coach by using standard tools. The time for replacing a module shall not exceed:

- 30 minutes for the advanced version (signal generation purely electric),
- 120 minutes for the conventional version (pneumatic interface and electric interface).

The number of fixing elements shall be minimized. The connectors shall be coded to avoid mistakes. If hoses are used for the pneumatic connections they shall be marked.

Maintainability

Conventional version (pneumatic and electric interface): the time intervals between overhauls shall be at least 7 years.

Advanced version (interface entirely electric): the device shall be virtually maintenance- free over the lifetime of the vehicle (30 years). Therefore, self-lubricating bushings and gear wheels - if any - shall be used. Parts which are subject to wearing shall remain in tact over the lifetime of the vehicle.

Safety

The coupling between the lever and the signal generators shall be of the form locking type, so that the signal generators will follow the movement of the lever without fail.

Under no circumstances shall a movement from the F position back into a position demanding a braking force be hindered.

The functional safety of signal generation, signal transfer and signal processing is the responsibility of the BCU manufacturer.

Scalability

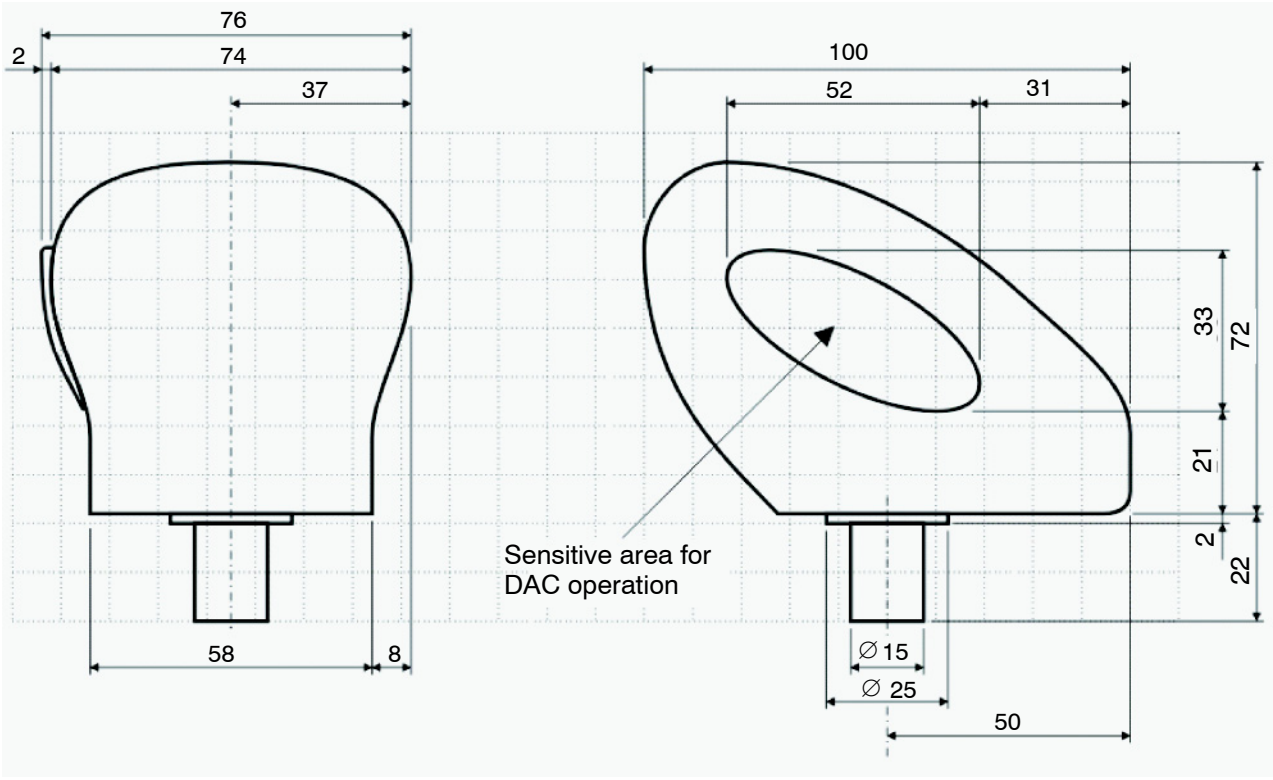
Not applicable.

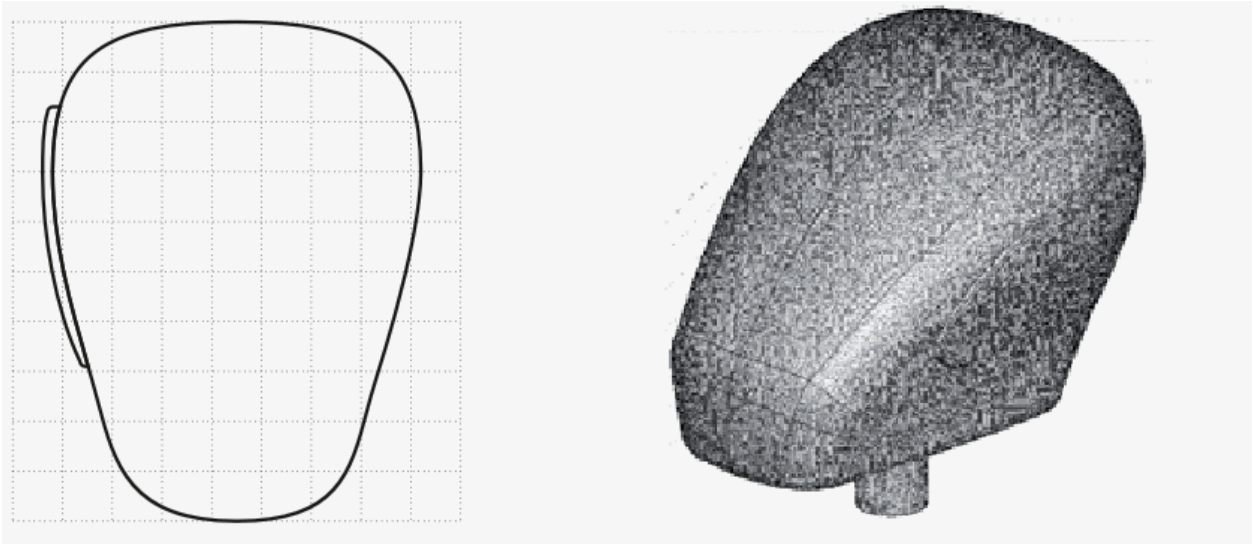
LCC

LCC shall be minimized by:

- reducing the number of moving and wearing parts to a minimum,
- using self-lubricating bushings and gears,
- using materials which are recyclable or biologically degradable,
- using materials which do not need to be painted,
- lengthening the time intervals for indispensable maintenance and repair work.

K.2 - Hand rest with DAC operational element (Pos. 031)





Permitted deviations for distances: 5 %

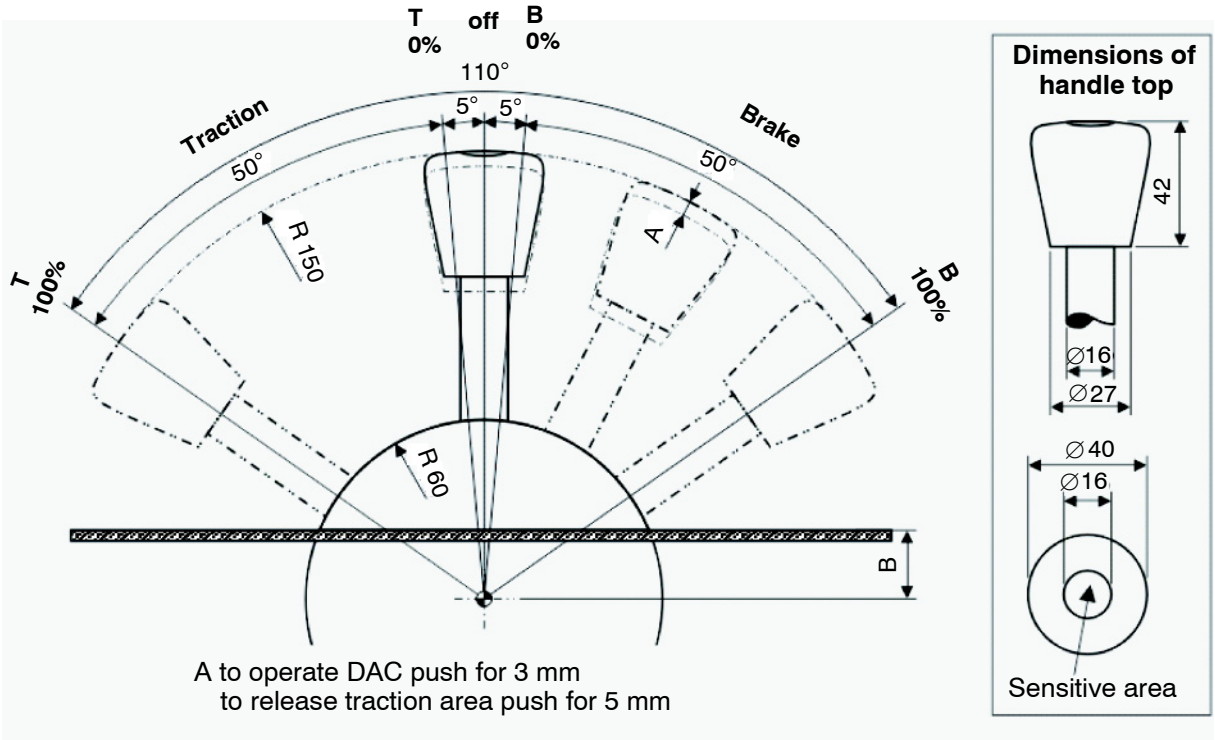
Possible rotation of hand rest:
A rotation of at least 60° shall be possible (30° right / 30° left)

K.3 - Traction / dynamic brake controller (Pos. 020)

Working angles (0° = vertical in relation to the desk surface / + stands for forward direction starting from the vertical position) and forces:

(See also drawing below)

Position	Angle	Force (Newton)	Description
T 100 %	+ 55°		
		↓↑10	
T 0 %	+ 5°		
		↓↑22	↑ traction can only be applied by pressing the knob on top of handle
Off	0°		
		↓↑22	
B 0 %	- 5°		
		↓↑10	
B 100 %	- 55°		



For DAC operation it is not necessary to push the handle top. Touching the top is sufficient for DAC.

Force to press the knob on top of handle (for distance A): 12 N

For distance B a minimum of 20 mm and a maximum of 24 mm is required to allow correct desk integration.

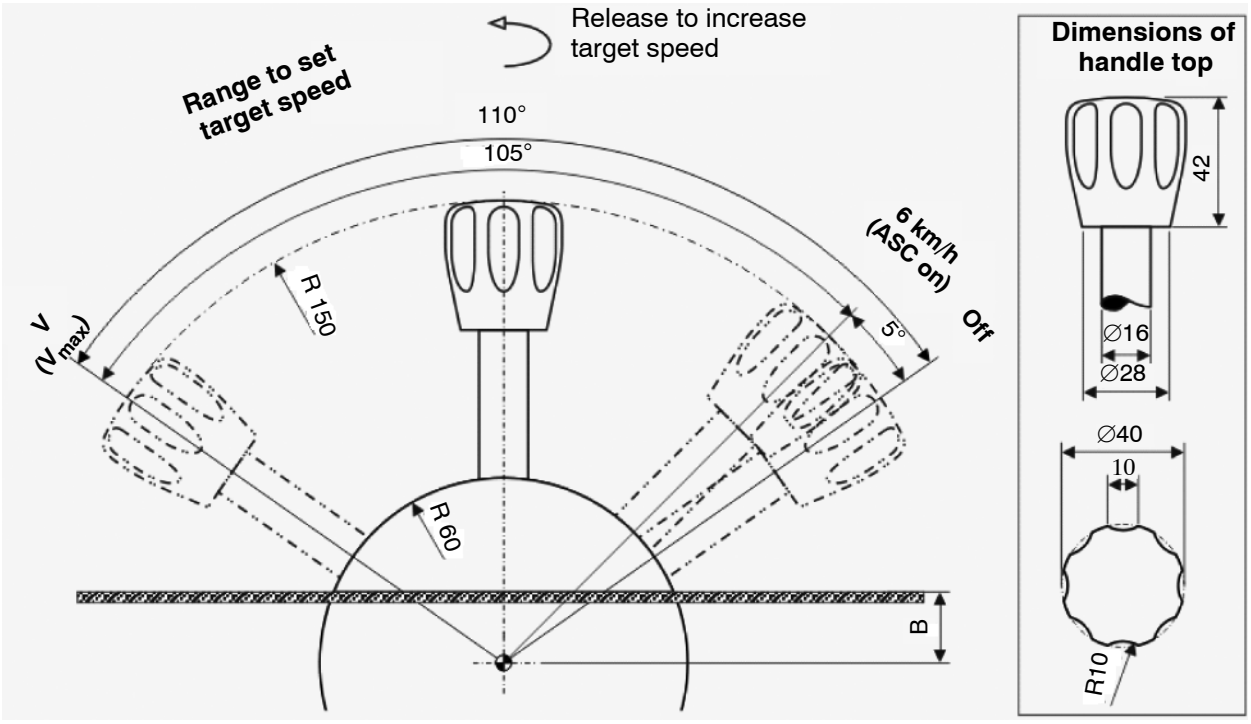
Permitted deviations for distances:	5 %
Permitted deviations for angles:	1°
Permitted deviations for forces:	± 1N

K.4 - V-target actuator for ASC (Pos. 019)

Working angles (0° = vertical in relation to the desk surface/+ stands for forward direction starting from the vertical position) and forces:

(see also drawing below)

Position	Angle	Force (Newton)
V (V_{max})	+ 55°	
See Appendix A - page 28, position 019		↓↑10 Setting a target speed -> Handle is self locked ↑ to increase target speed, rotate handle top for 45° with force 4 N in an anti-clockwise direction ↓ To decrease target speed, no specific action required
0 km/h (ASC on)	- 50°	
		↓↑20
Off	- 55°	



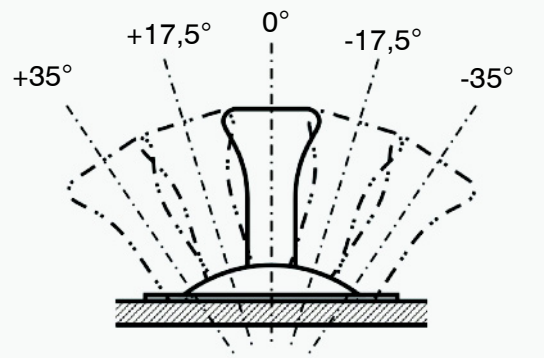
For distance B a minimum of 20 mm and a maximum of 24 mm is required to allow correct desk integration.

Permitted deviations for distances:	5 %
Permitted deviations for angles:	1°
Permitted deviations for forces:	± 1N

K.5 - Toggle switches

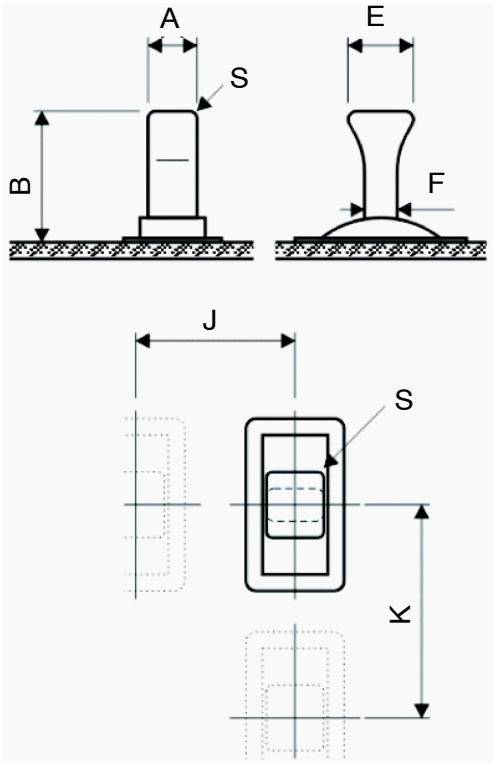
For all forms of toggle switches the following possible positions are valid:

Allowed deviations for angles: ± 1°



+: Operation away from the operator (forwards) or upwards
-: Operation towards the operator (backwards) or downwards

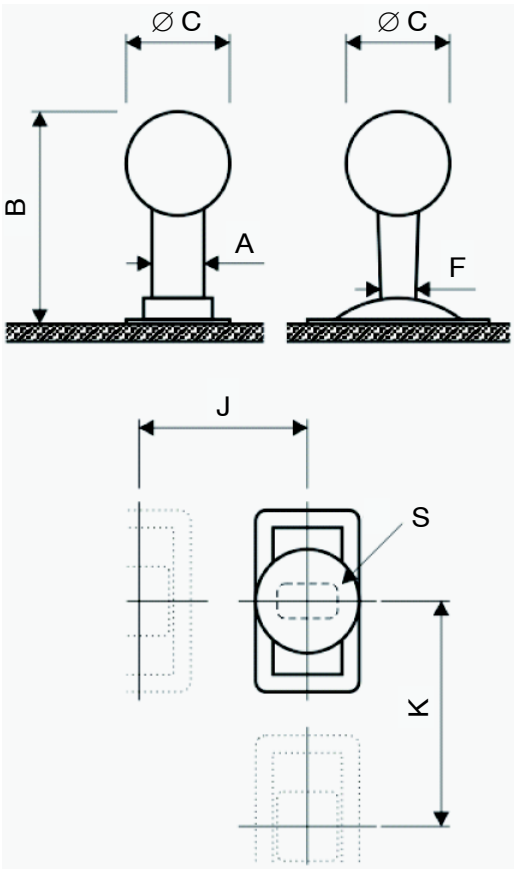
K.5.1 - Geometric form B



A	min. 10 mm/max. 15 mm
B	min. 32 mm/max. 40 mm
E [a]	min. 14 mm/max. 20 mm
F [a]	min. 8 mm/max. 12 mm
J [b]	min. 35 mm/max. 37 mm
K [b]	min. 65 mm/max. 75 mm
S	min. 1 mm/max. 2 mm

a. The proportion F to E should be 1:1,75
b. Min. corresponds to the minimum suitable distance between the switches.
Max. corresponds to the maximum distance permitted between the switches.

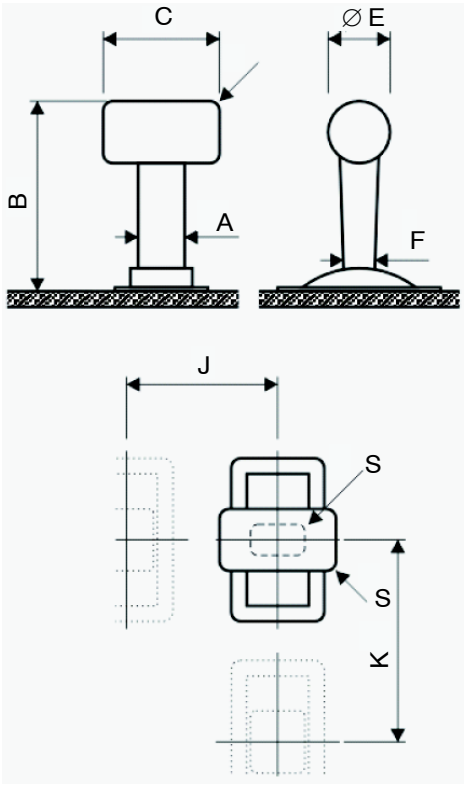
K.5.2 - Geometric form K



A	min. 10 mm/max. 15 mm
B	min. 42 mm/max. 55 mm
C	min. 20 mm/max. 30 mm
F [a]	min. 8 mm/max. 12 mm
J [b]	min. 35 mm/max. 37 mm
K [b]	min. 65 mm/max. 75 mm
S	min. 1 mm/max. 2 mm

a. The dimension should be the same as for form B.
b. Min. corresponds to the minimum suitable distance between the switches.
Max. corresponds to the maximum distance permitted between the switches.

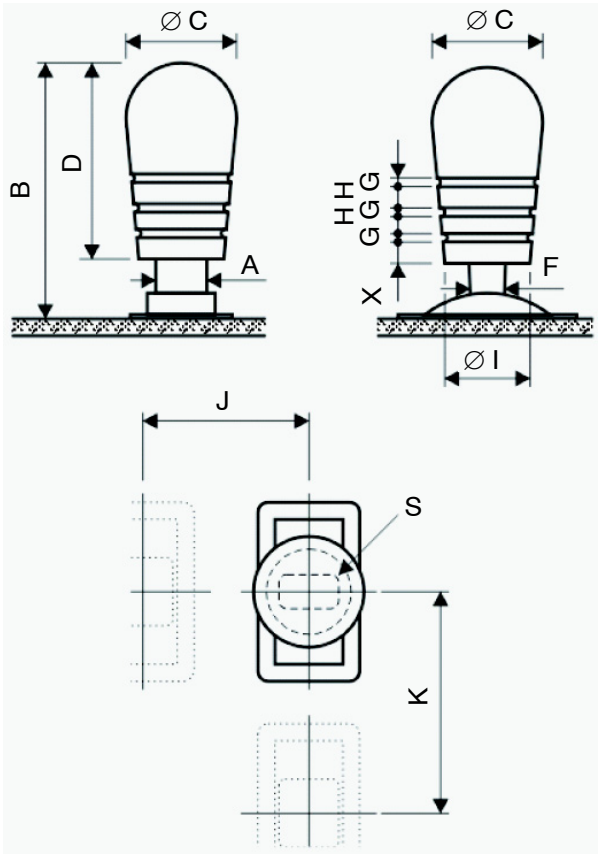
K.5.3 - Geometric form Z



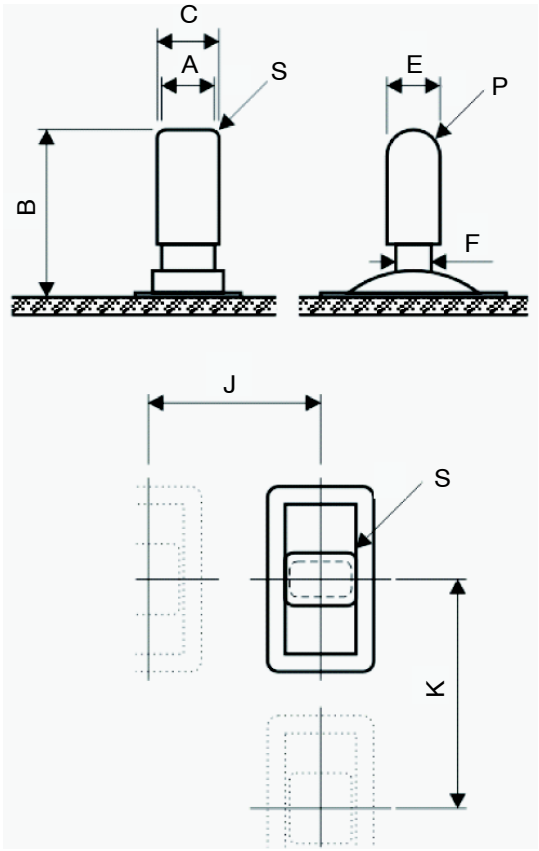
A [a]	min. 10 mm/max. 15 mm
B	min. 42 mm/max. 55 mm
C [a]	min. 25 mm/max. 30 mm
E [b]	min. 20 mm/max. 28 mm
F [b]	min. 8 mm/max. 12 mm
J [c]	min. 35 mm/max. 37 mm
K [c]	min. 65 mm/max. 75 mm
S	min. 1 mm/max. 2 mm

a. The proportion F to E should be from 1:2 to 1:2,3
b. Min. corresponds to the minimum suitable distance between the switches.
Max. corresponds to the maximum distance permitted between the switches.
c. The proportion A to C should be at least 1:2

K.5.4 - Geometric form M



K.5.5 - Geometric form W



A	min. 10 mm/max. 15 mm
B	min. 32 mm/max. 40 mm
C	min. 12 mm/max. 17 mm
E [a]	min. 12 mm/max. 17 mm
F [a]	min. 8 mm/max. 12 mm
J [b]	min. 35 mm/max. 37 mm
K [b]	min. 65 mm/max. 75 mm
P	0,5 x E
S	min. 1 mm/max. 2 mm

a. The proportion F to E should be 1:1,5
b. Min. corresponds to the minimum suitable distance between the switches.
Max. corresponds to the maximum distance permitted between the switches.

K.6 - Push buttons and indicators

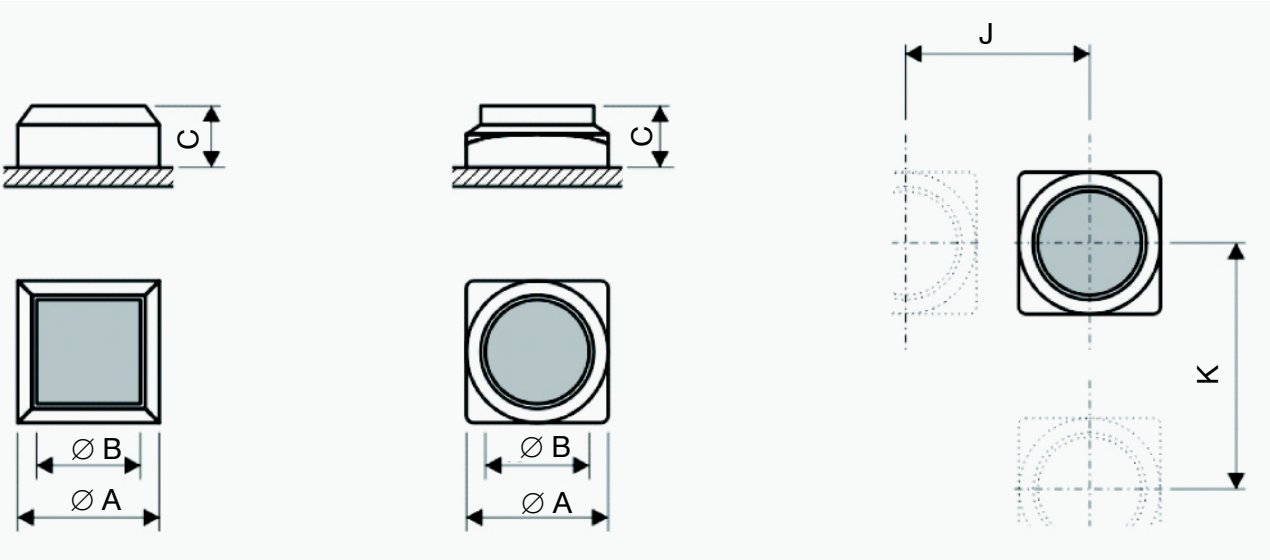
For ergonomic reasons all buttons shall have tactile feedback.

For safety reasons:

- The travel of the push button shall be at least 3 mm.
- The collar shall protect the button from inadvertent operation

K.6.1 - Geometric form Q (square form)

The shape of the push buttons or indicators shall be square. For the area B, which is to be pushed or/and illuminated, both round and square forms are permitted.

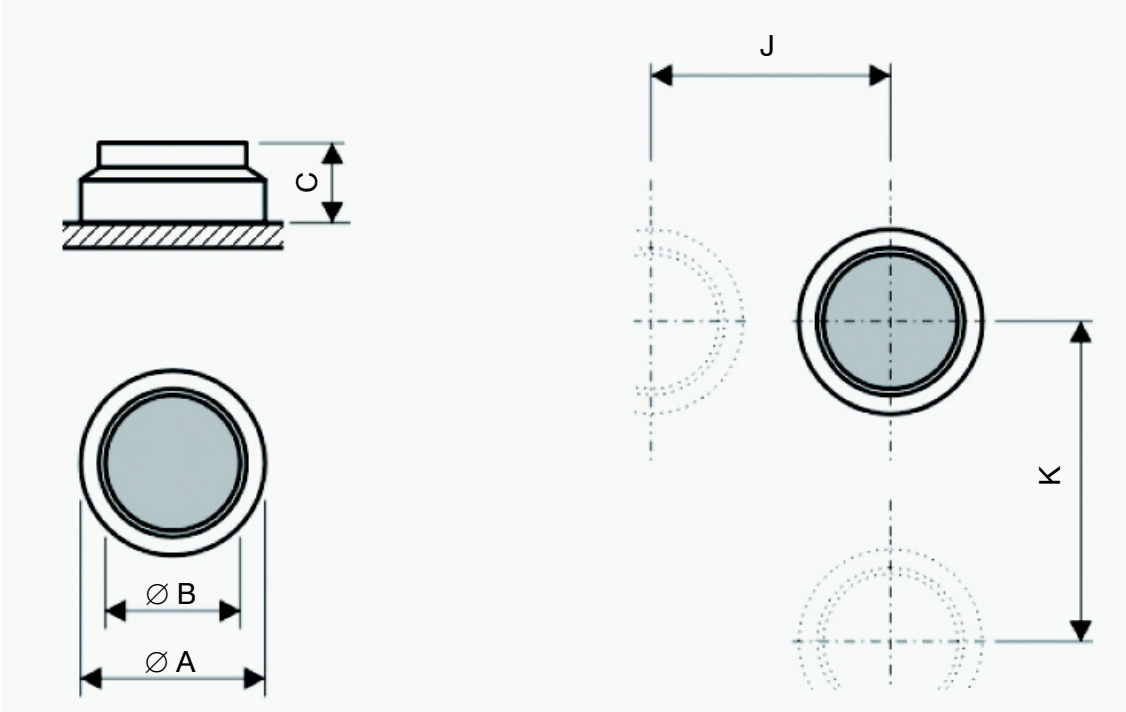


A [a]	min. 28 mm/max. 32 mm
B [a]	min. 22 mm/max. 32 mm
C	min. 1 mm/max. 13 mm
J [b]	min. 30 mm/max. 32 mm
K [b]	min. 40 mm/max. 60 mm

a. The distance B should be at least 75 % of distance A
b. Min. corresponds to the minimum suitable distance between the elements.
Max. corresponds to the maximum distance permitted between the elements.

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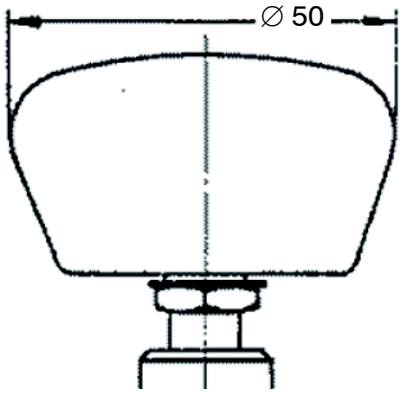
K.6.2 - Geometric form R (round form)



A [a]	min. 28 mm/max. 32 mm
B [a]	min. 22 mm/max. 32 mm
C	min. 1 mm/max. 13 mm
J [b]	min. 30 mm/max. 32 mm
K [b]	min. 40 mm/max. 60 mm

a. The distance B should be at least 75 % of distance A
b. Min. corresponds to the minimum suitable distance between the elements.
Max. corresponds to the maximum distance permitted between the elements.

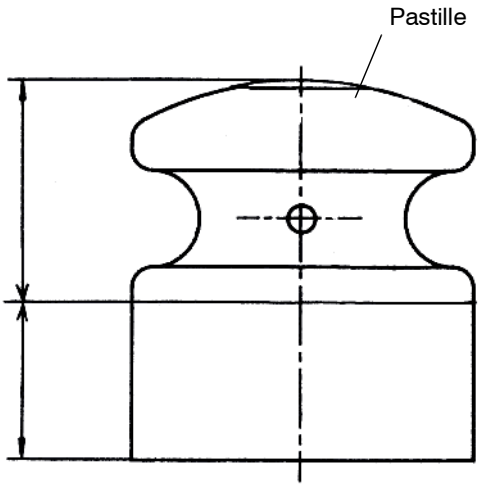
K.6.3 - Geometric form X (DAC push button)



Force to operate the push button: 12 N

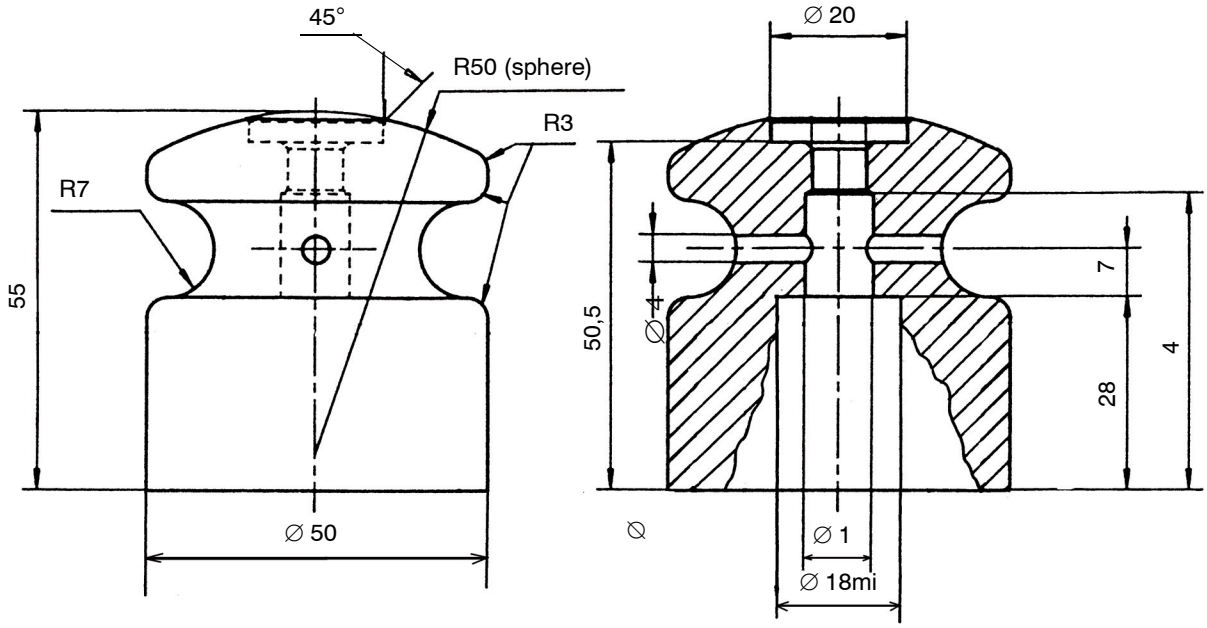
To operate DAC push for 5 mm

K.6.4 - Geometric form P (emergency stop valve - "mushroom button")



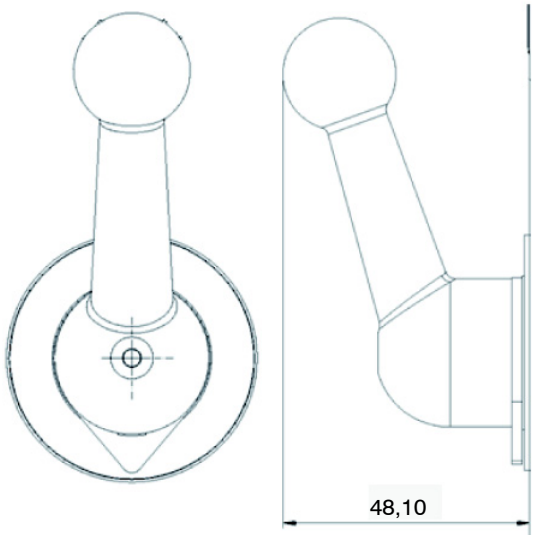
Epoxy paint, colour: vermillon red 605 in accordance with FI 01-007

White epoxy paint

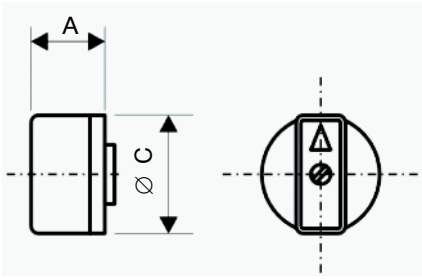


K.7 - Turn switches

K.7.1 - Geometric form A (Handle for auxiliary desk traction / brake controller)

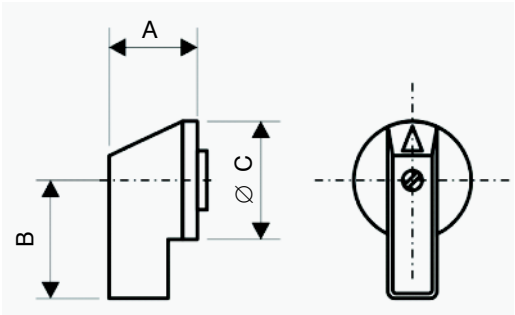


K.7.2 - Geometric form E



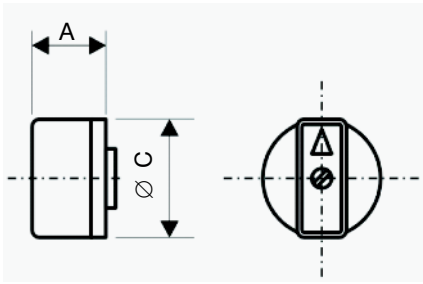
A min. 15 mm/max. 18 mm
C min. 24 mm/max. 28 mm

K.7.3 - Geometric form F



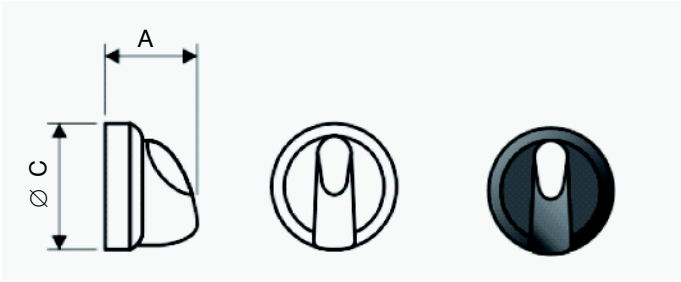
A min. 16 mm/max. 20 mm
B min. 20 mm/max. 23 mm
C min. 24 mm/max. 28 mm

K.7.4 - Geometric form G



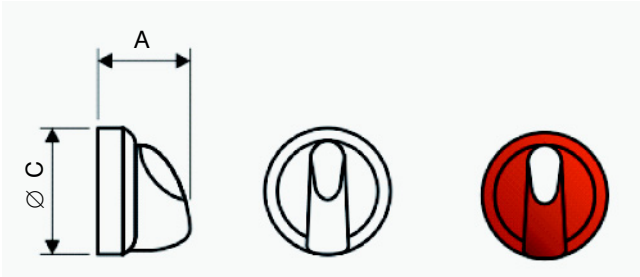
A	min. 21 mm/max. 25 mm
C	min. 36 mm/max. 40 mm

K.7.5 - Geometric form H



A	min. 20 mm/max. 24 mm
C	min. 28 mm/max. 33 mm

K.7.6 - Geometric form S

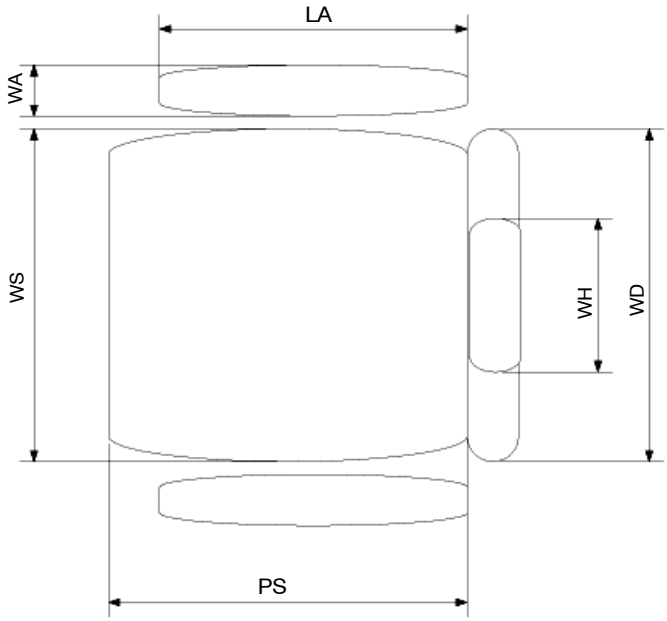


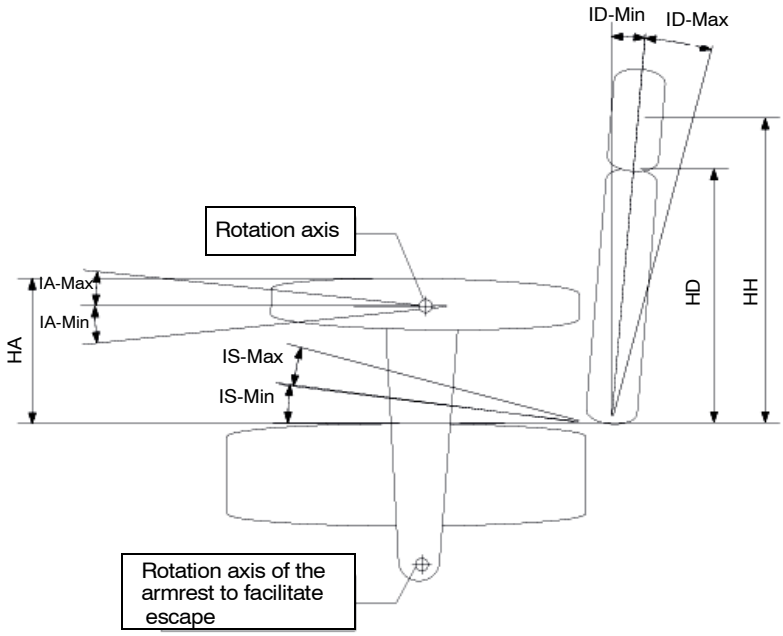
A	min. 20 mm/max. 24 mm
C	min. 28 mm/max. 33 mm

K.8 - "Swan neck" microphone (pos. 030)

(Stalk microphone)

K.9 - Driver's seat





N°	Function/Seat parameter	Requirements	Tolerances
1.	Fixation on the floor of the cab	Easily removable (e.g. fixation with screws)	
2.	Sturdiness of fixation with a maximum load of 1 500 N in the longitudinal axis	Acceleration 5 g	
3.	Sturdiness of fixation with a maximum load of 1 500 N in the transversal axis	Acceleration 1 g	
4.	Space for standing driving in front of the desk	The furthest backward position + 250 mm	± 10 mm

N°	Function/Seat parameter	Requirements	Tolerances
5.	Stability of the furthest backward position	Locked at this position	
6.	Height of the middle part of the head (HH)	Continuous adjustment from 700 mm to 855 mm	± 10 mm
7.	Height of the seat head	200 mm	± 10 mm
8.	Width of seat head (WH)	300 mm	± 10 mm
	Height of the back (HD)	600 mm	± 10 mm
9.	Width of the back (WD)	450 mm	± 10 mm
10.	Adjustment of the angle of the back (ID-Min/ID-Max)	Continuous adjustment from 5° to 15°	±1°
11.	Lumbar supports	Adjustment of the curve of the concavity seat back: 10 to 20 mm	
12.	Concavity (swell) seat back	Between 180 and 230 mm height above loaded seat surface	
13.	Width of the seat (WS)	450 mm	± 10 mm
14.	Length of the seat (PS)	Continuous adjustment from 350 mm to 480 mm	± 10 mm
15.	Shape of the seat	Shall be designed to avoid the compression of leg muscles	
16.	Adjustment of the angle of the seat (IS-Min/IS-Max)	Continuous adjustment from +6° to -6°	± 1°
17.	Number of armrests	2, one on each side of the seat	
18.	Minimum distance between the armrests	450 mm	± 10 mm
19.	Maximum load on the armrests	1 400 N at the front end	
20.	Adjustment of the height of the armrests	Continuous adjustment from 180 mm to 280 mm	± 10 mm
21.	Adjustment of the angle of the armrests	Continuous adjustment from +6° to -6°	± 1°
22.	Width of armrest (WA)	60 mm to 80 mm	± 10 mm
23.	Length of armrests (LA)	250 mm to 300 mm	± 10 mm
24.	Rotation of the seat	The rotation of the seat, when it is in the furthest forward position, shall not be blocked	
25.	Retractability of armrests	Easy and quickly: axis rotation below the seat	
26.	Shape of armrests	The rear part shall be designed to avoid tearing jackets (or trouser pockets)	
27.	Weight of the driver	45 kg to 140 kg	± 5kg

N°	Function/Seat parameter	Requirements	Tolerances
28.	Stability of driver seating for transversal acceleration	Enough for a cant insufficiency of 180 mm at 350 km/h	
29.	Damping for vertical vibrations	Automatic adjustment to the driver's weight	
30.	Effort to rotate the seat in case of emergency escape	First 45 Nm (+ 5 Nm) to unlock the seat and then 15 Nm (- 5 Nm) to continue the rotation	
31.	Rotation angle of the seat in case of emergency escape	At least 120° to the right and to the left	
32.	Activation of the adjustment	Easy, with minimal movement. Logical connection between direction of adjustment of controls and direction in which the seat moves.	
33.	Safety adjustment	The actuators for adjustment shall be safe for the driver	
34.	Identity of each actuator for adjustment	Clear and easily understandable	
35.	Possibility of hanging up a jacket	A system shall be integrated in the rear of the seat	
36.	Time of use: this time is given to help designer in the choice of the different components for the seat (e.g. textiles, suspension, …) but also to avoid compression of leg muscles	6 hours continuously	± 30 min
37.	Vertical adjustability of seat	Min 120 mm (if adjustable foot rest, reduced value of 80 mm)	
38.	Horizontal adjustability of seat	Min 200 mm	
39.	Comfortable lateral guidance	Designed under ergonomic aspects	
40.	Front edge of seat surface	Rounded	
41.	Automatic fixation in driving position	By notch, but without locking device	
42.	Seat materials	Breathable, cleanable, without electrostatic charge	

→ running direction of the EMU/DMU, locomotive or driving coach

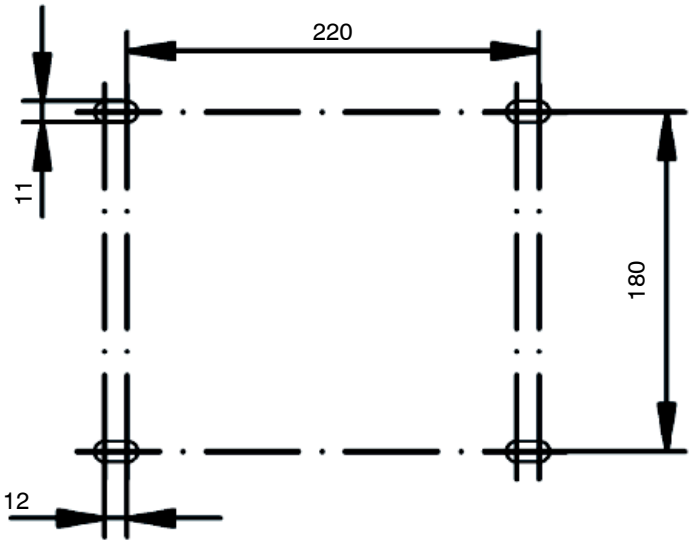


Fig. 38 - Mechanical interface between driver's seat and cab body frame

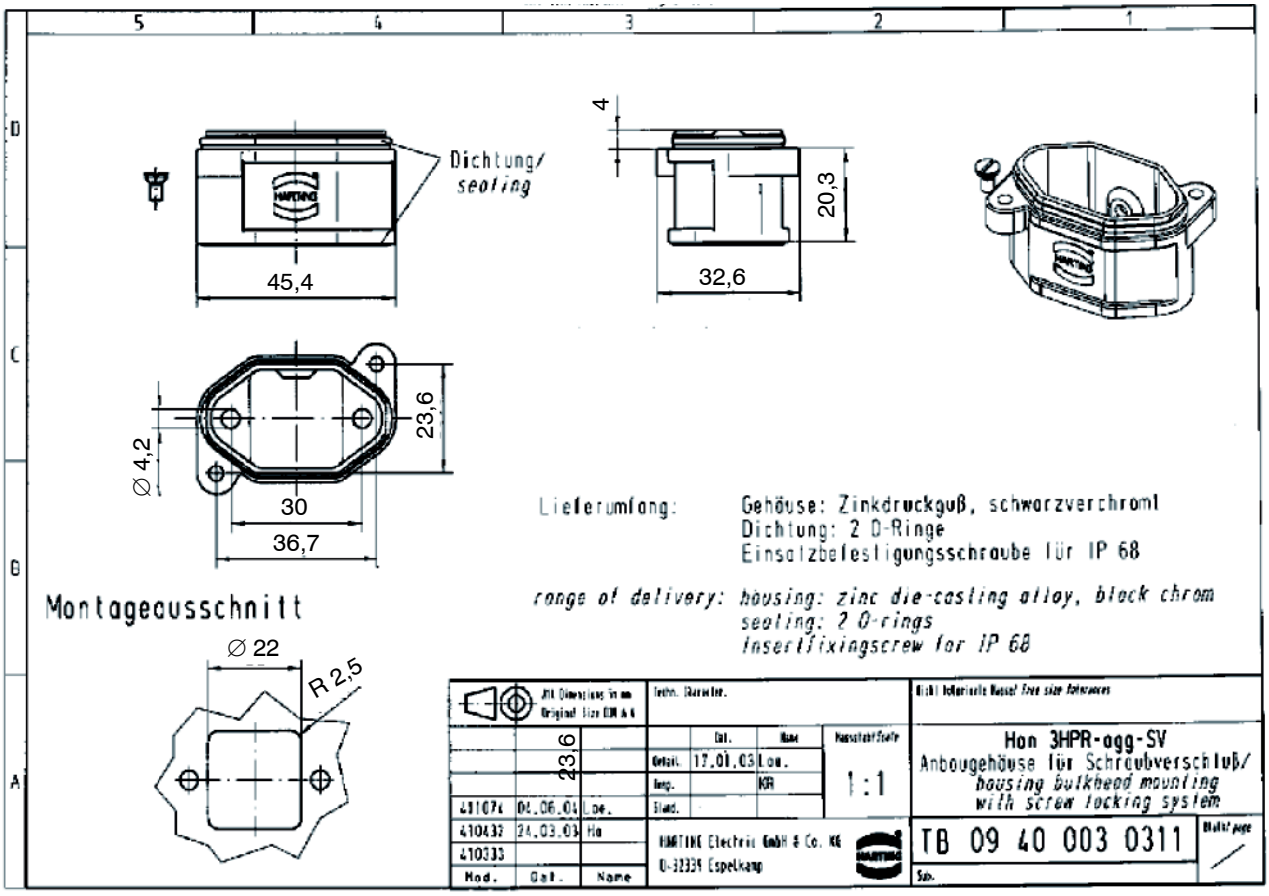


Fig. 39 - Electrical interface between driver's seat and vehicle

Appendix L - Automatic Speed Control (ASC)

ONLY for Trains using automatic speed control (ASC).

(This Appendix applies only to DB and ÖBB)

General

ASC (Automatic speed control) is an optional function for supporting the driver while driving a train in all operational situations:

- Starting,
- Keeping the speed constant,
- Change of speed,
- Braking to a stop

By operating pos. 019 the driver may switch the ASC "on" or "off" at any time while at standstill or running.

ASC activation criteria:

- driver has set pos. 019 to "V ≥ 0 km/h" **and**
- electronic brake pipe control is available **and**
- direction of travel has been chosen (pos. 033: "F", in case of pos. 035 ("R") a speed limit of 50 km/h applies - see *UIC Leaflet 612-0*, pos. 019)

Once all activation criteria have been fulfilled, ASC is ready for operation by shifting pos. 019 from "OFF"-position to "V >/= 0 km/h". ASC shall be available unless a fault message is displayed on the TDD (see *UIC Leaflet 612-03*).

Traction inhibition due to ASC shall apply after ASC activation if a brake command has been given from another function than ASC (e. g. pos. 032a, pos. 032b or pos. 020 has been set in braking position or an emergency brake applies). Traction inhibition is cancelled after the brake command has been released **and** pos. 020 has been set into "0"-position ("0-acknowledgement").

Independent from any traction inhibition, ASC shall continue to automatically control the automatic brake including all dynamic brakes but only if no higher brake application is requested by a function other than ASC.

ASC deactivation criteria:

- pos. 019 is set to "OFF" **and** pos. 020 is set to "0" for at least 1 s **or**
- no electronic brake pipe control is available **or**
- no direction of travel is recognised by TCMS.

However, if a brake operation has been commanded but not completed before deactivation occurs, ASC will continue to control the brake unless the command has been completed. ASC then shall become deactivated and no "V_{target}" information by ASC shall be displayed on the speedometer in the CCD.

If traction power is applied when setting pos. 019 to "OFF", this traction power will remain constant unless pos. 020 has been set to "0".

ASC will control the speed set manually by pos. 019 or automatically according to the permitted speed profile set by an on board signalling system. ASC shall therefore automatically control the traction power, the dynamic brake force and, if necessary, the automatic air brake (BP pressure).

TCMS shall provide available train and line-side data to ASC for generating a train calculation model. This data shall encompass:

- characteristic parameters for braking mode and the major friction brake elements (pads or blocks) of the train (e. g. provided by the train's nominal timetable data)
- the braking performance by braked weight percentage or deceleration function
- the maximum speed limit of the train, possibly also limited by the on board signalling system's speed profile
- the length of the train
- as an option: train mass (shall be cross-checked with train length in terms of plausibility)
- as an option: line characteristics (e. g. line gradients).

Entry or processing of data on train mass and/or type of friction brake elements in the train may not be requested for calculation of braking curves by the on-board signalling system, but it is recommended for improving the quality of ASC. However, ASC shall still be available if train data is incomplete or faulty, even if the quality of speed control is less reliable.

Once the driver has acknowledged the system, ASC shall accept any change of train data, whether it is done automatically or manually. ASC shall accept any change of line-side data without any driver acknowledgement.

The train data shall only be accepted by TCMS if it is within the following limits:

Braked weight percentage	30 - 250 %
Train length	20 - 2 000 m
Train mass	80 - 8 000 t

Regarding speed, ASC shall always accept the last available speed setting, even if the previous setting has not yet been fully realised. If an on-board signalling system provides a permitted speed profile, it shall immediately become valid for the ASC. As a consequence, the driver may set pos. 020 at maximum traction power and pos. 019 at maximum speed and then she/he simply needs to monitor if ASC is minimising the difference between permitted and realised speed without any over-speeding.

Manual speed setting other than by pos. 019, e. g. by an impulse lever (where the duration of the driver application is directly proportional to the speed difference selected) is not recommended as this would make the ASC "unpredictable" for the driver.

If the driver changes the maximum traction power by pos. 020 and/or the target speed by pos. 019, for example for the purpose of energy efficient driving, ASC shall select the minimum values amongst any available input command settings.

The ASC shall be capable of distributing the brake force to be applied by the train to the automatic (indirect) pneumatic train brake and the dynamic brake. Dynamic braking shall always be prior (see *UIC Leaflet 612-2*).

Driving with ASC

Full "start to stop cycle" summarised:

- At standstill the holding brake shall automatically and immediately be applied (see UIC 612-2).
- The driver may set pos. 019 to "V-target ≥ 5 km/h" and pos. 020 or pos. 032b to "T > 0"
- If T > 10 kN/loco becomes effective, the holding brake shall automatically be released
- Starting: Traction power T will increase in accordance with the requirements specified below.
- Keeping the speed constant: see below
- Braking: If pos. 019 has been set to "0 km/h" while driving, service braking will be enough to stop the train. If pos. 019 has been set from $V_{target1}$ to $V_{target2}$ with $V_{target2} < V_{target1}$, a service brake application shall meet $V_{target2}$. The same applies if an on-board signalling system provides a permitted speed profile. Braking shall follow the parameters specified below.

Starting in 3 phases (acceleration):

1. Traction power build-up gradient until the "lengthening force" is obtained
   - train with only one EMU/DMU or locomotive: 75 kN/s
   - train with more than one EMU/DMU or locomotive: 37,5 kN/s per EMU/DMU/locomotive
2. Max. traction power per train for lengthening the train after starting ("lengthening force"):
   - 50 kN for a train with a mass up to 250 t
   - 100 kN for a train with a mass higher than 2000 t
   - for a train with a mass between 250 t and 2000 t, ASC shall linearly interpolate the traction power build-up gradient. ASC may get the train mass from TCMS train data input or by modelling based on train length and selected brake mode
   - default value for a train mass remaining unknown: 60 kN

   "Lengthening force" shall be applied for duration depending on the type of train:
   train in brake mode "R" or "P" and braked weight percentage (BWP) more than 65 %":
   - 3 s for train length up to 100 m
   - 5 s for train length of more than 450 m
   - linear interpolation for train length between 100 m and 450 m

train in brake mode "G" or "P and BWP up to 65 %":

- 5 s for train length up to 100 m
- 10 s for train length of more than 800 m
- linear interpolation for train length between 100 m and 800 m

3. Increasing the traction power up to the limit.
The traction power upper limit shall be derived from the value set by pos. 020 and from the acceleration settings ("A_{target}", see below). The traction power shall be applied for reaching the maximum permitted speed without exceeding it. The traction power gradient shall depend on the type of train and shall apply for each unit in the train:

- Train in brake mode "R" or "P **and** BWP more than 65 %": 18 kN/s
- Train in brake mode "G" or "P **and** BWP up to 65 %": 10 kN/s
- Starting the train rapidly, if the above mentioned gradients are not sufficient: 30 kN/s
- In case of the train rolling back while starting (negative speed detected for a duration of at least 2 s): 37,5 kN/s

Keeping the speed constant:

Acceleration and deceleration shall be "0", i. e. ASC shall control the speed so that:
$(V_{target} - V_{actual}) < \pm 3$ km/h.

Braking:

Deceleration "A_{target}" (see below) shall be constant.

ASC shall initiate a decrease of the brake pipe (BP) pressure of at least:

Passenger trains	> 100 m	→	0,4 bar
Freight trains	< 85 % BRH	→	0,7 bar
Freight trains	> 85 % BRH	→	0,5 bar

Acceleration and deceleration "A_{target}":

For determining the traction power in "starting - phase 3" (see above), the following acceleration basic settings "A_{target}" may apply:

- Train in brake mode "R" **or** "BWP more than 99 %" **and** "length of train less than 400 m": 0,9 m/s^2

- Train in brake mode "P" **or** "BWP more than 99 %" **or** "length of train at least 400 m": 0,7 m/s^2

- Train in brake mode "G" **or** "BWP up to 99 %": 0,5 m/s^2

As an option in the TDD (see *UIC Leaflet 612-03*) it may be possible to select acceleration settings different from the basic setting. The selection should be possible in steps of 0,2 m/s^2 between 0,3 m/s^2 and 1,3 m/s^2. The selected acceleration shall remain valid until the next train data entry.

For braking the basic setting for the target **deceleration** "A_{target}" may be:

- Train in brake mode "R" **or** "BWP more than 99 %" **and** "length of train less than 400 m": 0,7 m/s^2
- Train in brake mode "P" **or** "BWP more than 99 %" **or** "length of train at least 400 m": 0,5 m/s^2
- Train in brake mode "G" **or** "BWP up to 99 %": 0,33 m/s^2

Transition from manual speed command (following the setting of pos. 019) to automatic speed command (following the permitted speed profile of the on board signalling system):

- no "0-acknowledgement" by pos. 020 shall be necessary
- traction power set by pos. 020 shall apply as maximum limit for both types of speed command

Transition from automatic speed command (following the permitted speed profile of the on board signalling system) to manual speed command (following the setting of pos. 019):

- When the on board signalling system can no longer provide a permitted speed profile or if transmission of signalling information from ground to train fails, then the CCD shall request first driver acknowledgement by operating pos. 014 within a time-frame of 10 s. After acknowledging, the driver accepts to mind the track side signalling and to control speed manually. Once the permitted speed profile of the on board signalling system is no longer available or no longer displayed on the CCD, ASC shall inhibit traction power, but still control the automatic brake. In case of missing driver acknowledgement by pos. 014 after 10 s, a service brake may be commanded by the on board signalling system until standstill or driver acknowledgement by pos. 014. ASC shall realise this service brake.
- After a second driver acknowledgement by setting pos. 019 first in "V = 0 km/h" position and then in a "V > 0 km/h" position, ASC shall become effective again. In case of missing driver acknowledgement by pos. 019, the following procedure shall apply:
  - Keeping the speed constant for 4 s, based on the actual speed at the end of speed monitoring by the on board signalling system (this means for ASC applying traction power in an ascending slope and braking in a downward slope).
  - After 4 s and still missing driver acknowledgement by pos. 019, ASC shall apply a traction inhibition and only control the braking in order not to exceed the speed mentioned above.

In the degraded mode of no on board signalling system being effective, the speed shall automatically be limited to V_{max} = 100 km/h.

In the degraded mode of ASC, TDD shall display a request for "0 acknowledgement" by pos. 020. The failure management shall advise the driver what to do, if no automatic mitigation of the failure is possible. If ASC is not able to become deactivated, a traction inhibition shall apply in addition.

- In case of keeping the speed constant or accelerating, ASC shall immediately become deactivated.
- In case of braking, ASC shall continue the braking and not become deactivated unless the braking has been accomplished. Only if ASC can no longer control the braking, it shall become deactivated immediately.

- ASC shall be considered to be degraded in case of "V_{target}" being constant for more than 30 s and the actual speed exceeding "V_{target}" more than 4 km/h for at least 10 s.

- ASC may be re-activated by manually setting pos. 019 first in "V = 0 km/h" position and then in a "V > 0 km/h" position.

Holding brake application due to ASC:

- In braking to standstill, whether automatically by ASC or manually by the driver, the holding brake shall apply below a certain speed (around 12 km/h, depending on train data) such that it becomes fully effective at standstill.

- While braking to standstill ASC shall decrease the BP pressure to 4,6 bars at a speed of 2,0 km/h and then fully release the automatic train brake at 0,5 km/h. Due to parallel decrease of the dynamic brake, the train shall come to a standstill with the residual brake force of the automatic train brake and the holding brake. If necessary, ASC shall decrease the brake pipe pressure again in order to provide the necessary brake force.

- If the holding brake force is not sufficient and subsequently the train is unintentionally moving, then ASC shall reduce the BP pressure continuously until TCMS recognises standstill. Unintentional movement is defined as a movement > 5 m forwards or backwards.

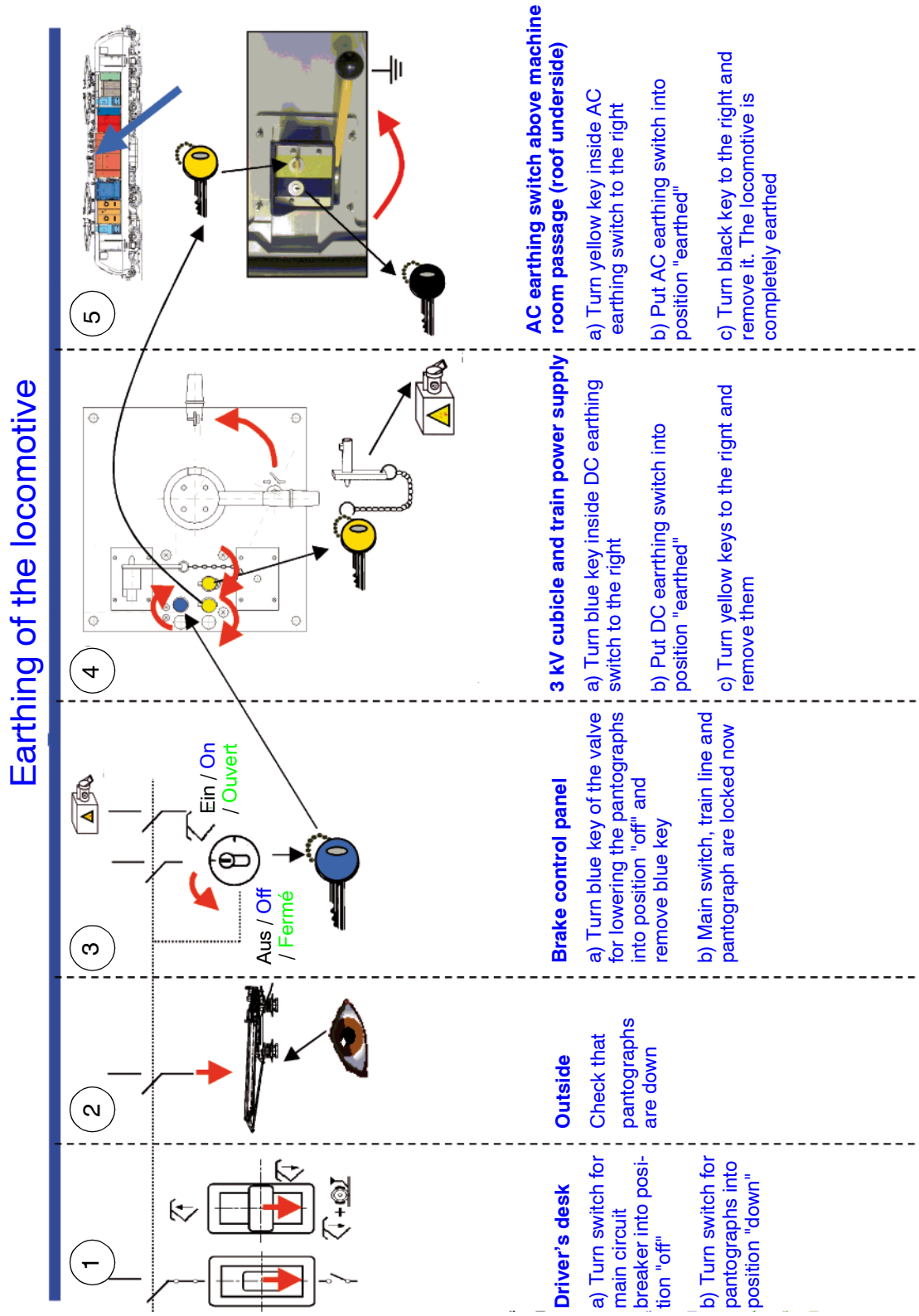
Holding brake release due to ASC:

- If pos. 019 has been set in a "V > 0 km/h" position or if ASC is deactivated or degraded **and** "0 acknowledgement" by pos. 020 has been done **and** the actual traction power has reached at least 10 kN/(EMU/DMU or locomotive), then ASC shall immediately and rapidly release the holding brake.

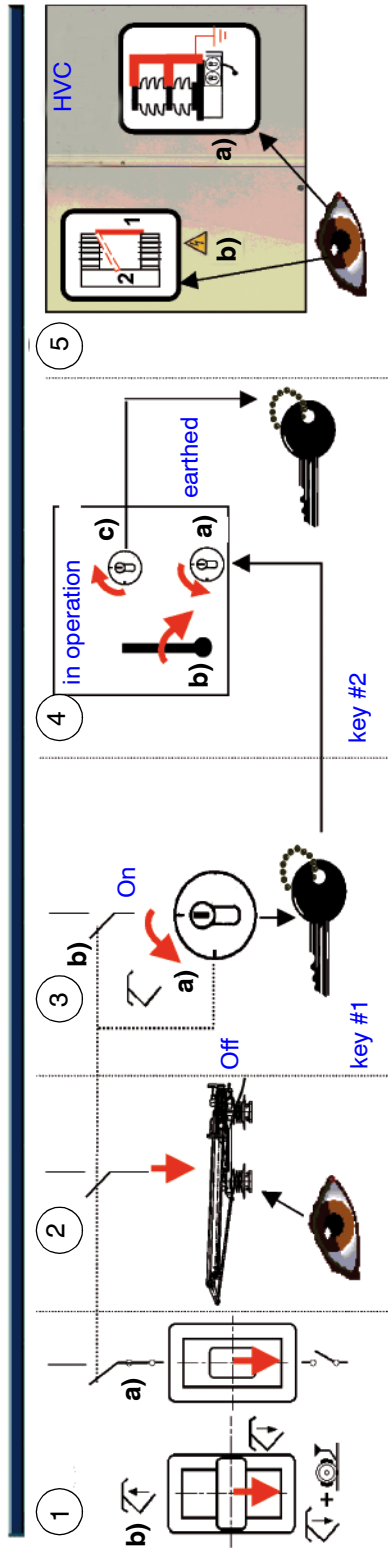
- If ASC is deactivated at standstill, it shall not release the holding brake. TCMS shall release the holding brake, if the actual traction power has reached at least 10 kN/(EMU/DMU or locomotive).

- The holding brake shall also be released by setting pos. 037 to the middle position and then to the release position.

Appendix M - Examples for earthing for electric locomotives and EMUs



Earthing of the locomotive



Driver's desk

a) Turn main switch into position "off"

b) Turn switch for pantographs into position "down"

Outside

Check that pantographs are down

Brake control panel

a) Turn key#1 of the valve for lowering the pantographs into position "off". Remove key #1

b) Main switch and pantograph are locked now

High voltage cubicle (HVC)

a) Push key #1 into the lock of the earthing switch and turn left

b) Switch ezrthing switch into position "earthed"

c) Turn key #2 of HV cubicle to the right and remove. Earthing switch is locked now

Opening of HVC by authorized staff only

High voltage cubicle (HVC)

a) Check, that the main switch is earthed

b) If present the disconnector for the pantographs is in one of the two end positions (1 or 2)

Appendix N - Specific cases

N.1 - Specific cases "DB, ÖBB"

1.

Operating element Position and [Form]		Symbol/ indication	Positions and function		Explanations Mandatory preconditions Miscellaneous	Operational constraint for				
						Master cab status	MCB	v=0	v>0	MT
003	**Only for STM "LZB":** **shall be replaced by STM information in CCD (Icon "Ü")** Illuminated push switch [Q]		**S**	- Indicator light of the STM "LZB": lights up if data is transmitted between route (trackside CCS) and vehicle ("Übertragung") - Dimming the control lamp by pressing the switch, requirement for light: 0,3 lx + 20 lx/s, max. 150 lx						

2.
Auxiliary desk (Appendix D - page 56): Where two auxiliary desks are installed in the same cab it shall be possible to change from one desk to another (e. g. from left side window to right side window) at speeds of up to 40 km/h.

N.2 - Specific cases "SNCF"

1. The outside end cocks for the hose couplings of the main air reservoir pipe (MP) shall be marked in white to comply with the French law.

2. Emergency stop valve with "Emergency stop" function - Pos. 010: **The emergency stop valve shall comply with** the French law SAM-F-301.

3. Sanding function - Pos. 025: **The sanding function shall comply with** the French law SAM-S-901 **to be used on the French national railway network**.

4. "Release brake" function - Pos. 026: **The "release brake" function shall be activated only if the speed V ≤ 5 km/h**.

5. Driver's automatic brake controller - Pos. 032: The driver's automatic brake controller shall be a time dependent controller and shall fulfil the French specification STM F 301 and 302. In this case, the BP and Pilot Chamber (PC) pressures of the driver's brake valve information shall be given to the driver. This information and other pressure information can be displayed on a screen or on a gauge. The BP and PCP pressures shall be displayed together for easy comparison by the driver. The requirements for the PC pressure information are identical to the BP pressure information requirements given in Pos. 007. The MP pressure information is displayed alone. The position dependent combined traction/brake Monomaster controller is also a possible option for SNCF for EMU/DMUs.

6. Pressure information and EMU/DMU, locomotive or driving coach shut down - Pos. 007 and 008 and PC pressure information: It is not mandatory **to have or display the BP, MP, Brake Cylinder and Pilot Chamber Pressures information in case of an EMU/DMU, locomotive or driving coach in parking mode and in slave operation mode**.

7. Dynamic brake - Pos. 020 and 032a: **The regenerative dynamic brake shall comply with the French law SAM-F-006 to be used on the French national railway network. The operating criteria for the use of the regenerative dynamic brake depend on national infrastructure register and not only on the rolling stock**. If the EMU/DMU or locomotive is equipped with brake blocks, before the activation of the electro-dynamic brake the mechanical brake (brake blocks) shall be applied to clean the wheels to improve the adhesion between the wheel and the rail.

8. Direct brake controller - Pos. 037: It is not mandatory to have at least 30 KN per set of wheels at an average wheel diameter. The direct brake force per wheelset can be lower than 30 KN but the direct brake force shall be determined in accordance with the braking performance requested and at least in accordance with the French law SAM F 005. The direct brake commands shall not be transmitted to slave locomotives.

9. The activation of an auxiliary desk shall be done when the train is at standstill. Traction shall be automatically cut off as long as the speed is higher than 30 Km/h (tolerance to be obtained from BB 475000) and a status message shall be given to the driver according to *UIC Leaflet 612-03*.

10. Connecting HV to earth device - Pos. 510: **In addition to the requirements of the Pos. 510 "Connecting HV to earth device"**, the French specification **for safety requirements for electrical hazards** STM E 014 **shall be fulfilled. This document is the** French implementation of the *EN 50153*.

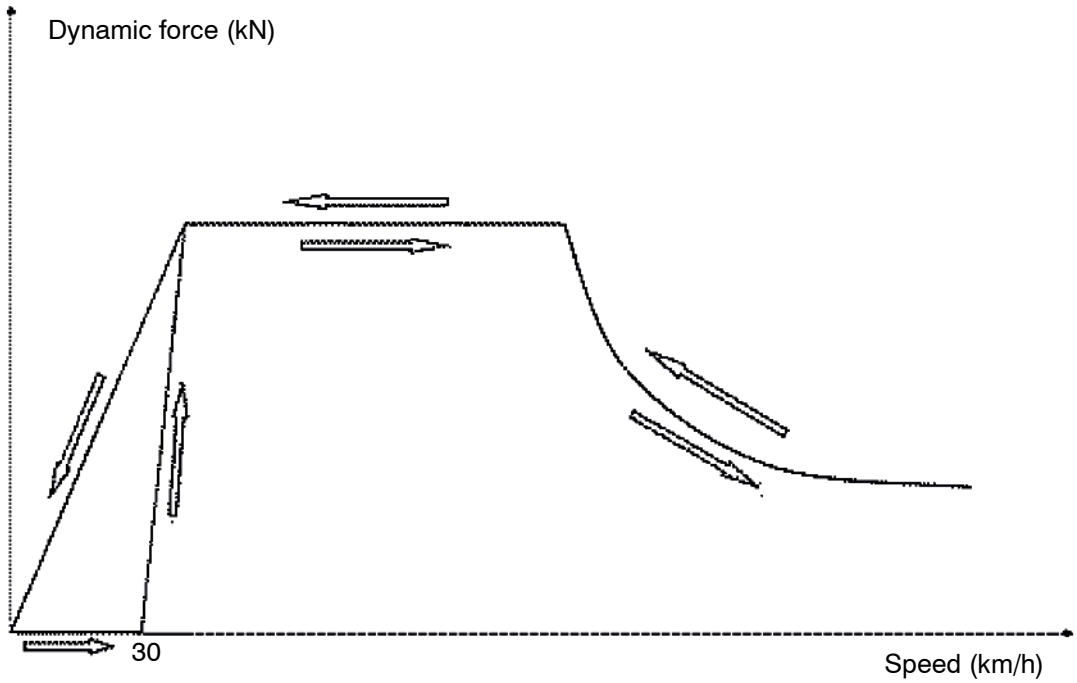
11. Automatic adjustment function - *UIC Leaflet 612-2*: **The following automatic adjustment function "During all manual and automatic braking processes with the automatic brake, a BP excess pressure ("offset") that corresponds with the previous BP reduction is added to the original BP pressure before braking, after the complete release of the brakes. This offset should be at least 0,1 bar, increase in proportion with the braking position and shall not exceed the maximum value of 0,2 bar after full application of the brakes (BP = 3,5 bar). The offset applies after reaching the regular BP-pressure (5 bar) in addition and shall not be implemented in an SNCF locomotive**.

12. Electrical traction (pantograph) or diesel-units (diesel-engine) - Pos. 016: It is not mandatory to have a soft key for this function: **A) "Choose current value" ➔ additional hardware device on the main desk requested and B) "Choose local pantograph" ➔ additional hardware device on the main desk requested**.

N.3 - Specific cases "Trenitalia"

1. Pos. 015: yellow illuminated push-button instead of yellow lever
2. Pos. 026 activation criteria: V ≤ 5 km/h
3. Pos. 032a: The reduction of the BP pressure under 5 bar will be converted into a set point value for the dynamic brake of the locomotive: 5 bar = 0 %, full service brake = 100 %, linear interpolation in between. For Trenitalia 100 % is already achieved at BP = 4,2 bar.
4. Manual application of the parking brake in addition to the holding brake is still requested. The driver shall check the efficiency of the parking brake by applying traction power.
5. For the parking brake an independent external indicator shall be provided.
6. When a parking brake cylinder is unlocked by the handle placed on the bogie two different situations can occur:

   - If every axle equipped with parking brake has its own parking brake indicator, the latter becomes green instead of red;
   - If there is only one parking indicator for each bogie, it becomes green only when the second parking brake cylinder is also unlocked.



Appendix O - Guideline for the revision of *UIC Leaflet 612 series* - Human factors aspects and standardisation benefits

O.1 - General introduction

Driving cabs of rail vehicles shall be designed and maintained so that they provide safe and efficient working environments in which drivers and other authorised staff can carry out their duties safely and effectively. Providing the driver with an environment that is safe and efficient will reduce the likelihood of errors occurring and therefore reduce the likelihood of incidents. This will reduce costs due to incidents associated with human errors - both for the railways and for the whole environment. Driver cabs which comply with TSI and EN are deemed to be safe. Driver cabs which in addition comply with this UIC-standard can be considered to provide efficient working conditions.

Basic requirements for driving cabs have been provided by *UIC Leaflet 651*, last update in 07/2002, that has served not only as a recommendation but moreover as the legal reference for rolling stock homologation regarding essential driving aspects for many national safety authorities.

Within the principle framework set by *UIC Leaflet 651* various different driving cab designs have been developed and operated. Meeting the objectives set out in point 1 of this standard requests further progress in standardisation of functional driving requirements and a number of key interfaces to the driver.

Where those interfaces have not already been mandated by European law (e. g. for ETCS) or unambiguously recommended by European standards, then *UIC Leaflet 612 series* paves the way towards converging operators' requirements on a European level.

Future human factors analyses therefore should be based on design according to *UIC Leaflet 612 series* requirements and support the migration from this common starting point to a common upgrade of *UIC Leaflet 612 series*, where requirements are increasingly driven by future operational tasks and rolling stock compatibility and less by heritage constraints of rolling stock operation. Then different requirements may be justified by different operational tasks of the rolling stock. Human factors analyses may be understood as a contributor to both risk analysis and cost-benefit analysis in the process of introducing technical innovation and its adverse impact on the upgrade of operational requirements. However, the mere availability of a technical innovation should not supersede standardisation, if its cost-benefit has not been quantitatively demonstrated. The revision process shall demonstrate improvements in costs and increasing degree of standardisation following the revision loop "check - improve - test - apply - check - …".

Human Factors involves the application of a wide range of tools and techniques. This is to take account the complexity of humans and the systems in which they work, as it is rarely possible for there to be a "one-answer-fits-all" solution. Human Factors good practice enables assessments to be made, e.g. of the humans, their environment, new systems and equipment, rules and procedures, training requirements etc. so that the final design is appropriate to the users and is relevant to the railway application. This will enable the humans to perform optimally and minimise the likelihood of errors that occur when there is a conflict between a design and the humans who are expected to operate it.

O.2 - Human Factors / Ergonomics Specialists

Due to the criticality of the interface between the driver and the cab controls it is essential to ensure that all Human Factors work is carried out effectively. The specialist shall carry out Human Factors analyses; integration of Human Factors design requirements into the design process and testing of the design to assess its suitability and make recommendations for further improvement. The Human Factors Specialist should be qualified to at least degree level in a relevant subject, e.g. ergonomics, occupational psychology, or applied psychology.

O.3 - Use of tools and techniques

Human Factors is a discipline that uses tools and techniques to ensure that any design meets the needs of the end user, i.e. it is more than asking end users for their opinions. It is important to make this explicit in the standard as a means of ensuring that those with no human factors experience understand what is expected. Any development project aiming on introduction of innovation should use Human Factors tools and techniques to take into account the following:

- Task / functional analysis taking into account all area and traction-specific tasks as well as those required by the Operational Rules and general train operation;
- Definition of user requirements (defined with/by end users);
- Definition of good practice ergonomic design requirements;
- Identification and management of workload issues (overload & underload);
- Identification and mitigations of human error and the associated risks;
- Interfaces that allow the driver, whilst at the driving position, to observe and interpret the display(s)/indications and operate the associated controls (where provided) that are required to be used whilst the train is being driven or the intended vehicle movements being performed;
- Identification of any item which most likely will cause long-term health effects, which may arise from the ergonomics of the workplace;
- Iterative application of human factors testing, i.e. the design/development process should be iterative as more is learned about the design in relation to the human users through testing.

O.4 - Drivers' task / functional analysis

Driver's tasks are specified in *TSI Operations, Annex H*, and in *UIC Leaflet 612 series*.

The definition of the tasks should be done in normal and degraded conditions.

The task description defines the link between operational scenarios and operating elements taking into account the location of elements and the succession of the driver's movements and actions. This task description is supposed to avoid a sub-optimal desk/cab layout.

O.5 - User requirements

The driver in and outside the cab is the primary user; however the needs of all end users need to be considered to ensure that the DMI's do not pose any risks to other users. Users shall include drivers, shunting staff, guards, maintenance staff, driver managers & trainers, train preparers etc. For any revision of this standard the needs of all end users should be:

- Identified by carrying out suitable analyses, e.g. task/functional analyses, modelling users' characteristics, ways of working, reasoning and acting;

- Identified through consultation with end users (this consultation should continue throughout the project to ensure that the final solution is acceptable).

O.6 - Workload assessments

Workload assessments should be performed for any proposed changes to the cab environment that are likely to impact on the driving task, before it enters service (changes to the cab include new systems, displays, controls, alarms, automation, environmental controls etc.). The workload assessment shall take into account the four main demands upon the human user, i.e. Visual, Auditory, Cognitive and Psychomotor (VACP):

- Visual - requirement to view or monitor visual information sources, e.g. view the aspect of signals;

- Auditory - requirement to listen to, or monitor auditory information sources, e.g. listen to general broadcasts from line controller;

- Cognitive - requirement to mentally process information, e.g. process vision of red signal and make decision on correct course of action;

- Psychomotor - requirement to perform physical tasks, e.g. apply brake to slow or stop train, and speech activity.

Systems and equipment within the cab should not overload or underload the driver in terms of VACP workload requirements. Should this occur, the likelihood of human error, and subsequently the risk will be increased. The likelihood of mental overload may be reduced by understanding the tasks to be performed by the driver, and by careful consideration of how information (both visual and auditory) be presented to the driver.

A qualified and experienced Human Factors Specialist shall carry out any workload assessment. The following steps are suggested for inclusion in the process to determine the most effective method.

- Identify the objective of the workload assessment,

- Carry out the relevant analyses, e.g. task, system, functional etc,

- Assess constraints and resource availability,

- Select the appropriate types of workload measures,

- Select the specific workload techniques,

- Formulate the proposed modifications of *UIC Leaflet 612 series* for the evaluation.

O.7 - Human Error Analysis

To ensure that the likelihood of Human Error is minimised, Human Error Analysis shall be carried out using the Human Error Identification and Quantification tools. Especially after accidents the identification of human errors should be made available for further update of design recommendations.

List of abbreviations

ASC	Automatic Speed Control
ATC	Automatic Train Control
ATP	Automatic Train Protection
ATSS	STM used on Swedish infrastructure (Banverket)
BP	Brake Pipe
CCD	Control Command Display, (application for display of ETCS-information, available on any screen in the driver's cab, primarily on the central one, see *UIC Leaflet 612-02*)
CCS	Control Command System
CCTV	Closed Circuit Television (video surveillance inside passenger compartment)
CR	Conventional Rail
DAC	Driver Activity Control, TCMS function for monitoring driver machine interactions.
DDS	Driver display system, (a set of displays with standardized DMI, mounted in the main desk and interfacing with TCMS. The DDS displays all potential display-based applications (such as ETCS, diagnosis, train-radio, electronic timetable, video surveillance) to the driver on any of the available displays. For details see *UIC Leaflets 612-01 to 612-05*).
DDU	Driver Display Unit
DMI	Driver Machine Interface
DMU	Diesel Multiple Unit
EMU	Electric Multiple Unit
EN	European Norm
EOA	End of authority (ETCS)
ep	electro pneumatic (brake)
ERTMS	European Rail Traffic Management System

ETCS	European Train Control System
ETD	Electronic Timetable Display, (application for display of not ETCS redundant timetable information, available on any screen in the driver cab, primarily on the left one, *see UIC Leaflet 612-05*)
F	Travel direction "forwards"
G (mode)	brake mode "G"
ID	Identity (card)
JRU	Juridical Recording Unit
KVB/TVM	STM used on French infrastructure (RFF)
LCC	Life Cycle Cost
LED	Light Emitting Diode
LSS	Line Safety Switch
LV	Low voltage
LZB/PZB	Heritage ATP used on German infrastructure (DB)
MCB	Main Circuit Breaker (in UK: VCB, ABB)
MMI	Man Machine Interface
MP	Main (air) pipe
MSS	Motor safety switch
MT	Multiple Traction
MTBF	Mean Time Between Failure
N	No travel direction, but cab activated (driver's ID inserted)
OTM	On-track machinery (track construction, inspection or maintenance vehicle)
P (mode)	Brake mode "P"
PIS	Passenger Information System
PTR	Plateforme Technique & Recherche (UIC - Technology & Research Platform)
R (mode)	Brake mode "R"
R	Travel direction "reverse"

RAMS	Reliability Availability Maintainability Safety
RPS	STM used on French infrastructure (RFF)
RST	Rolling Stock
STM	Specific transmission module (ETCS)
TCMS	Train Control and Monitoring System, i. e. all means for rolling stock internal control of functions, whether by software or hardware
TDD	Technical & Diagnostic Display (application for display of TCMS information, available on any screen in the driver cab, primarily on the right one, *see UIC Leaflet 612-03*)
TFR	Target Failure Rate
TFT	Thin Film Transistor
TRD	Train Radio Display, (application for display of GSM-R- and optionally national analogous radio-information, available on any screen in the driver cab, primarily on the very left one, *see UIC Leaflet 612-04*)
TSI	Technical Specification for Interoperability
WSP	Wheel slide protection
WTB	Wired train bus
0	No travel direction and cab not activated

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